

Allosteric modulation of the CXCR4: CXCL12 axis by targeting receptor nanoclustering via the TMV-TMVI domain

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Abstract

CXCR4 is a ubiquitously expressed chemokine receptor that regulates leukocyte trafficking and arrest in both homeostatic and pathological states. It also participates in organogenesis, HIV-1 infection and tumor development. Despite the potential therapeutic benefit of CXCR4 antagonists, only one, plerixafor (AMD3100), which blocks the ligand-binding site, has reached the clinic. Recent advances in imaging and biophysical techniques have provided a richer understanding of the membrane organization and dynamics of this receptor. Activation of CXCR4 by CXCL12 reduces the number of CXCR4 monomers/dimers at the cell membrane and increases the formation of large nanoclusters, which are largely immobile and are required for correct cell orientation to chemoattractant gradients. Mechanistically, CXCR4 activation involves a structural motif defined by residues in TMV and TMVI. Using this structural motif as a template, we performed *in silico* molecular modeling followed by *in vitro* screening of a small compound library to identify negative allosteric modulators of CXCR4 that do not affect CXCL12 binding. We identified AGR1.137, a small molecule that abolishes CXCL12-mediated receptor nanoclustering and dynamics and blocks the ability of cells to sense CXCL12 gradients both *in vitro* and *in vivo* while preserving ligand binding and receptor internalization.

Significance Statement

The chemokine receptor CXCR4 and its ligand CXCL12 are key for development, hematopoiesis, neutrophil homeostasis and lymphocyte trafficking. The only commercially available CXCR4 antagonist currently approved for clinical use is plerixafor (AMD3100), a small compound that blocks the ligand-binding site. Unfortunately, its clinical use is limited by poor pharmacokinetics and adverse effects associated with long-term administration. Here, we performed *in silico* analyses of a small aromatic compound library and *in vitro* screening to identify allosteric CXCR4 modulators. These compounds abolish the ability of cells to sense chemoattractant gradients without affecting other ligand-mediated functions such as cAMP production or receptor internalization. The selected compounds were also effective *in vivo*, as demonstrated by reduced tumorigenesis and metastasis in a zebrafish tumor model. Our study describes a new approach to selectively alter some GPCR functions while maintaining other functions.

eLife assessment

This is an **important** study that describes an elegant modelling driven approach to design of allosteric antagonists for CXCR4 that have a selective effect on receptor nanocluster formation, cell polarisation and chemotaxis, but spare binding of CXCL12 to the receptor and inhibition of adenylate cyclase. This enables selective targeting of processes dependent upon cell polarisation and chemotaxis without impacting signalling effects and may avoid some of the toxicity associated with antagonists that target CXCL12 binding and thus block all CXCR4 signalling. The revised manuscript offers **convincing** evidence to support the claims. The modelling work is better described and additional data has been presented that better illustrates the unique features of the new antagonist. The *in vivo* studies in the zebrafish model open a path to studies in mammalian models.

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Introduction

CXC chemokine receptor 4 (CXCR4) is a homeostatic G protein-coupled receptor (GPCR) that is widely expressed in both embryonic and adult tissues (1 [↗](#)). It is also ubiquitously expressed in the hematopoietic system, where it plays a pivotal role in leukocyte trafficking and arrest in specific niches in homeostasis and disease. CXCR4 is essential for adaptive and innate immune responses, as well as in the organization and maintenance of the bone marrow (BM) (2 [↗](#)). Indeed, CXCR4 and its unique chemokine ligand CXC motif chemokine 12 (CXCL12) are largely responsible for the migration (3 [↗](#)), homing (4 [↗](#)) and survival (5 [↗](#)) of hematopoietic stem cells in the BM.

Results from studies in conditional *Cxcr4*-knockout mice point to a relevant role for CXCR4 in several non-hematopoietic tissues, for example, in regulating central nervous system development (6 [↗](#)), and in vasculature development in the gastrointestinal tract (7 [↗](#)) and the kidney (8 [↗](#)). In addition, CXCR4, together with CCR5, serve as primary co-receptors (with CD4) for HIV-1 fusion and entry into target cells (9 [↗](#)).

CXCR4 expression is frequently elevated in many cancers, including breast (10), ovarian (11), prostate (12), melanoma (13) and neuroblastoma (14), where it participates in tumor growth, tumor cell interactions with the microenvironment (15), vasculogenesis, angiogenesis (16) and metastasis (10). Indeed, increased CXCR4 expression in metastatic lesions correlates with tumor progression and with preferential metastatic sites of the primary tumor (17). Studies in mice have suggested that CXCR4 is a good target for cancer therapy, as blockade of its signaling impairs metastasis in several models (10, 18). Despite the essential role of the CXCR4/CXCL12 axis in physiology and pathology, the only commercially available CXCR4 antagonist approved for clinical use is plerixafor (AMD3100), which is indicated for the mobilization of stem cells from the BM to the peripheral blood in autologous transplantation (19).

Single particle tracking in total internal reflection fluorescence (SPT-TIRF) experiments and stimulated emission depletion super resolution microscopy (STED) have demonstrated that CXCR4 is organized at the cell membrane as monomers, dimers and small aggregates (groups of ≥ 3 receptors) termed nanoclusters, and that CXCL12 binding decreases the percentage of monomers/dimers and increases the formation of large nanoclusters (20). This mechanism is essential for CXCR4 signaling, which allows cells to orient themselves correctly in response to CXCL12 gradients (21). A triple-mutant CXCR4 (K239L/V242A/L246A; CXCR4mut), located in the N-terminal region of transmembrane (TM) VI, dimerizes but neither forms nanoclusters in response to CXCL12 nor supports CXCL12-induced directed cell migration, although it can induce some Ca^{2+} flux and is internalized after ligand binding (20). A cluster has been identified in CXCR4, defined by TMV and TMVI residues that form a hydrophobic lock connecting the signal from the orthosteric ligand-binding site to the downstream conserved domains involved in G protein coupling and signaling (22, 23).

Here, we used this signal domain of CXCR4 as a base to search for new CXCR4 antagonists. We performed an *in silico* screening of a small aromatic compound library for the identification of allosteric modulators that block directed cell migration, without interfering with ligand binding and, thus, allowing other CXCL12-mediated functions. Using molecular modeling analyses, we identified a novel class of structurally related small molecules that modulate some CXCR4 functions by binding to a novel regulatory cleft formed by the TMV and TMVI transmembrane helices. One of the selected compounds, AGR1.137, abolished CXCL12-mediated receptor nanoclustering and dynamics and the ability of the cells to specifically sense CXCL12 gradients, but did not alter ligand binding or receptor internalization. Notably, AGR1.137 had a minimal effect on ERK1/2 and AKT phosphorylation, which may reduce the side effects associated with full CXCR4 inhibition. Finally, using a zebrafish tumor model, we observed that AGR1.137 treatment of HeLa cells reduced tumorigenesis and metastasis, suggesting that AGR1.137 also impairs cell sensitivity to CXCL12 gradients by blocking CXCR4 nanoclustering *in vivo*. In conclusion, our results establish the importance of residues within the TMV–VI cleft for CXCR4 nanoclustering and CXCL12-mediated directional migration, and identify a negative allosteric modulator that exhibits both *in vitro* and *in vivo* activity.

Results

Screening for small compounds targeting CXCR4 that block CXCL12-induced CXCR4 nanoclustering

The propagation domain of CXCR4, which links ligand binding to the residues involved in G-protein association, includes residues bounded by the transmembrane helices TMV and TMVI, which enclose a cleft of 870 Å³ (22) is surface-exposed for interactions with the plasma membrane, as it contains residues involved in lipid binding (24), and also includes residues implicated in receptor activation and in the transmission of conformational changes through the

TM helix domains (22 [↗](#)) (**Fig. 1A** [↗](#)). Additionally, the cleft contains the residues K239, V242, and L246, the mutation of which prevents CXCL12-mediated receptor oligomerization (20 [↗](#)). We thus screened for allosteric modulators that fit into this cleft and the cavity formed by TMV and TMVI but do not interfere with the CXCL12 binding site. Virtual screening was performed using the Medicinal and Biological Chemistry (MBC) library (CIB-CSIC) (25 [↗](#)) comprises more than 2,000 small heterocyclic compounds with drug-like properties. As a template for modeling CXCR4 we used the atomic structure of the receptor solved in complex with the inhibitor IT1t (PDB code: 3ODU). As this included T4 lysozyme inserted between helices TMV and TMVI to increase the stability of the protein (**Fig. 1A** [↗](#)), a common strategy to facilitate crystallogenes of GPCRs (26 [↗](#)), we first generated (24 [↗](#), 27 [↗](#)) a CXCR4 homology model using the SWISS_MODEL server (28 [↗](#)). The compounds were then prepared and docked against the modeled CXCR4 structure using “Glide”, and those identified were ranked by docking score, that is, by the minimal interaction energy given by Glide and visualization of the poses. The results were confirmed using a second strategy based on the CXCR4 model predicted by AlphaFold (29 [↗](#)) and the sequence available under UniProt entry P61073.

The top 40 candidates with the best docking score in the region of interest were chosen and were selected for functional analysis (**Supplementary Table 1** [↗](#)). We first screened their ability to block CXCL12-induced cell dependent migration in Transwell chambers.

Jurkat cells that were either untreated or treated with different concentrations (1–100 μ M) of the selected small compounds (30 min, 37°C, 5% CO₂) were allowed to migrate towards CXCL12. To enhance their possible inhibitory effects, the compounds were maintained in the upper chamber containing the cells throughout the chemotaxis experiment (**Supplementary Fig. 1A, B** [↗](#)).

Based on these results, we selected two compounds, AGR1.135 and AGR1.137, which showed reproducible dose-dependent inhibitory effects on CXCR4-induced cell migration (**Fig. 1B** [↗](#)) and had the highest docking score on the cleft between TMV and TMVI (–51.4 and –37.2 kcal/mol, respectively). As a control for further analysis, we used AGR1.131, which shares a common scaffold with the other compounds and interacts with TM domains of CXCR4, but did not affect CXCL12-promoted cell migration. It also theoretically targeted the same motif on CXCR4 as AGR1.135 and AGR1.137 (–39.8 kcal/mol), but showed a better docking score on a pocket between TMI and TMVII (–43.6 kcal/mol). Under these experimental conditions, none of the selected compounds were toxic against Jurkat cells, as shown by propidium iodide incorporation and cell cycle analysis (**Supplementary Fig. 2** [↗](#)). The structures of the three selected compounds was verified by NMR analysis (Supplementary information) and were characterized by the presence of a common core (4-(1-benzyl-1H-1,2,3-triazol-4-yl)phenyl) methanol, linked through a benzamide group to a lateral chain bearing amines of different length and chemical nature. AGR1.131 bears a simple benzylamine, AGR1.135 contains a complex lateral chain due to the presence of the 1-(2-nitro-4-(trifluoromethyl)phenyl)piperazine, and AGR1.137 incorporates ethyl piperidine-4-carboxylate (**Fig. 1C** [↗](#)).

AGR1.135 and AGR1.137 block CXCL12-mediated CXCR4 nanoclustering and dynamics

We next evaluated whether the selected compounds altered CXCL12-mediated CXCR4 nanoclustering and dynamics. To do this, we utilized SPT-TIRF on CXCR4-deficient Jurkat cells (JK^{–/–}) transiently transfected with CXCR4 fused to the AcGFP monomeric protein (CXCR4-AcGFP) and pretreated or not with 50 μ M of the selected small compounds (Supplementary videos 1–8), a concentration that inhibited CXCL12-induced cell migration by 50–75% (**Fig. 1B** [↗](#)). The compounds had no effect on DMSO-stimulated cells, and we observed mainly receptor monomers and dimers (~82–98%), with a very low percentage of complexes with >3 receptors (~2–18%) (**Fig. 2A** [↗](#)). Accordingly, the basal mean spot intensity (MSI) of CXCR4 was similar in all cases (~1,500 a.u.) (**Fig. 2B** [↗](#)). Under these experimental conditions, most of the particles corresponded to

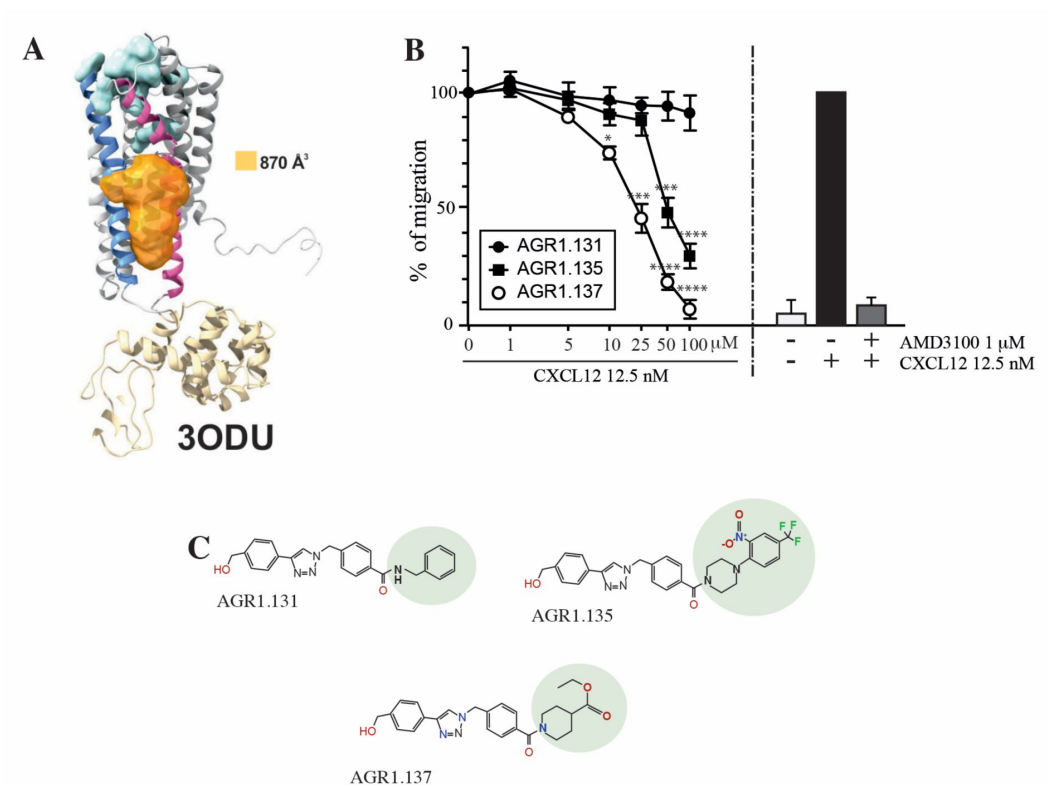


Figure 1.

Screening for small compound antagonists acting at the oligomerization site of CXCR4.

A) Cartoon and surface representation of CXCR4 and the cleft identified between TMV and TMVI. The protein structure is shown in gray, with TMV and TMVI colored in blue and pink, respectively. In green, residues involved in CXCL12 binding. The cavity identified by SurfNet is shown in orange. Volume for the cavity is measured and shown in Å³. T4 lysozyme inserted between TMV and TMVI in the crystallized version of CXCR4 (PDB: 3ODU) is also shown in yellow. **B)** Dose-response curve of the selected antagonists in Jurkat cell migration experiments in response to 12.5 nM CXCL12. Data are shown as percentage of migrating cells (mean ± SD; n = 5; * ≤ 0.05, *** p ≤ 0.001, **** p ≤ 0.0001). **C)** Chemical structure of the selected compounds (AGR1.131, AGR1.135 and AGR1.137). The differences between the lateral chains of the three compounds are shaded in green.

mobile particles (~88%) (**Fig. 2C**), and none of the selected compounds affected the short time-lag diffusion coefficient (D_{1-4}) for CXCR4 trajectories, with a median value of $\sim 0.02 \mu\text{m}^2/\text{s}$ (**Fig. 2D**).

We next examined how pre-treatment with the selected compounds affected receptor dynamics upon CXCL12 stimulation. Pre-treatment with AGR1.135 and AGR1.137, but not with AGR1.131, significantly altered CXCL12-mediated receptor nanoclustering (32–69% of nanoclusters with ≥ 3 receptors/particle in untreated and in AGR1.131-treated cells *versus* 7–22% in cells pretreated with AGR1.135 or AGR1.137) (**Fig. 2A**), with an MSI of 3,219 a.u. for CXCR4 in untreated cells and 3,618 a.u. in AGR1.131-treated cells *versus* 1,611 a.u. in AGR1.135- and 1,403 a.u. in AGR1.137-treated cells (**Fig. 2B**). This effect was also evident when we evaluated the dynamic parameters of CXCR4. Under control conditions (untreated or AGR1.131-treated cells), CXCL12 significantly increased the percentage of immobile particles (12% in untreated cells *versus* 25% in untreated cells + CXCL12 and 18% in AGR1.131-treated cells + CXCL12) (**Fig. 2C**), whereas AGR1.135 and AGR1.137 pretreatment had no effect on the percentage of immobile particles after CXCL12 stimulation (14% for both AGR1.135 and AGR1.137). Moreover, the expected reduction in CXCR4 diffusivity triggered by CXCL12 in untreated (median $D_{1-4} = 0.0088 \mu\text{m}^2/\text{s}$) or AGR1.131-treated (median $D_{1-4} = 0.0099 \mu\text{m}^2/\text{s}$) cells was abolished with AGR1.135 ($D_{1-4} = 0.0169 \mu\text{m}^2/\text{s}$) and AGR1.137 ($D_{1-4} = 0.0165 \mu\text{m}^2/\text{s}$) treatments (**Fig. 2D**).

Altogether, these results suggest that AGR1.135 and AGR1.137 behave as negative allosteric modulators of CXCR4 and alter the CXCL12-mediated receptor nanoclustering and dynamics.

AGR1.135 and AGR1.137 incompletely abolish CXCR4-mediated responses in Jurkat cells

CXCR4 internalizes in response to CXCL12, a process that mediates receptor desensitization (30). This process is associated with clathrin-mediated endocytosis of chemokine receptors (31), and requires receptor aggregation. Our previous results indicate that CXCR4 nanoclustering and the aggregates involved in receptor internalization are independent processes (20). Indeed, CXCL12 triggers CXCR4^{mut} internalization but fails to promote receptor nanoclustering (20). We thus tested whether the selected antagonists affected CXCL12-mediated CXCR4 internalization. Jurkat cells treated with vehicle or the selected compounds were stimulated with CXCL12 and receptor internalization was evaluated by flow cytometry using anti-CXCR4 antibodies. Neither AGR1.135 nor AGR1.137 treatment altered the internalization of CXCR4 that was observed in untreated Jurkat cells or in AGR1.131-treated cells (**Fig. 3A**). These data suggest that none of the allosteric modulators block CXCL12 binding to CXCR4, supporting the *in silico* screening strategy of preserving ligand binding integrity. Indeed, using flow cytometry analysis and CXCL12-ATTO 700 (32) we confirmed that none of the selected compounds blocked ligand binding, whereas this was blocked by the ligand-binding inhibitor AMD3100 (**Fig. 3B**, **Supplementary Fig. 3**). Taken together, the data indicate that the selected compounds do not abrogate ligand binding or receptor internalization, but blocked other functions such as cell migration.

Because CXCL12 promotes Gi-protein activation, we analyzed the effects of the selected compounds on CXCL12-mediated inhibition of cAMP production, a canonical signaling pathway downstream of CXCR4 activation. In contrast to AMD3100, an orthosteric antagonist of CXCR4, none of the compounds affected CXCL12-mediated Gi protein activation (**Fig. 3C**). We next evaluated their effects on other signaling pathways such as ERK1/2 phosphorylation and PI3K activation (33). Jurkat cells treated with vehicle or the selected compounds were activated with CXCL12 (50 nM) for different time periods and cell lysates were analyzed by western blotting using anti-P-ERK1/2 and -P-AKT antibodies. Neither AGR1.135 nor AGR1.137 nor the control AGR1.131 treatment inhibited CXCL12-mediated activation of the two signaling pathways (**Fig. 3D–F**, **Supplementary Fig. 4**), whereas AMD3100 treatment as a positive control did so

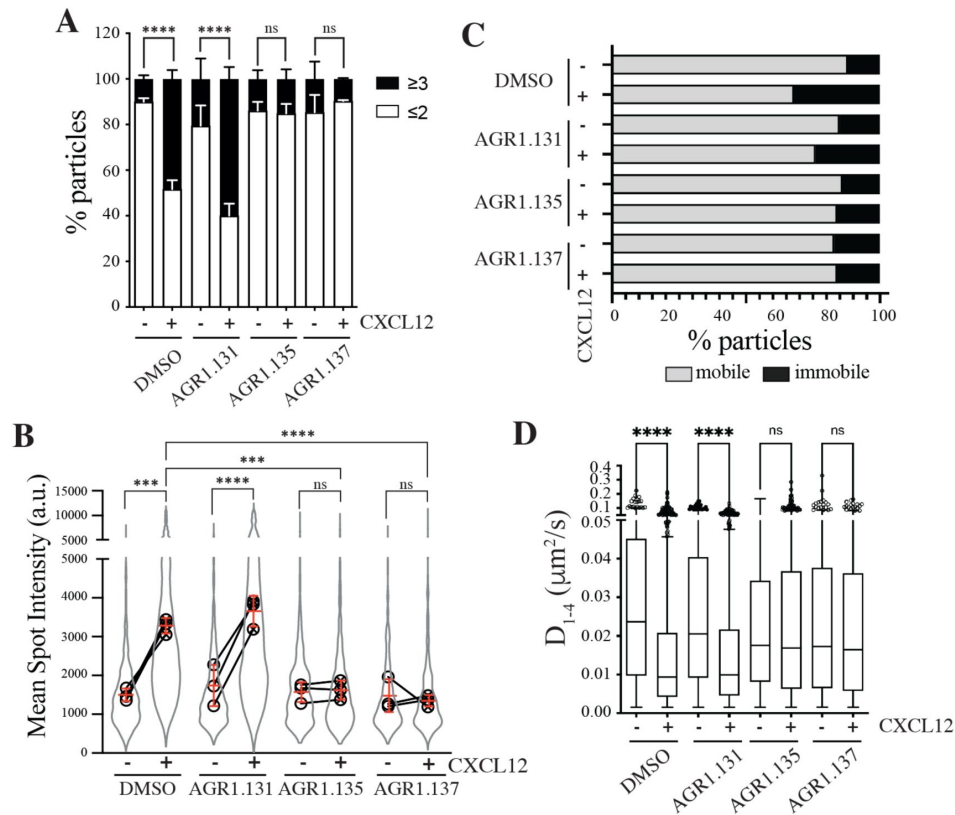


Figure 2.

AGR1.135 and AGR1.137 alter CXCL12-mediated CXCR4 dynamics and nanoclustering.

Single-particle tracking analysis of $JK^{-/-}$ cells transiently transfected with CXCR4 ($JK^{-/-}$ -X4) treated with DMSO (control), AGR1.131, AGR1.135 or AGR1.137 on fibronectin (FN)- or FN+CXCL12-coated coverslips (DMSO: 581 particles in 59 cells on FN; 1365 in 63 cells on FN+CXCL12; AGR1.131: 1019 particles in 71 cells on FN; 1291 in 69 cells on FN+CXCL12; AGR1.135: 862 particles in 70 cells on FN; 1003 in 77 cells on FN+CXCL12; AGR1.137: 477 particles in 66 cells on FN; 566 in 64 cells on FN+CXCL12) $n = 3$. **A**) Frequency of CXCR4-AcGFP particles containing monomers plus dimers (≤ 2) or nanoclusters (≥ 3), $\pm$ SEM, calculated from mean spot intensity values of each particle as compared with the value of monomeric CD86-AcGFP ($n = 3$; n.s., not significant; $****p \leq 0.0001$). **B**) Intensity distribution (arbitrary units, a.u.) from individual CXCR4-AcGFP trajectories on unstimulated and CXCL12-stimulated $JK^{-/-}$ -X4 cells pretreated or not with the indicated compounds or vehicle (DMSO). Graph shows the distribution of all trajectories (gray outline), with the mean value of each experiment (black circles) and the mean of all experiments $\pm$ SD (red lines) ($n = 3$; n.s., not significant; $***p \leq 0.001$, $****p \leq 0.0001$). **C**) Percentage of mobile and immobile CXCR4-AcGFP particles at the membrane of cells treated as indicated. **D**) Diffusion coefficients (D_{1-4}) of mobile single particle trajectories at the membrane of cells treated as indicated, represented by box-and-whiskers plots (Tukey method) with median (black line) indicated. (n.s., not significant, $****p \leq 0.0001$).

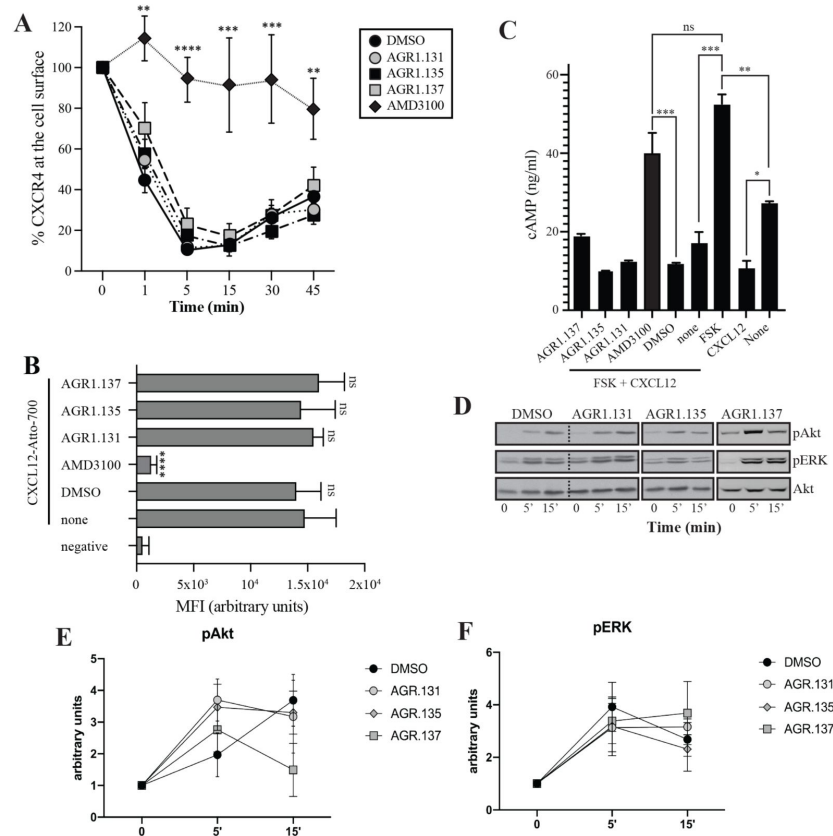


Figure 3.

AGR1.135 and AGR1.137 treatments do not block CXCL12-mediated signaling pathways.

A) Cell surface expression of CXCR4 in Jurkat cells after stimulation with CXCL12 (12.5 nM) at different time points and analyzed by flow cytometry using an anti-CXCR4 antibody in nonpermeabilized cells. Results show mean $\pm$ SEM of the percentage of CXCR4 expression at the cell surface ($n = 4$; n.s. not significant; $**p \leq 0.01$, $***p \leq 0.001$, $****p \leq 0.0001$). **B)** CXCL12-Atto-700 binding on untreated or Jurkat cells pre-treated with DMSO (vehicle), AGR1.131, AGR1.135, AGR1.137 or with AMD3100 as a control, followed by flow cytometry analysis. Results are expressed as mean fluorescence intensity (MFI) values (arbitrary units). Negative corresponds to basal cell fluorescence in the absence of CXCL12-Atto-700 (mean $\pm$ SD, $n = 3$ n.s. not significant; $****p \leq 0.0001$). **C)** Cells untreated or pretreated with the small compounds (50 μ M, 30 min 37°C) or AMD3100 (1 μ M, 30 min 37°C) were stimulated with CXCL12 (50 nM, 5 min, 37°C) followed by forskolin (10 μ M, 10 min, 37°C). Cells were then lysed and cAMP levels were determined (mean $\pm$ SD; $n = 3$; n.s. not significant; $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$). **D)** Western blot analysis of phospho (p)Akt and pERK in Jurkat cells pre-treated with DMSO, AGR1.131, AGR1.135 or AGR1.137, in response to CXCL12. As a loading control, membranes were re-blotted with an anti-Akt antibody. Representative experiments are shown ($n = 4$). **E, F)** Densitometry evaluation of the specific bands in D) (mean $\pm$ SD; $n = 4$).

(**Supplementary Fig. 4** [↗](#)). Overall, the data indicate that the selected compounds inhibit CXCL12-induced chemotaxis in Transwell assays, but do not affect G protein activation or other ligand-mediated signaling pathways such as ERK1/2 or PI3K.

Alterations in actin cytoskeletal dynamics are linked to deficiencies in ligand-mediated receptor nanoclustering and to defects in the ability of cells to sense chemoattractant gradients. Cells treated with latrunculin A, an actin polymerization inhibitor ([20](#) [↗](#)), and cells expressing CXCR4WHIM^{R334X}, which fail to properly control actin dynamics ([21](#) [↗](#), [34](#) [↗](#)), have defective CXCL12-mediated nanoclustering and are unable to properly sense chemoattractant gradients. Using flow cytometry and phalloidin staining of Jurkat cells treated with vehicle or the selected compounds, we detected altered actin polymerization in cells treated with AGR1.135, with AGR1.137, and with AMD3100 (**Fig. 4A** [↗](#)). These data were corroborated by immunostaining and confocal analysis (**Fig. 4B, C** [↗](#)). Whereas Jurkat cells treated with vehicle or AGR1.131 were correctly polarized after CXCL12 activation, AGR1.135 and AGR1.137 treatments promoted a reduction in the number of polarized cells. As a control, AMD3100 treatment also abolished CXCL12-mediated cell polarization (**Fig. 4B, C** [↗](#)). None of the compounds affected anti-CD3 mediated actin polymerization, ruling out a direct effect of the modulators on actin or actin-binding proteins (**Supplementary Fig. 5** [↗](#)).

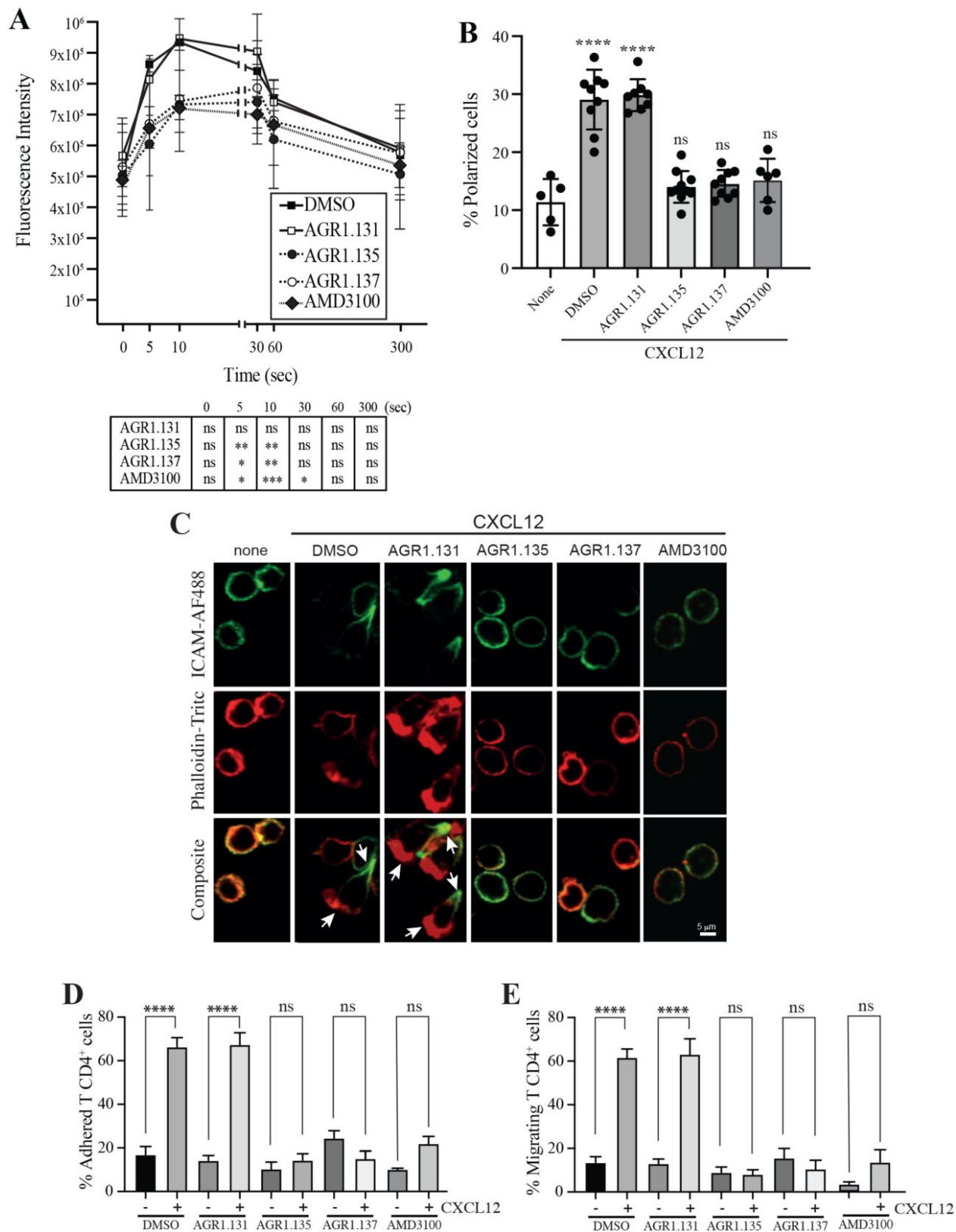


Figure 4.

AGR1.135 and AGR1.137 treatment alter CXCL12-mediated actin polymerization.

A) Actin polymerization in response to CXCL12 as determined by F-actin (phalloidin-TRITC) staining in Jurkat cells treated with DMSO (vehicle) or the indicated modulators. Statistical significance of the different time points in comparison with the control (DMSO) is shown in the table (mean $\pm$ SD; $n = 3$; n.s. not significant; * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$). **B)** Percentage of polarized T cell blasts adhered to fibronectin and treated or not with CXCL12 in the presence of the indicated antagonists, as analyzed by immunostaining with anti-ICAM3-Alexa fluor 488 and phalloidin-TRITC. More than 500 cells were analyzed in each condition. Data are presented as percentage of polarized cells (mean $\pm$ SD; $n = 3$; n.s. not significant; **** $p < 0.0001$). **C)** Representative images of T cell blasts adhered to fibronectin and treated or not with CXCL12 in the presence of the indicated antagonists, as analyzed by immunostaining with anti-ICAM3-Alexa fluor 488 and phalloidin-TRITC as in B). **D)** CD4⁺ T cells pretreated with AGR1.131, AGR1.135 or AGR1.137 were perfused in flow chambers coated with ICAM-1-containing lipid bilayers, alone or CXCL12-coated, and analyzed for cell contacts with the substrate. Data are presented as percentage of adhered cells (mean $\pm$ SD; $n = 3$; n.s. not significant; **** $p \leq 0.0001$). **E)** Cells in D) were analyzed for cell migration. Data are presented as percentage of migrating cells (mean $\pm$ SD; $n = 3$; n.s. not significant; **** $p \leq 0.0001$).

CXCR4 nanoclustering has been recently associated with the migratory phenotype of T cell blasts (21 [4](#), 34 [4](#)). To explore how the allosteric modulators affect the phenotype of migrating cells, we added T cell blasts previously treated with vehicle or with the selected compounds to a 2D lipid bilayer system with embedded ICAM-1, alone or with CXCL12. We observed that treatment with AGR1.135, AGR1.137, or AMD3100 (control), but not with AGR1.131, abolished CXCL12-induced cell migration and cell adhesion (Fig. 4D, E [4](#)).

As the antagonistic effects of AGR1.135 and AGR1.137 are also compatible with a partial agonist behavior, we used Transwell chambers to evaluate the ability of the selected compounds to promote Jurkat cell migration. None of the compounds induced cell migration (Supplementary Fig. 6 [4](#)). Overall, our results suggest that AGR1.135- and AGR1.137-driven modulation of CXCR4 nanoclustering blocks certain receptor-associated functions, including actin dynamics, directional cell migration, integrin-mediated adhesion and migration, while leaving other functions unaffected (receptor internalization, inhibition of cAMP production and ERK1/2 activation).

AGR1.135 and AGR1.137 antagonists act by direct binding on CXCR4

We developed a fluorescence spectroscopy strategy and FRET assays to formally examine the ability of the compounds to bind CXCR4, but our efforts were unsuccessful. AGR1.135 exhibited a yellow color that interfered with these techniques, and AGR1.137 did not alter the FRET efficiency of CXCR4 dimers (Supplementary Fig. 7 [4](#)). Therefore, we designed an *in silico* approach including a final step of molecular dynamics (MD) analysis, which allowed us to define the CXCR4 residues involved in the binding of the compounds, and to generate point mutant receptors to determine the inhibitory effect of AGR1.135, AGR1.137 and AGR1.131 (control) on CXCL12-mediated chemotaxis. We initially performed a binding-site search using PELE software, a Monte Carlo-based technique (35 [4](#)), to identify the most promising CXCR4 binding sites of the selected compounds. AGR1.135 and AGR1.137 exhibited one of the most stable trajectories (less than 0.25 Å² RMSD fluctuation) upon binding to the cleft formed by TMV and TMVI (Fig. 5A, B [4](#)). By contrast, the most stable trajectories for AGR1.131 corresponded to the binding to a contiguous region located between TMI and TMVII (Fig. 5C [4](#)). Once the binding site of the compounds was confirmed, we performed docking studies to identify the best poses as a starting point for further studies.

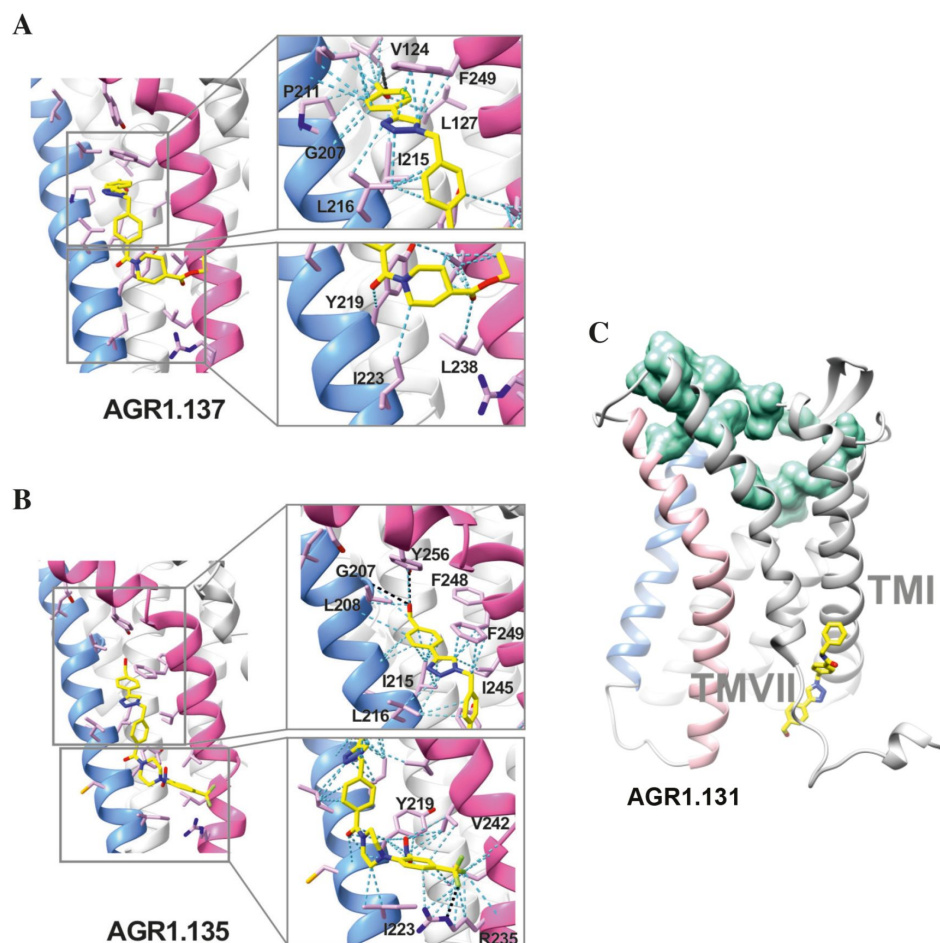


Figure 5.

Ribbon and sticks representation of the CXCR4 modulators bound to the receptor.

In the ribbon and sticks representations, the TMV helix is colored in blue and TMVI in pink. Figures show the binding of ligands AGR1.135 (**A**) and AGR1.137 (**B**) Zoom images showing the residues involved in the interaction of both compounds are also shown (right panels). **C**) Binding of AGR1.131 represented as sticks with carbon atoms in yellow, oxygen in red, nitrogen in blue and fluorine in green. CXCR4 residues involved in CXCL12 engagement and initial signal transmission are represented as green spheres.

Results showed that AGR1.135 and AGR1.137 shared a similar binding mode (**Fig. 5A, B**) and might interact with CXCR4 in a molecular cavity between TMV and TMVI. Specifically, AGR1.135 formed hydrogen bonds with residues G207, Y256 and R235, although a pi-cation interaction was also possible in the latter case (**Fig. 5A**). Additionally, AGR1.135 could interact hydrophobically with several residues in TMV and TMVI, comprising two independent cavities of 434 and 1,381 Å³, respectively (Supplementary Fig. 8A), which were not connected to the orthosteric site as indicated by SURFNET analysis (36) included in the PDBsum server (37) (**Supplementary Fig. 8A**). AGR1.137 could use the carboxyl group of V124 in TMIII and overlap with AGR1.135 binding in the cavity, interacting with the other 19 residues scattered between TMV and VI to create several interaction surfaces of 790 and 580 Å³ (**Fig. 5B**; **Supplementary Fig. 8B**). AGR1.137 did not have the phenyl ring present in AGR1.135, likely explaining why it did not interact with residue R235 (**Fig. 1C**, **Fig. 5B**). Regarding AGR1.131, its best binding site was the pocket between helices TMI and TMVII (**Fig. 5C**), which is likely the reason why this compound had no effect on receptor oligomerization. These results were also confirmed using the CXCR4 model predicted by AlphaFold (29) and the sequence available under UniProt entry P61073.

To confirm the putative binding sites for AGR1.135 and AGR1.137, we generated several point mutations in CXCR4 using the identified residues that were not critical for signal transduction processes. These included the G207I, L208K, R235L, F249L and Y256F mutants. We also included the mutations I204K, S260A as controls, as they are located in the same region (TMV-TMVI cleft) but did not interfere with compound binding or the CXCL12 binding site. All CXCR4 mutants were normally expressed at the cell surface, as demonstrated by flow cytometry (**Supplementary Fig. 9A**), and were fully functional in CXCL12-mediated cell chemotaxis assays (**Supplementary Fig. 9B**).

We next used transient transfection of JK^{-/-} cells with each of the mutants or with wild-type CXCR4 treated with AGR1.131, AGR1.135 or AGR1.137 (50 µM) prior to assessing migration to CXCL12 gradients in Transwell chambers. The antagonistic effect of AGR1.135 was reversed in cells transfected with CXCR4 carrying the specific point mutations L208K or Y256F (**Fig. 6A**). Similarly, the antagonistic effect of AGR1.137 was reversed in cells expressing G207I, L208K, R235L, F249L or Y256F mutants (**Fig. 6A**). Collectively, these data indicate that the selected compounds bind directly to CXCR4, that the L208 and Y256 residues in the CXCR4 TMVI domain are critical for the allosteric modulation activities of AGR1.135 and AGR1.137, and that other residues, G207, R235 and F249, are also important for the effect of AGR1.137 (**Fig. 6A-D**).

AGR1.137 treatment reduces tumor volume and dissemination in a zebrafish xenograft model

Finally, to determine the *in vivo* activity of the selected compounds, we considered the role of CXCR4 in tumor growth and metastasis (10, 15) and the transparency of zebrafish larvae to develop a tumor xenograft (ZTX) model. ZTX models have proven to be a powerful complementary *in vivo* system for research in oncology and tumor biology (38, 39), particularly for early tumor invasion and dissemination.

We used HeLa cells, which are derived from a human cervical tumor that expresses functional CXCR4 (**Fig. 7A, B**; **Supplementary Fig. 10A**), and which have been used previously in ZTX models (40, 41). HeLa cells respond to CXCL12 gradients in directional cell migration assays (**Fig. 7A, B**) and, as in Jurkat cells, their treatment with AGR1.137 but not with AGR1.131 (control) abolished directional cell migration toward a CXCL12 gradient (**Fig. 7A, B**, **Supplementary videos 9–12**). As a control for specificity, CXCL2-driven HeLa cell migration was unaffected by AGR1.137 treatment but was blocked by the specific CXCR2 inhibitor AZD5069 (**Fig. 7C, D**).

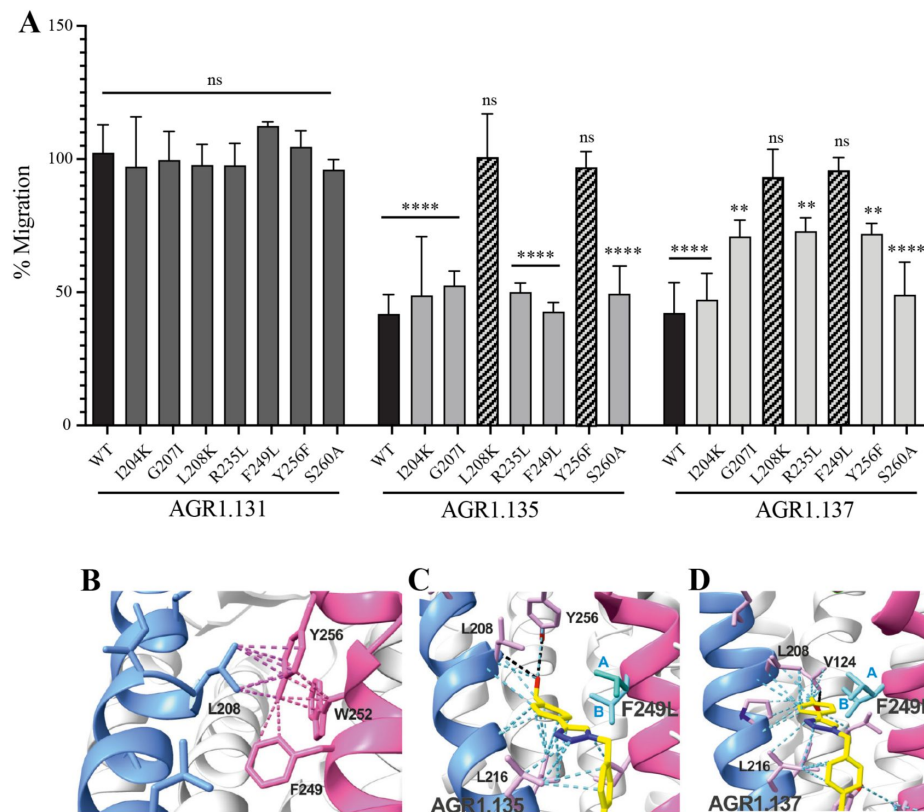


Figure 6.

The antagonistic behavior of AGR1.135 and AGR1.137 depends on specific residues of CXCR4.

A) CXCL12-induced migration of AGR1.131-AGR1.135- or AGR1.137-pretreated $JK^{-/-}$ cells transiently transfected with CXCR4wt or the different mutants described. Data are shown as the mean percentage $\pm$ SD of input cells that migrate ($n = 4$; n.s. not significant, $**p \leq 0.01$, $****p \leq 0.0001$). **B)** Cartoon representation of Y256 and its intramolecular interactions in the CXCR4 X-ray solved structure 3ODU. **C, D)** Cartoon representation of the interaction of CXCR4 F249L mutant with AGR1.135 and AGR1.137, respectively. The two most probable conformations of leucine rotamers are represented in cyan. Van der Waals interactions are depicted in cyan dashed lines and hydrogen bonds in black dashed lines.

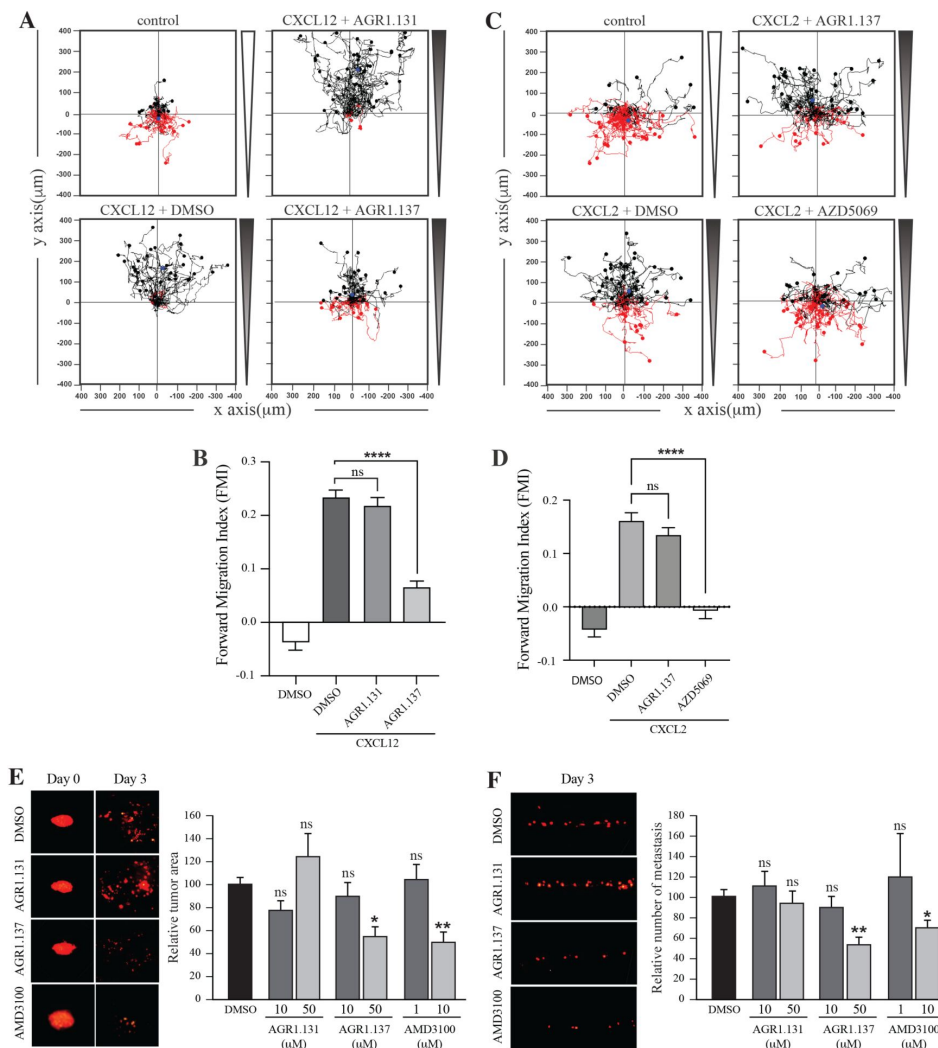


Figure 7.

AGR1.137 reduces tumorigenesis and metastasis in a zebrafish model.

(A–D) Migration of HeLa cells treated with vehicle (DMSO) or with the selected small compounds or inhibitor as indicated, on μ -chambers in response to a CXCL12 (A, B) or CXCL2 (C, D) gradient ($n = 2$, in duplicate, with at least 50 cells tracked in each condition). Figures A) and C) show representative spider plots with the trajectories of tracked cells migrating along the gradient (black) or moving in the opposite direction (red). Black and red dots in the plots represent the final position of each single tracked cell. (B, D) Quantification of the Forward Migration Index of experiments performed in A) and C) (mean $\pm$ SD; $n = 3$; n.s. not significant, **** $p \leq 0.0001$). (E) Representative fluorescent images of DiI-labeled HeLa cells in zebrafish larvae treated with vehicle (DMSO), AGR1.131 50 μ M, AGR1.137 50 μ M or AMD3100 10 μ M at 0 or 3 days post-implantation and treatment. Quantitation of the relative tumor size at day 3 compared with that of day 0 normalized to the relative tumor size in the DMSO control group, is shown for each experimental group (mean $\pm$ SD; $n = 20$; n.s. not significant, ** $p \leq 0.01$). (F) Representative fluorescent images of the caudal hematopoietic plexus of larvae from the same groups as shown in E) at 3 days postimplantation. Quantitation of the relative amount of metastasized cells in each group relative to DMSO-controls is shown (mean $\pm$ SD; $n = 20$, n.s. not significant, * $p \leq 0.05$, *** $p \leq 0.001$).

DiI-labeled HeLa cells treated with vehicle, AMD3100, AGR1.135, AGR1.137 or AGR1.131 (control) were implanted subcutaneously into the perivitelline space of 2-day-old zebrafish larvae. Images of each tumor-bearing larva were taken immediately after implantation and 3 days later, and the relative change in tumor size was determined. AGR1.135 was discarded because most of the treated larvae died, indicating a toxic effect. By contrast, AGR1.137 was not toxic to embryos, as shown by visual inspection of the larvae at the end of the experiment, and no other non-lethal toxic phenotypes (e.g., pericardial edema, head and tail necrosis, head or tail malformation, brain hemorrhage or yolk sac edema) were evident. AGR1.137 10 μ M had no effect but 50 μ M significantly reduced the size of high-intensity red fluorescent tumors (~40%), whereas the control AGR1.131 had no effect on tumor size as compared with vehicle-treated fish, and AMD3100 reduced tumor size to a similar extent as AGR1.137 (**Fig. 7E**). These data demonstrate an antiproliferative effect of AGR1.137 on HeLa cells. As a control, AGR1.137 also reduced HeLa cell proliferation in *in vitro* assays mimicking the *in vivo* experimental conditions in zebrafish (addition of compounds every 24 hrs for 3 days) (**Supplementary Fig. 10B**). We next evaluated the ability of the compounds to block cell dissemination by determining the number of labeled HeLa cells that appeared in the main metastatic niche at the caudal hematopoietic plexus. AGR1.137 50 μ M treatment of HeLa cells reduced cell dissemination by ~40% (**Fig. 7F**), strongly suggesting that this compound exerts anti-metastatic effects. As control AMD3100 (10 μ M) treatment also reduced cell dissemination (~30%). The combined ability of AGR1.137 to reduce proliferation and dissemination *in vivo* is likely to have therapeutic value.

Discussion

The CXCL12/CXCR4 axis is involved in myriad functions, including leukocyte recruitment, embryogenesis, vascular development, hematopoiesis, heart development, nervous system organization, tumor progression, autoimmune diseases and, together with CD4, CXCR4 is a co-receptor used by HIV-1 to infect immune cells (42). This evidence supports the clinical interest in developing specific antagonists to modulate or directly block CXCR4 functions.

Plerixafor (AMD3100), which blocks ligand binding, is the first FDA-approved CXCR4 antagonist, and is used in peripheral blood stem cell transplantation regimens (43). Unfortunately, its clinical use is limited by poor pharmacokinetics and adverse effects associated with long-term administration (44, 45). These limitations and the poor clinical success of other chemokine receptor antagonists have prompted the search for alternative strategies to block chemokine receptor functions.

As for other GPCRs, CXCR4 exists in multiple conformations on the plasma membrane and interacts with other chemokine receptors (46) and with cell surface proteins such as the T-cell receptor (47), CD4 and other receptors (20, 42). Advanced imaging-based techniques such as SPT-TIRF and super-resolution microscopy have revealed the existence of CXCR4 monomers, dimers and oligomers at the cell membrane, diffusing into the lipid bilayer. CXCL12 binding induces receptor nanoclustering and reduces the proportion of monomers and dimers, which is essential for the full activation of CXCR4 signaling and the correct orientation of the cell towards chemokine gradients (20, 21, 34). CXCR4R^{334X}, a truncated mutant receptor that causes WHIM syndrome, fails to form nanoclusters in the presence of CXCL12 and is unable to sense chemoattractant gradients, although it remains capable of triggering Ca²⁺ flux and other signaling pathways (21). Due to the inability of CXCR4 to associate with the actin cytoskeleton, β -arrestin-1-deficient cells do not exhibit CXCL12-mediated receptor nanoclustering (21). Similar observations are seen CXCR4-expressing cells treated with latrunculin A, an inhibitor of actin polymerization (20, 42). Collectively, these observations suggest that receptor nanoclustering is a ligand-mediated process that requires activation of a signaling pathway involving β -arrestin-1 and proper actin cytoskeletal dynamics.

CXCR4 contains a pocket formed by two TM domains, TMV and TMVI, and mutations in this pocket can affect the dynamics of the conformational changes triggered by ligand binding. The mechanism involves the initial interaction of CXCL12 with the extracellular region of CXCR4, which promotes conformational changes at a series of hydrophobic residues located mainly on TMV and TMVI that continue downward in the TM domains of the receptor, ultimately allowing G-protein interaction and activation (22 [↗](#)). Interestingly, a complete alignment of GPCR class A chemokine receptors reveals that hydrophobic residues in this region that are not in highly conserved positions can have any side-chain length at any site depending on the receptor, likely conferring a specificity that could be exploited for the discovery and/or design of new antagonists (48 [↗](#)) (**Supplementary Fig. 11** [↗](#)). A recent report using meta-dynamic approaches highlights the importance of the TM regions for dimerization and oligomerization of CXCR4 and CCR5 (24 [↗](#)).

The majority of GPCR-targeted drugs are functionally active by binding to the orthosteric site of the receptor. Nonetheless, these orthosteric compounds have off-target effects and poor selectivity due to highly homologous receptor orthosteric sites and abrogation of spatial and/or temporal endogenous signaling patterns. The use of negative allosteric modulators can tune receptor-associated functions without affecting the orthosteric binding site. For example, the negative allosteric modulator 873140 blocks CCL3 binding to CCR5 but does not alter CCL5 binding (49 [↗](#)). Maraviroc, an anti-HIV-1 molecule is an allosteric modulator of CCR5 that regulates CCR5 dimer populations, their localization to the cell membrane, and their subsequent subcellular trafficking (50 [↗](#)). Allosteric modulators of the kappa opioid receptor have been demonstrated to have desirable antinociceptive and antipruritic effects without sedative and dissociative effects in rodent models (51 [↗](#)).

To identify new allosteric modulators, we focused on an *in silico* screen targeting the pocket in CXCR4 created by TMV and TMVI, without affecting the residues involved in CXCL12 binding (20 [↗](#), 42 [↗](#)). Subsequent functional analysis of the selected compounds identified AGR1.135, AGR1.137, and AGR1.131. The latter, although selected with the same *in silico* criteria, showed a better docking score on a cleft between TMI and TMVII and did not block CXCL12-mediated chemotaxis. In addition, the compounds had no effect on CXCR2-mediated cell migration, supporting their specificity for CXCR4. The three compounds had a common core of (4-(1-benzyl-1H-1,2,3-triazol-4-yl)phenyl) methanol but different side chains carrying amines of different length and chemical nature, which may explain their different biological activity. AGR1.131 seems to be more flexible than the other two compounds. It has an amide group and an additional CH₂ linking the phenyl ring. The absence of the piperidine or piperazine rings in AGR1.131 may be related to its lack of activity on CXCR4. AGR1.137 is expected to be less hydrophilic than AGR1.135, as it contains only one tertiary amine while AGR1.135 contains two. These tertiary amines are susceptible to protonation to form a quaternary ammonium salt. In addition, the ethyl ester group present in AGR1.137 is less polar than the nitro group in AGR1.135, a feature that could affect the solubility of the compounds as well as their interactions on the receptor.

SPT-TIRF analysis showed that AGR1.135 and AGR1.137 abolished the ability of CXCL12 to induce CXCR4 nanoclustering and altered receptor dynamics at the cell membrane. Moreover, as expected from the selection criteria, these compounds had no effect on CXCL12 binding, receptor internalization, cAMP-production, or ERK1/2 or PI3K activation. The results also suggested the existence of different CXCR4 conformational states responsible for the activation of different signaling pathways. It is well known that GPCRs are thought to reside on the plasma membrane in equilibrium between distinct states, depending on complex allosteric interactions and ligand-induced conformational changes, as well as cell-specific parameters (52 [↗](#), 53 [↗](#)). The emerging view is that GPCRs are highly dynamic proteins, and ligands with varying pharmacological properties differentially modulate the balance between multiple conformations. In the absence of agonists, the receptor population is dominated by conformations closely related to those observed in inactive-state crystal structures (54 [↗](#)). While agonist binding drives the receptor population towards conformations similar to those in active-state structures, a mixture of inactive and active

conformations remains, reflecting “loose” or incomplete allosteric coupling between the orthosteric and transducer pockets (55 [↗](#)). Surprisingly, for some GPCRs, and under some experimental conditions, a substantial fraction of unliganded receptors are already in an active-like conformation, which may be related to their level of basal or constitutive signaling (56 [↗](#), 57 [↗](#)). These basal activities can be modulated by ligands of varying potency. Full agonists can induce the maximal signaling response, whereas partial agonists and inverse agonists promote submaximal signaling and decrease basal activity, respectively. In addition, some ligands are known to be biased by selectively activating specific receptor-associated pathways at the expense of others, supporting the existence of distinct receptor conformations (58 [↗](#)). Studies on the β 2-adrenergic receptor support the idea that different agonists stabilize different receptor conformations (59 [↗](#)), raising the possibility that allosteric ligands shift the balance to favor a particular receptor conformation. Biased ligands or allosteric modulators may thus achieve their distinctive signaling profiles by modulating this distribution of receptor conformations (60 [↗](#)). For example, some analogs of angiotensin II do not appreciably activate Gq signaling (e.g., increase IP3 and Ca^{2+}), but still induce receptor phosphorylation, internalization, and mitogen-activated protein kinase (MAPK) signaling (61 [↗](#)). Some of these ligands activate Gi and G_{12} in bioluminescence resonance energy transfer experiments (62 [↗](#)). A similar observation has been described for CCR5, where some chemokine analogs promote G protein subtype-specific signaling bias (63 [↗](#)). Structural analyses of various GPCRs in the presence of different ligands vary considerably in the extent and nature of the conformational changes in the orthosteric ligand-binding site upon agonist binding (64 [↗](#)). However, these changes modify conserved motifs in the interior of the receptor core and induce common conformational changes in the intracellular site involved in signal transduction.

We failed to detect a direct interaction between the compounds and CXCR4. Therefore, we employed an indirect strategy using docking and MD to confirm this interaction and to predict the most promising CXCR4 binding residues involved in the binding of the selected compounds. This methodology was combined with a mutation strategy to confirm the binding specificity of the allosteric modulators. AGR1.135 and AGR1.137 lost antagonism on cells expressing CXCR4L208K or CXCR4Y256F. LigPlot+ analysis (65 [↗](#)) indicated that AGR1.135 interacts directly with the receptor through a hydrogen bond with Y256 (**Supplementary Fig. 12** [↗](#)). When this residue was mutated to F, one of the anchors for the compound was lost, weakening the potential interaction in the region of the upper anchor point. It is not clear how the Y256F mutation affects the binding of AGR1.137, but other potential contacts cannot be ruled out since this part of the compound is identical in both AGR1.135 and AGR1.137. This is particularly important for its neighboring residues in the alpha helix, F249, L208, as shown in the 3ODU structure, which appear to be directly involved in the interaction of both compounds with CXCR4. Alternatively, we cannot rule out an interaction between Y256 and other TMs or lipids stabilizing the overall structure, which could reverse the effect of the mutant at a later stage.

The inhibitory behavior of AGR1.137 was also lost in cells expressing CXCR4F249L. Both allosteric modulators showed stable trajectories bound to CXCR4, being able to occupy the cavity formed by TMV and TMVI and maintaining specific hydrogen-bond interactions and hydrophobic contacts along the interaction surface. Moreover, there were differences between the two modulators in terms of binding; whereas AGR1.135 showed a more elaborated interaction due to three hydrogen bonds that curved the position of the molecule as it fitted into the cavity formed by TMV and TMVI, AGR1.137 bound to the carboxyl group of V124 in TMIII, penetrating deeper into the receptor and then following the TMV–TMVI cleft down to the bottom of the cavity. In both cases, hydrophobic interactions with several residues of the receptor allowed the compounds to attach to the cleft and partially abolish the movement of TMVI required for complete signal transmission (22 [↗](#)). AGR1.137 allowed for a different conformation of the TMVI helix, one that is more shifted towards an active conformation (**Supplementary Fig. 13** [↗](#)). By contrast, predictions indicated that AGR1.135 established hydrogen bonds with residues G207, Y256 and R235, complemented by hydrophobic interactions in TMV and TMVI, comprising a total surface area of 400 Å² with a

length of 20 Å. AGR1.135 appeared in a more stretched conformation than AGR1.137 due to its ability to share a hydrogen bond with R235. AGR1.137 could use the carboxyl group of V124 in TMIII and overlap with AGR1.135 binding in the cavity, interacting with the other 19 residues scattered between TMV and VI to create an interaction surface of 370 Å² along 20 Å. AGR1.137 did not have the phenyl ring present in AGR1.135, resulting in a shorter moiety with greater difficulty in reaching the lower part of TMVI where R235 sits. These differences may explain the different behavior of the two compounds in cells expressing CXCR4F249L and suggest a closer interaction between the triazole group of AGR1.137 and the F249 residue on CXCR4. The main difference between AGR1.135 and AGR1.137 is the way in which the triazole group interacts with F249 and L216. For AGR1.137, the three groups are aligned in a parallel organization, which appears to be more effective. This is because there is only one hydrogen bond with V124, which significantly contributes to a more flexible interaction. Contrastingly, AGR1.135 has two anchor points on either side of the link, which, together with the interaction with F248, results in a more rigid position in the cleft. When F249 is changed to L and the two most likely rotamers resulting from the mutation are shown, rotamer B appears in close proximity to the compound, which may cause the binding to either be displaced or to adopt an alternative conformation that is easier to bind into the cleft. As mentioned above, it is likely that AGR1.135 can displace the mutant rotamer and bind more easily to the cleft due to its higher affinity. This region has also been implicated in the attachment of the orthosteric chemokine binding pocket to the lipid bilayer. F249 in CXCR4 as well as L132, V214 and L216 are involved in cholesterol binding (24). Thus, AGR1.137 may also affect cholesterol binding to CXCR4, thereby altering receptor oligomerization. The x-ray structure of CXCR4 with PDB code 3ODU (27) shows experimentally the presence of two fatty acid molecules in contact with both TMV and TMVI. Although further experiments are needed to clarify the mechanism, our results suggest that cholesterol and/or other lipids might also play an important role in CXCR4 oligomerization and function, as seen for other GPCRs (66). We have recently shown that the lipid composition of the cell membrane modulates CXCR4 oligomerization (67). The chemokine field is very complex, and most receptors can form dimers (homo- and heterodimers) as well as oligomers (68). It is therefore possible that the compounds might also affect the function of other chemokine receptors that heterodimerize with CXCR4. Nonetheless, it should be noted that the residues involved in CXCR4 oligomerization might differ from those involved in dimeric structures (24). These observations highlight the flexibility of the cavity and its potential for modulation. They also suggest that this cavity could be used to guide the development of a new generation of antagonists that, without affecting CXCL12 binding, allosterically modulate some GPCR-mediated functions without altering others.

CXCR4 overexpression contributes to tumor growth, invasion, angiogenesis, metastasis, relapse, and therapeutic resistance (69). As might be expected, CXCR4 antagonism has been shown to disrupt tumor-stromal interactions, sensitize cancer cells to cytotoxic drugs, and reduce tumor growth and metastatic burden (69). We used a ZTX model with HeLa cells expressing functional CXCR4 to evaluate the *in vivo* relevance of the selected antagonists. This is a well-studied model to assess tumor progression and metastasis (40). Our results demonstrate that AGR1.137 treatment of HeLa cells induce a dose-dependent reduction tumor size by 40% and limits cell dissemination in the absence of toxicity, suggesting that this compound has both anti-proliferative and anti-metastatic properties. While the effect of AGR1.137 on cell dissemination may be a consequence of its effect on tumorigenesis, it also abolished directed HeLa cell migration *in vitro*, supporting the inhibition of cell metastasis. The data support that AGR1.137 reduces tumor size and metastasis to a similar extent as AMD3100, but the allosteric modulator is highly hydrophobic and therefore it will be very important to improve solubility, increase affinity and optimize delivery, to facilitate its use, particularly *in vivo*. AMD3100, which is FDA-approved for clinical use, is an orthosteric inhibitor of CXCR4 that blocks the entire signaling cascade triggered by CXCL12. Indeed, we observed that AMD3100 treatment blocked CXCL12 binding, cAMP inhibition, calcium flux, cell adhesion and cell migration, whereas the effect of AGR1.137 was limited to CXCL12-mediated directed cell migration. Although AMD3100 was well tolerated by healthy volunteers in a single-dose study, it also promotes some mild and reversible events such as increases in white blood cell

counts and variations in urinary calcium just outside the normal range (70 [↗](#)). For the treatment of viral infections, continuous daily dosing of AMD3100 was impractical due to severe side effects including cardiac arrhythmias (71 [↗](#)). It is therefore critical to control the timing of administration of AMD3100 in the context of therapeutic use. In addition, side effects after long-term administration are a potential problem. Shorter-term usage and lower doses would be fundamental keys to success in clinical use (72 [↗](#)). The use of new drugs, such as the negative allosteric modulator AGR1.137, which block cell migration but do not affect other CXCL12-mediated signaling pathways, would, at least in theory, be more specific and produce fewer side effects.

Our data support the notion that CXCR4 is a flexible protein that can adopt a spectrum of conformations depending on several factors such as the presence of a bound ligand, the lipid composition of the cell membrane (67 [↗](#)) and the presence of other interacting proteins. As with many GPCRs, the stabilization of distinct CXCR4 states is a key element to modulate its function (73 [↗](#), 74 [↗](#)). Our data show that by targeting the pocket between TMV and TMVI in CXCR4, AGR1.137 can block (both *in vitro* and *in vivo*) CXCL12-mediated receptor nanoclustering and cell sensitivity to chemoattractant gradients without altering ligand binding, thus preserving other signaling events. Ultimately, these results demonstrate the value of stabilizing specific CXCR4 conformations and considering targets other than the ligand-binding site for the design of negative allosteric modulators to block a specific set of CXCR4 functions.

Materials and Methods

Cells and reagents

HeLa cells (CCL-2) and HEK293T cells (CRL-11268) were obtained from the American Type Culture Collection (Rockville, MD). Jurkat human leukemia CD4⁺ cells were kindly provided by Dr. J. Alcamí (Centro Nacional de Microbiología, Instituto de Salud Carlos III, Madrid, Spain). Where indicated, Jurkat cells lacking endogenous CXCR4 (JK^{-/-}) (21 [↗](#)) were transiently transfected with plasmids expressing wild-type or mutant CXCR4-AcGFP receptors (20 µg), as described (21 [↗](#)). CXCR4 mutants were generated by PCR using the QuikChange site-directed mutagenesis kit (Stratagene, La Jolla, CA) with full-length CXCR4-AcGFP serving as a template, and specific primers (Supplementary Table 2 [↗](#)).

Human peripheral blood mononuclear cells isolated from buffy coats were activated *in vitro* for 1 week with 20 U/ml of IL-2 (Teceleukin; Roche, Nutley, NJ) and 5 µg/ml phytohemagglutinin (Roche) to generate T cell blasts (34 [↗](#)).

The following antibodies were used: monoclonal mouse anti-human CXCR4 (clone 44717) and phycoerythrin-conjugated human CXCR4 (clone 12G5; both from R&D Systems, Minneapolis, MN); purified mouse anti-human CD3 (clone HIT3a, BD Biosciences, Franklin Lakes, NJ); goat F(ab')₂ anti-mouse IgG-PE (Southern Biotech, Birmingham, AL); and anti-phospho-AKT (Ser473), anti-AKT (#9272), anti-phospho-ERK1,2 (#9191) and anti-ERK (#9102) (all from Cell Signaling Technology, Danvers, MA). Human CXCL12, CXCL2 and CCL17 were obtained from PeproTech (Rocky Hill, NJ). Human CXCR4 was cloned into the pAcGFPm-N1 plasmid (Clontech Laboratories, Palo Alto, CA), as described (20 [↗](#)).

Unless otherwise indicated, cells were pre-treated with 50 µM of the selected small compounds, 10 µM AMD3100 (Merck, Darmstadt, Germany), 1 µM AZD5069 (MedChemExpress, Monmouth Junction, NJ) or vehicle (DMSO) as control (30 min, 37°C, 5% CO₂).

Binding experiments were performed using CXCL12-Atto-700 ([32](#)), kindly donated by Prof. Marcus Thelen (Institute for Research in Biomedicine, Università della Svizzera italiana, Bellinzona, Switzerland).

Compounds database

All the compounds used in the present study were synthesized in the Centro de Investigaciones Biológicas Margarita Salas (CIB-CSIC, Madrid) following previously described procedures ([75](#)). Synthetic and analytical data are shown as Supplementary information. All the compounds have a purity $\geq 95\%$ determined by high-performance liquid chromatography. Compounds are collected in the MBC library (CIB-CSIC) ([25](#)), which contains more than 2,000 drug-like compounds.

Computational studies of CXCR4 modulators

Protein modeling

The CXCR4 model was built on the SWISS-MODEL server ([28](#)) using the human CXCR4 sequence and the crystallographic structure of a CXCR4 dimer in complex with the small molecule IT1t as template (PDB code: 3ODU ([27](#))). At the time we performed these analyses, several crystallographic structures of CXCR4 in complex with different molecules and peptides were deposited in the PDB. None of them were fully resolved because the N- and C-terminal ends of CXCR4 are very flexible loops. In addition, the CXCR4 constructs contained T4 lysozyme inserted between helices TMV and TMVI to increase the stability of the protein; this was a common strategy to facilitate crystallogenesis of GPCRs ([26](#)). Therefore, a CXCR4 homology model was generated using the SWISS-MODEL server ([28](#)). This program reconstructed the loop between TMV and TMVI, a domain of particular importance in this study and not present in any of the crystal structures available in the PDB.

The model was further optimized by adding hydrogens, ionizing the structure at pH 7.2, and adjusting side chain positions using the Maestro Protein Preparation Wizard tool included in the Schrödinger software package ([76](#)).

Virtual Screening

Virtual screening of ligands was performed from the compounds included in the MBC library using the “Glide” module as a docking tool on the CXCR4 model using standard-position (SP) and extra-position (XP) scoring functions. The center of the catalytic pocket was selected as the centroid of the grid. In the grid generation, a scaling factor of 1.0 in van der Waals radius scaling and a partial charge cutoff of 0.25 were used. A rescoring of the SP poses of each compound was then performed with the XP scoring function of Glide. The XP mode in Glide was used in the virtual screening, the ligand sampling was flexible, epik state penalties were added, and an energy window of 2.5 kcal/mol was used for ring sampling. In the energy minimization step, the distance dependent dielectric constant was 4.0 with a maximum number of minimization steps of 100,000. In the clustering, poses were considered as duplicates and discarded if both RMS deviation is less than 0.5 Å and maximum atomic displacement is less than 1.3 Å.

Simulations were calculated with an Asus 1151 h170 LVX-GTX-980Ti workstation, with an Intel Core i7-6500 K Processor (12 M Cache, 3.40 GHz) and 16 GB DDR4 2133 MHz RAM, equipped with a Nvidia GeForce GTX 980Ti available for GPU (Graphics Processing Unit) computations.

The computational sequential workflow applied in this project included: i) Protein model construction, ii) Virtual screening (Glide), iii) PELE, iv) Docking (autoDock and Glide) and v) MD (AMBER).

PELE

A Protein Energy Landscape Exploration (35 [↗](#)) study was carried out for the control molecule IT1t, which is a small druglike isothiourea derivative that has been crystallized in complex with CXCR4 (PDB code: 3ODU). PELE is a method to explore the energy landscape of a protein that can perform unconstrained ligand exploration to search its binding site. PELE combines a Monte Carlo stochastic approach with protein structure prediction algorithms. The script “Unconstrained ligand exploration and binding search” was used to calculate 20 PELE trajectories for each ligand of the PELE program, yielding 20 PELE trajectories for each ligand. From the best five trajectories, two of them presented clearly better binding energies, and corresponded to almost the same predicted pose of the molecule. Although the predicted binding mode was not exactly the same as that in the crystal structure, the approximation was very good, allowing the approach validation. Although PELE is a suitable technique to find potential binding sites, the predicted poses must be subsequently refined using docking programs.

Docking studies

All compounds were adjusted to a pH of 7.2 ± 0.2 using the Maestro LigPrep tool (77 [↗](#)). The processed geometry of these ligands was optimized using Gaussian09 software (78 [↗](#)), which calculates RESP (Restrained ElectroStatic Potential) charges and other molecular properties. The method used for calculations was Hartree-Fock/6-311++G(2d,2p), using the tight option to ensure adequate convergence. Subsequently, all ligands were parameterized using Antechamber (79 [↗](#)). In this work, docking calculations were performed using Autodock v4.2 (80 [↗](#)).

All docking assays were performed using a box that contained the intracellular half of the protein. The CXCL12 binding pocket was excluded from the search based on the experimental results showing that none of these molecules prevent CXCL12 binding (unpublished results). The grid was formed by $126 \times 126 \times 126$ points with a spacing of 0.310 Å ($\sim 39 \times 39 \times 39$ Å³). Ligand posing was performed using a stochastic search with Lamarckian Genetic Algorithms (GAs). GAs use a population of poses and combine them to generate offspring that are evaluated with the scoring function to select the best ones to create the next generation. The GA parameters were 200 runs, a population size of 150 and 2,500,000 evaluations. The results were analyzed clustering the best 200 poses with a Root Mean Square Deviation (RMSD) of 2.0 Å.

Molecular Dynamics

The most promising docking poses from the docking studies were further studied using MD simulations to determine their stability and interactions with the receptor. The CXCR4-IT1t complex was used as a control to validate the protocol. Simulations were performed using AMBER14 (81 [↗](#)) with ff14SB (82 [↗](#)) and lipid14 (83 [↗](#)) force fields in the NPT thermodynamic ensemble (constant pressure and temperature). All bonds involving hydrogen atoms were constrained with the SHAKE algorithm (84 [↗](#)). To stabilize CXCR4 and more accurately reproduce the real environment in the MD simulation, the system was embedded in a lipid bilayer using the Membrane Builder tool (85 [↗](#)) from the CHARMM-GUI server. The membrane was composed of 175 molecules of the fatty acid 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC) in each leaflet. The protein-membrane complex was solvated with TIP3 water molecules. Chloride ions were added up to a concentration of 0.15 M in water, and sodium ions were added to neutralize the system. Minimization was performed using 3500 Steepest Descent steps and 4500 Conjugate Gradient steps three times, firstly considering only hydrogens, then considering only water molecules and ions, and finally minimizing all atoms. Equilibration raises system temperature from 0 to 300 K at a constant volume, fixing everything but ions and water molecules. After thermalization, several density equilibration phases were performed. In the production phase 50 ns MD simulations without position restraints were calculated using a time step of 2 fs. Trajectories of the most interesting poses were extended to 150 ns. “Stable trajectories” have been

defined to refer to those trajectories of the ligand-protein complexes whose RMSD does not fluctuate more than 0.25 Å. To estimate the affinity of the ligands for CXCR4, the binding energy of each representative structure obtained with the principal components analysis was calculated using the PRODIGY-LIGAND server (86 [↗](#)).

Virtual screening validation

We also used a strategy based on the model of CXCR4 predicted by AlphaFold (87 [↗](#)) and the sequence available under UniProt entry P61073 to validate our studies. We prepared the ligands using the OpenBabel (88 [↗](#)), with a gasteiger charge assignment, and generated 10 conformers for each input ligand using OpenBabel. We then prepared the target structure with Openmm, removing all water and possible heteroatoms, and adding all missing atoms. We next predicted the target binding pockets with fPocket (89 [↗](#)), p2rank (90 [↗](#)), and AutoDock autosite (91 [↗](#)). We chose only those pockets between TMV and TMVI. We merged the results of the three programs into so-called consensus pockets, as two pockets are said to be sufficiently similar if at least 75% of their surfaces are shared (92 [↗](#)). From the consensus pockets, one pocket was significantly larger than the others and was, therefore, selected. We then docked the ligand conformers in this pocket using AutoDock GPU (93 [↗](#)), LeDock (94 [↗](#)), and Vina (95 [↗](#)). The number of dockings varied from 210 to 287 poses. We scored each pose with the Vina score using ODDT (96 [↗](#)). Then, we clustered the different solutions into groups whose maximum RMSD was 1 Å. This resulted in 40 clusters; the representative of each cluster was the one with maximum Vina score, and confirmed that the selected compounds bound this pocket. When required, we calculated the binding affinity using Schrodinger's MM-GBSA procedure (97 [↗](#)) in two ways: first, assuming that the ligand and target are fixed; second, with an energy minimization of all the atoms within a distance of 3 Å from the ligand.

Cleft volume calculations

Once we detected that several compounds bound CXCR4 in the target region, the cleavage properties were calculated by subtracting the compound structure. The resulting PDB was analyzed using the PDBsum server (98 [↗](#)). Volume calculations were obtained using the SURFNET server to analyze surface clefts (36 [↗](#)). The theoretical interaction surface between the selected compounds and CXCR4 and the atomic distances between the protein residues and the compounds was calculated using the PISA server (99 [↗](#)).

Transwell migration assay

Cells (3×10^5) in 0.1 ml of RPMI medium containing 10 mM HEPES and 0.1% BSA were placed in the upper wells of 5-μm pore size transmigration chambers (Transwell, Costar, Corning, NY). CXCL12 (12.5 nM) or CCL17 (50 nM) in 0.6 ml of the same medium was added to the lower well. Cell migration was evaluated as described (120 min, 37°C, 5% CO₂) (21 [↗](#)).

Cell cycle analysis

Cells (5×10^5 cells/well) were collected from microplates and washed twice in PBS and then resuspended in 50 μl of detergent (DNA-Prep Reagent Kit; Beckman Coulter, Brea, CA) containing 10 ng/ml propidium iodide (DNA-Prep Reagent Kit; 30 min, 37°C). Cell cycle phases were analyzed by flow cytometry on a Beckman Coulter FC500 flow cytometer and results were expressed as the percentage of stained cells. H₂O₂ (10%) and staurosporine (10 μM) treatments were used as positive controls of cell death.

Single molecule TIRF imaging and analysis

Transfected cells expressing 8,500–22,000 receptors/cell (<4.5 particles/μm²) were plated on glass-bottomed microwell dishes (MatTek Corp., Ashland, MA) coated with fibronectin (Sigma-Aldrich, St. Louis, MO: 20 μg/ml, 1 hr, 37°C) as a substrate for cell adhesion and to facilitate image

acquisition, as TIRF determinations are restricted to the cell membrane in contact with the coverslip. These are conditions corresponding to the unstimulated cells. To observe the effect of the ligand, we coated dishes with CXCL12 (100 nM, 1 hr, 37°C); stimulated cells were incubated (20 min, 37°C, 5% CO₂) before image acquisition. Experiments were performed at 37°C with 5% CO₂ using a total internal reflection fluorescence microscope (Leica AM TIRF inverted microscope; Leica Microsystems, Wetzlar, Germany). Image sequences of individual particles (500 frames) were then acquired at 49% laser power (488-nm diode laser) with a frame rate of 10 Hz (100 ms/frame). The penetration depth of the evanescent field was 90 nm. Particles were detected and tracked using the U-Track2 algorithm (100) implemented in MATLAB, as described (101). Mean spot intensity (MSI), number of mobile and immobile particles and diffusion coefficients (D_{1-4}) were calculated from the analysis of thousands of single trajectories over multiple cells (statistics provided in the respective figure captions) using described routines (101). The receptor number along individual trajectories was determined as reported (20), using the intensity of the monomeric protein CD86-AcGFP as a reference (102). Briefly, for each particle detected we measured the intensity of pixels around the particle (in a 3×3 pixel matrix whose central position is given by the coordinates x,y obtained by U-Track2), in each frame along its trajectory. We also estimated the particle background (K0) for each frame. K0 for each frame was calculated by locating the cell in the video and analyzing the intensities of the cell background in each frame. K0 was chosen as the gray value at a given quantile of this distribution (80%). The intensity value for each particle is then given by the difference between particle intensity and the background in each frame. To minimize photon fluctuations within a given frame, we considered the particle intensity as the average value (background subtracted) obtained over the first 20 frames. To ensure that no photobleaching events occurred within this range of frames that would affect the quantitation of the data, we measured the photobleaching times (s) from individual CD86-AcGFP (electroporated in Jurkat CD4⁺ cells, (103)) particles that exclusively showed a single photobleaching step. Fitting of the distribution to a single exponential decay renders a τ_0 value of ~5 s, which corresponds to 50 frames (100 ms/frame). The total number of receptors/particles was finally estimated by dividing the average particle intensity by the particle intensity arising from individual AcGFP molecules. To unequivocally identify the intensity emitted by an individual AcGFP, we used as a calibrator Jurkat CD4⁺ cells electroporated with the monomeric CD86-AcGFP. We performed SPT experiments on CD86-AcGFP and analyzed the data in similar way as described above. Distribution of monomeric particle intensities was analyzed by Gaussian fitting, rendering a mean value of 980 ± 89 a.u. This value was then used as the monomer reference to estimate CXCR4-AcGFP particle size.

Internalization and flow cytometry studies

Cells (5×10^5 cells/well) were activated with CXCL12 (50 nM) at the indicated time points at 37°C. After incubation, cells were washed twice with staining buffer and receptor internalization was determined by flow cytometry using an anti-CXCR4 mAb (clone 44717, 30 min, 4°C), followed by secondary staining with PE-coupled goat anti-mouse IgG (30 min, 4°C). Results are expressed as a percentage of the mean fluorescence intensity of treated cells relative to that of untreated cells. When required, cells were pretreated with the small compounds (50 μ M, 30 min, 37°C) or AMD3100 (1 μ M, 30 min, 37°C) before activation with CXCL12.

Western blotting

Cells (3×10^6) were activated with CXCL12 (50 nM) at the time points indicated and then lysed in RIPA detergent buffer supplemented with 1 mM PMSF, 10 μ g/ml aprotinin, 10 μ g/ml leupeptin and 10 μ M sodium orthovanadate, for 30 min at 4°C. Extracts were analyzed by western blotting using specific antibodies. Densitometric evaluation of blots was performed using ImageJ (NIH, Bethesda, MD). When required, cells were pretreated with the small compounds (50 μ M, 30 min, 37°C) or AMD3100 (1 μ M, 30 min, 37°C) before activation with CXCL12.

cAMP determination

cAMP levels were determined using the cAMP-Glo™ Max Assay (Promega, Madison, WI, in cells (5×10^5 cells/well) untreated or pre-treated with CXCL12 (50 nM, 5 min, 37°C) followed by the addition of forskolin (10 μ M, 10 min, 37°C). When required, cells were pretreated with the small compounds (50 μ M, 30 min, 37°C) or AMD3100 (1 μ M, 30 min, 37°C) before activation with CXCL12.

CXCL12 binding

Cells (5×10^5 cells/well) were incubated with CXCL12 ATTO-700 ([32](#)), (30 min, 37°C) and maintained at 4°C before analyzing the bound fluorescence in a Beckman Coulter Cytoflex flow cytometer. Results are expressed as mean fluorescence intensity. When required, cells were pretreated with the small compounds (50 μ M, 30 min, 37°C) or AMD3100 (1 μ M, 30 min, 37°C) before activation with CXCL12.

Fluorescence Resonance Transfer Analysis by Sensitized Emission

HEK-293T cells transiently transfected at a fixed CXCR4-YFP: CXCR4-CFP ratio (15 μ g and 9 μ g, respectively) were treated or not with the compounds (50 μ M, 30 min, 37°C) or their vehicle (DMSO), and FRET efficiency was evaluated ($n = 3$, mean $\pm$ SEM, $***p \leq 0.001$). Emission light was quantified using the Wallac Envision 2104 Multilabel Reader (Perkin Elmer, Foster City, CA) equipped with a high-energy xenon flash lamp (donor: receptor fused to C-CFP, 8-nm bandwidth excitation filter at 405 nm; acceptor: receptor fused to YFP, 10 nm bandwidth excitation filter at 510 nm).

Actin polymerization

Cells (5×10^5 cells/well) were incubated with CXCL12 (50 nM, 37°C) or anti-CD3 (1 μ g/ml, 37°C) at the indicated time periods and then fixed with 2% paraformaldehyde and transferred to ice for 10 min. Fixed cells were permeabilized with 0.01% saponin (10 min, 4°C) and labeled with Phalloidin-TRITC (Merck, 30 min, 4°C). After washing, bound fluorescence was analyzed by flow cytometry on a Beckman Coulter FC500 flow cytometer. When required, cells were pretreated with the small compounds (50 μ M, 30 min, 37°C) or AMD3100 (1 μ M, 30 min, 37°C) before activation with CXCL12.

Immunofluorescence analyses

Cells on fibronectin (20 μ g/ml, Sigma)-coated glass slides were stimulated or not with 100 nM CXCL12 (5 min at 37 °C), fixed with 4% paraformaldehyde (10 min), permeabilized with 0.25% saponin (10 min), and stained with phalloidin-TRITC (Sigma-Merck; 30 min), all at room temperature (RT). Preparations were analyzed using a Leica TCS SP8 confocal multispectral microscope. When required, cells were pretreated with the small compounds (50 μ M, 30 min, 37°C) or AMD3100 (1 μ M, 30 min, 37°C) before activation with CXCL12.

Cell adhesion/migration on planar lipid bilayers

Planar lipid bilayers were prepared as reported ([104](#)). Briefly, unlabeled GPI-linked intercellular adhesion molecule 1 (ICAM-1) liposomes were mixed with 1,2-dioleoyl-phosphatidylcholine. Membranes were assembled in FCS2 chambers (Biopetech, Butler, PA), blocked with PBS containing 2% FCS for 1 hr at RT, and coated with CXCL12 (200 nM, 30 min, RT). Cells (3×10^6 cells/ml) in PBS containing 0.5% FCS, 0.5 g/l D-glucose, 2 mM $MgCl_2$ and 0.5 mM $CaCl_2$ were then injected into the pre-warmed chamber (37°C). Confocal fluorescence, differential interference contrast (DIC) and interference reflection microscopy (IRM) images were acquired on a Zeiss Axiovert LSM 510-META inverted microscope with a 40 $\times$ oil-immersion objective. Imaris 7.0 software (Bitplane, Zurich, Switzerland) and ImageJ 1.49v were used for qualitative and quantitative analysis of cell dynamics parameters, fluorescence and IRM signals. The fluorescence signal of the planar bilayer in each case was established as the background fluorescence intensity.

The frequency of adhesion (IRM⁺ cells) per image field was estimated as [n° of cells showing IRM contact/total n° of cells (estimated by DIC)] × 100; similarly, we calculated the frequency of migration (cells showing and IRM contact and moving over time). When required, cells were pretreated with the small compounds (50 μM, 30 min, 37°C) or AMD3100 (1 μM, 30 min, 37°C) before activation with CXCL12.

Directional cell migration

Pre-treated cells were diluted to 10×10^6 cell/ml in RPMI medium containing 1% BSA and 20 mM HEPES (chemotaxis medium) and then seeded into the channel of a chemotaxis slide (μ-Slide Chemotaxis System, 80326; Ibidi, Munich, Germany) (1 hr, 37°C, 5% CO₂). The reservoirs were then filled with chemotaxis medium containing 50 μM AGR1.137 or AGR1.131, or vehicle (DMSO) as a control condition, and 100 nM CXCL12 or CXCL2 was added to the right reservoir. Phase-contrast images were recorded over 20 hrs with a time lapse of 15 min using a Microfluor inverted microscope (Leica) with a 10× objective and equipped with an incubation system set to 5% CO₂ and 37°C. Single-cell tracking was evaluated by selecting the center of mass in each frame using the manual tracking plug-in tool in ImageJ. Spider plots, representing the trajectories of the tracked cells, forward migration index (FMI), and straightness values, were obtained using the Chemotaxis and Migration Tool (ImageJ).

Drug efficacy assay in zebrafish

HeLa cells were labeled with 8 μg/ml Fast-DiI™ oil (Cat #D3899; ThermoFisher Scientific, Waltham, MA) in RPMI medium supplemented with 2% FBS for 30 min at 37°C. Cells were then washed and filtered, and cell viability was determined using trypan blue-exclusion. Transgenic zebrafish Tg(fli1a:EGFP)y1 (105 [DOI](#)) were bred naturally and maintained in E3 embryo medium (deionized water containing 0.5 mM NaCl, 0.4 mM CaCl₂, 0.7 mM MgSO₄ and 0.2 mM KCl, pH 7.2) supplemented with 0.2 mM PTU (E3/PTU) at 28.5°C. DiI-labeled HeLa cells were implanted into the dorsal perivitelline space of 2-day-old embryos, as described (38 [DOI](#)). Tumors were imaged within 2 hrs of implantation and tumor-bearing embryos were treated with either vehicle (DMSO), AMD-3100 (1 and 10 μM) or with AGR1.131 or AGR1.137 (10 and 50 μM) for 3 days, followed by re-imaging. Changes in tumor size were evaluated as tumor area at day 3 divided by tumor area at day 0, and metastasis was evaluated as the number of cells disseminated to the caudal hematopoietic plexus.

Statistical analyses

All data were analyzed with GraphPad Prism software version 9 (GraphPad Inc., San Diego, CA). Cell migration in Transwell assays and in planar lipid bilayers and directional cell migration assays and cell polarization under the various conditions were analyzed to determine significant differences between means using one-way analysis of variance (ANOVA) followed by Tukey's multiple comparison test. A two-tailed Mann-Whitney non-parametric test was used to analyze the diffusion coefficient (D_{1-4}) of single particles. We used contingency tables to compare two or more groups of categorical variables, such as the percentages of mobile or immobile particles, and these were compared using a Chi-square test with a two-tailed p-value. Statistical differences were reported as n.s. = not significant $p > 0.05$, * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ and **** $p \leq 0.0001$.

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We also acknowledge the technical help of the Advance Light Microscopy Unit at the CNB/CSIC. Compounds AGR1.31, AGR1.35 and AGR1.37 are included in the patent PCT/ES2022/070379.

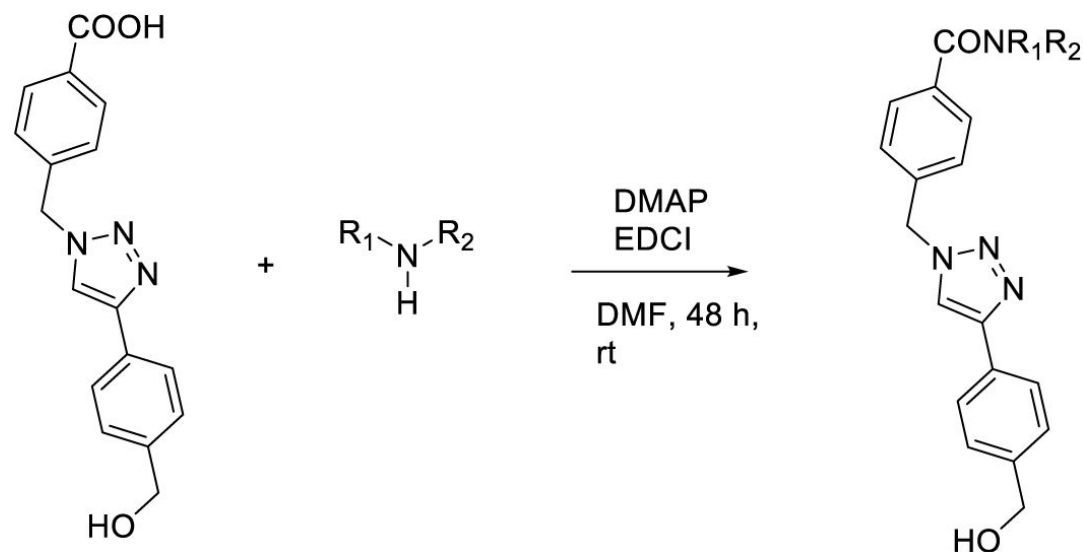
Supplementary Information

Compound synthesis and characterization

1. General Experimental Details

All solvents were purchased from Sigma-Aldrich (anhydrous solvents) and commercially available reagents were used as received. All reactions were followed by TLC analysis (TLC plates GF254, Merck) or LC-MS (liquid chromatography mass spectrometry). Melting points were determined in a Büchi Melting Point M-560 apparatus. NMR spectra were recorded at ambient temperature unless otherwise stated using standard pulse methods on one of the following spectrometers and signal frequencies: Bruker AV-300 (1H = 300 MHz, 13C = 75 MHz). Chemical shifts are reported in ppm and are referenced to tetramethylsilane (TMS) or to the following solvent peaks: CDCl₃ (1H = 7.27 ppm, 13C = 77.00 ppm), DMSO-d₆ (1H = 2.50 ppm, 13C = 39.51 ppm). Coupling constants are quoted to the nearest 0.1 Hz, and multiplicities are given by the following abbreviations and combinations thereof: s (singlet), d (doublet), t (triplet), q (quartet), m (multiplet), and br (broad). Column chromatography was performed on prepacked silica gel columns using a Biotage SP4 or Isolera One system. High resolution mass spectra (HRMS-ESI) were recorded on a Micromass Q-Tof Ultima hybrid quadrupole time-of-flight mass spectrometer, with analytes separated on an Agilent 1100 liquid chromatograph equipped with a Phenomenex Luna C18(2 [μm](#)) reversed phase column (100 mm × 2.1 mm, 3 μm packing diameter).

2. Synthesis and Characterization of Triazol derivatives General Procedure (GP)



Over a solution of 4-((4-(4-(hydroxymethyl)phenyl)-1H-1,2,3-triazol-1-yl)methyl) benzoic acid (123 mg, 0.4 mmol) in 4 ml of DMF, DMAP (97 mg, 0.8 mmol) and EDC.HCl (92 mg, 0.48 mmol) were added and stirred for 15 min at room temperature. Subsequently, the corresponding amine (0.4 mmol) was added and stirred at room temperature for 48 hrs. Next, H₂O (10 ml) was added and extracted with DCM (25 ml $\times$ 3). The organic phase was dried over MgSO₄, filtered and evaporated under reduced pressure. The crude sample obtained was purified by column chromatography, eluting with a DCM-MeOH solvent mixture (from 99:1 to 90:10).

N-Benzyl-4-((4-(4-(hydroxymethyl)phenyl)-1H-1,2,3-triazol-1-yl)methyl)benzamide (AGR1.131)

The reaction was performed according to GP using benzylamine (44 μ l, 0.4 mmol), and the title product was obtained after purification by silica gel column chromatography as a white solid (35% yield). Pf: 193 – 195 $^{\circ}$ C.

¹H NMR (300 MHz, DMSO-*d*₆) δ 9.09 (t, *J* = 6.0 Hz, 1H), 8.65 (s, 1H), 7.98 – 7.89 (m, 2H), 7.86 – 7.78 (m, 2H), 7.46 (d, *J* = 8.2 Hz, 2H), 7.40 (d, *J* = 8.1 Hz, 2H), 7.36 – 7.20 (m, 5H), 5.73 (s, 2H), 5.25 (t, *J* = 5.7 Hz, 1H), 4.53 (d, *J* = 5.7 Hz, 2H), 4.49 (d, *J* = 6.0 Hz, 2H). ¹³C NMR (75 MHz, DMSO) δ 165.69, 146.69, 142.29, 139.56, 139.01, 134.08, 128.99, 128.25, 127.83, 127.74, 127.12, 126.88, 126.71, 124.90, 121.42, 62.61, 52.58, 42.56.

HRMS (ESI⁺) *m/z* Calc. for C₂₄H₂₃N₄O₂ [M+H]⁺: 399.1816. Found: 399.1809.

(4-((4-(4-(Hydroxymethyl)phenyl)-1H-1,2,3-triazol-1-yl)methyl)phenyl)(4-(2-nitro-4-(trifluoromethyl)phenyl)piperazin-1-yl)methanone (AGR1.135)

The reaction was performed according to GP using (2-nitro-4-(trifluoromethyl)phenyl)piperazine (110 mg, 0.4 mmol), and the title product was obtained after purification by silica gel column chromatography as a yellow solid (37% yield). Pf: 184 – 186 $^{\circ}$ C.

¹H NMR (300 MHz, DMSO-*d*₆) δ 8.65 (s, 1H), 8.19 (d, *J* = 2.3 Hz, 1H), 7.89 (dd, *J* = 8.9, 2.3 Hz, 1H), 7.81 (d, *J* = 8.0 Hz, 2H), 7.53 – 7.35 (m, 7H), 5.71 (s, 2H), 5.23 (t, *J* = 5.7 Hz, 1H), 4.52 (d, *J* = 5.4 Hz, 2H), 3.73 (s, 2H), 3.48 (s, 2H), 3.21 (s, 4H). ¹³C NMR (75 MHz, DMSO) δ 169.12, 147.70, 147.08, 142.67, 140.04,

137.84, 135.74, 130.59, 130.54, 129.38, 128.27, 128.00, 127.25, 125.65, 125.29, 124.12, 124.07, 122.06, 121.92, 121.82, 120.57, 120.12, 62.99, 62.87, 52.97, 50.33, 47.01. HRMS (ESI⁺) Calc. for C₂₈H₂₆F₃N₆O₄ [M+H]⁺: 567.1962. Found: 567.1959.

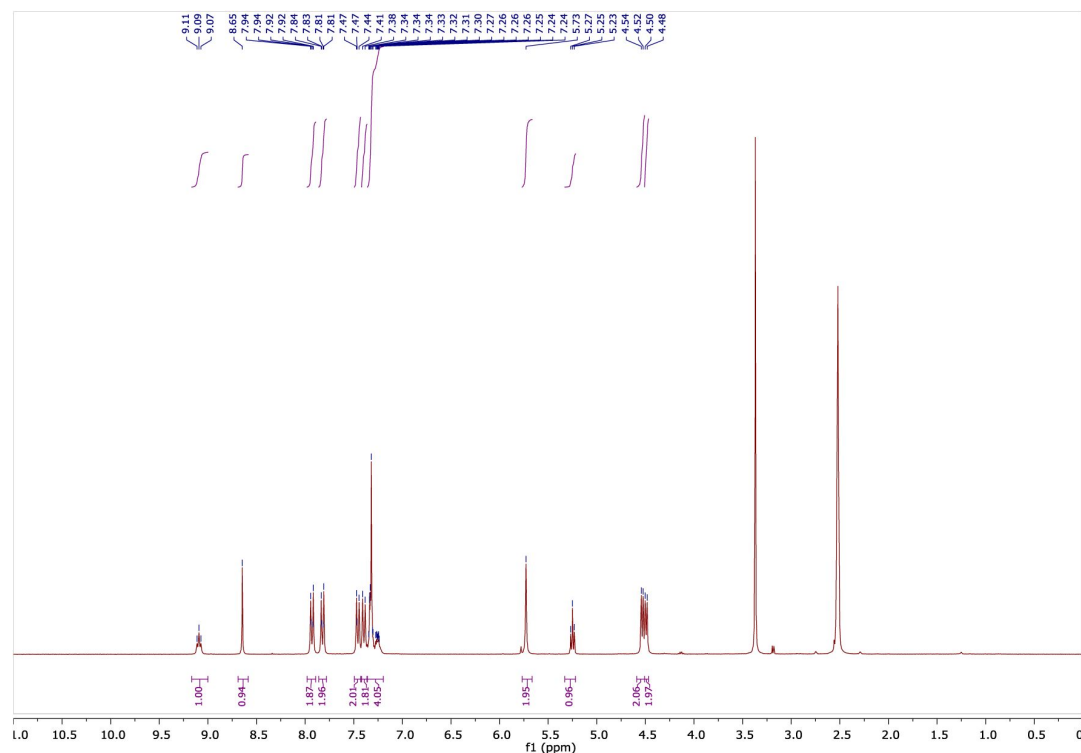
Ethyl 1-(4-((4-(hydroxymethyl)phenyl)-1H-1,2,3-triazol-1-yl)methyl)benzoyl) piperidine-4-carboxylate (AGR1.137)

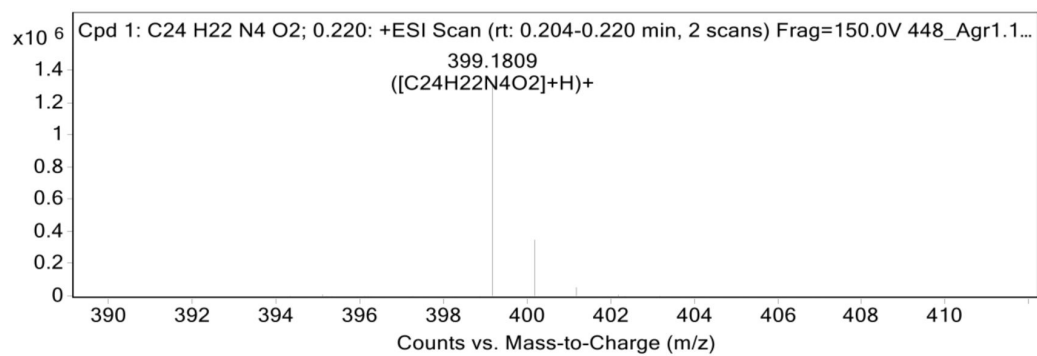
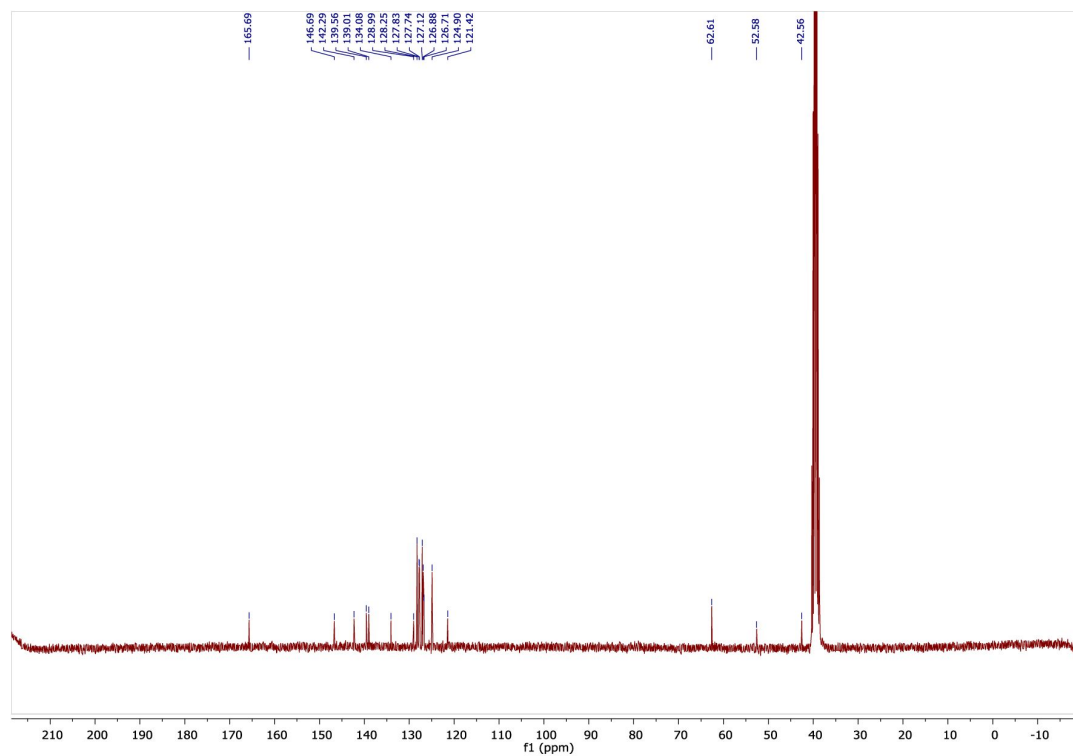
The reaction was performed according to GP using ethyl piperidine-4-carboxylate (63 mg, 0.4 mmol) and the title product was obtained after purification by silica gel column chromatography as a white solid (30% yield). Pf: 119 – 121 °C.

¹H NMR (300 MHz, Methanol-*d*₄) δ 8.40 (s, 1H), 7.89 – 7.74 (m, 2H), 7.56 – 7.31 (m, 6H), 5.73 (s, 2H), 4.66 (s, 2H), 4.48 (d, *J* = 13.2 Hz, 1H), 4.16 (q, *J* = 7.1 Hz, 2H), 3.69 (d, *J* = 13.0 Hz, 1H), 3.24 – 2.96 (m, 2H), 2.78 – 2.52 (m, 1H), 2.15 – 1.52 (m, 4H), 1.27 (t, *J* = 7.1 Hz, 3H). ¹³C NMR (75 MHz, MeOD) δ 175.82, 171.87, 149.21, 143.15, 138.81, 137.25, 130.56, 129.43, 128.59, 128.56, 126.71, 122.39, 64.88, 61.79, 54.58, 42.65, 41.96, 29.67, 28.96, 14.51. HRMS (ESI⁺) Calc. For C₂₅H₂₉N₄O₄ [M+H]⁺: 449.2183. Found: 449.2160.

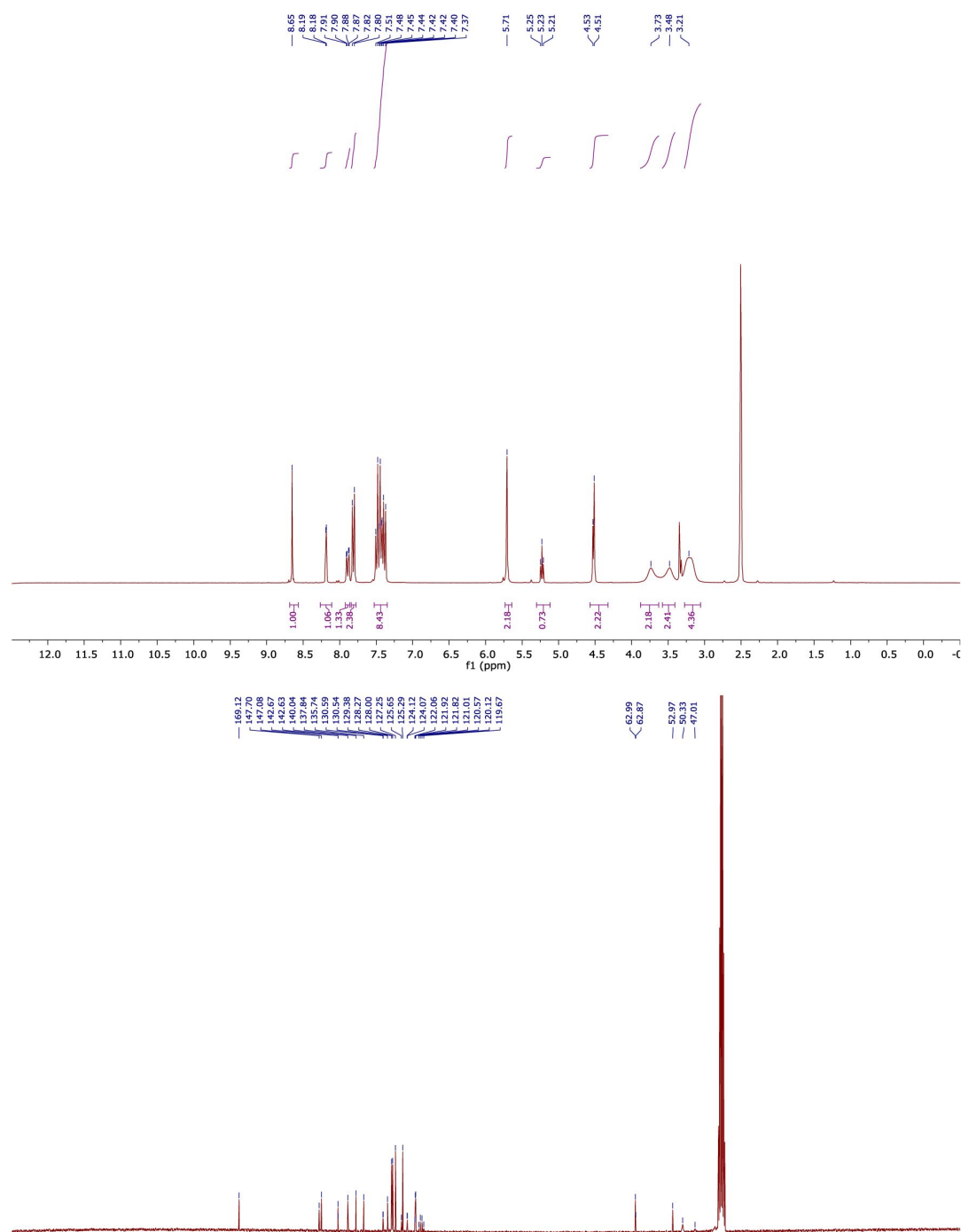
3. NMR Spectra and HRMS Analysis

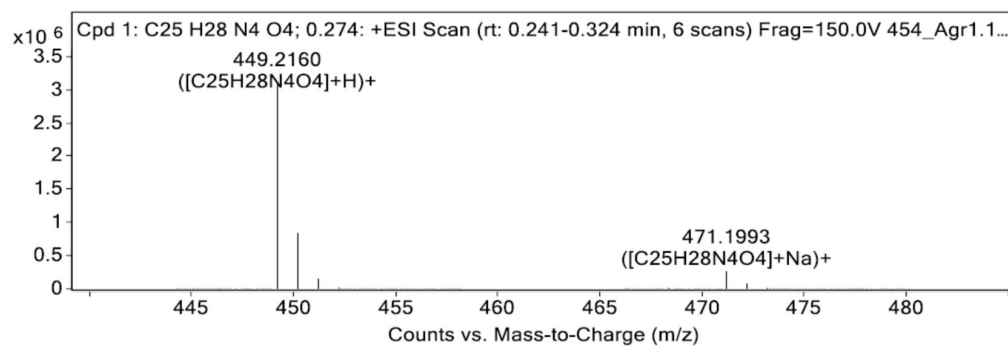
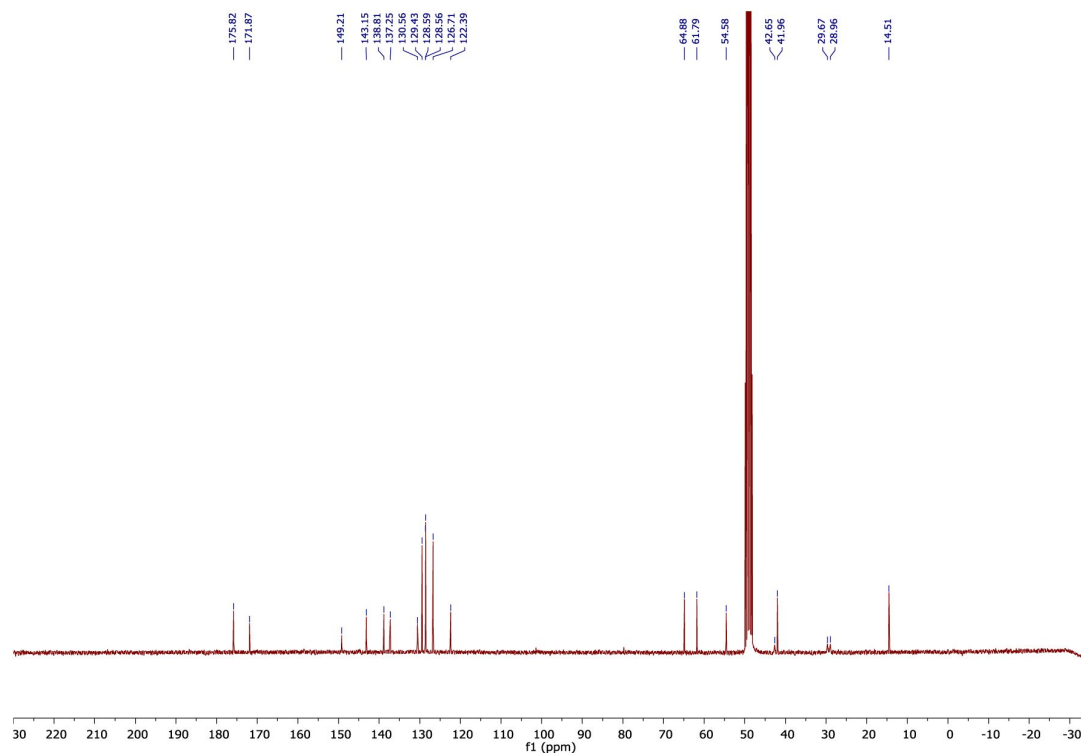
(AGR1.131)

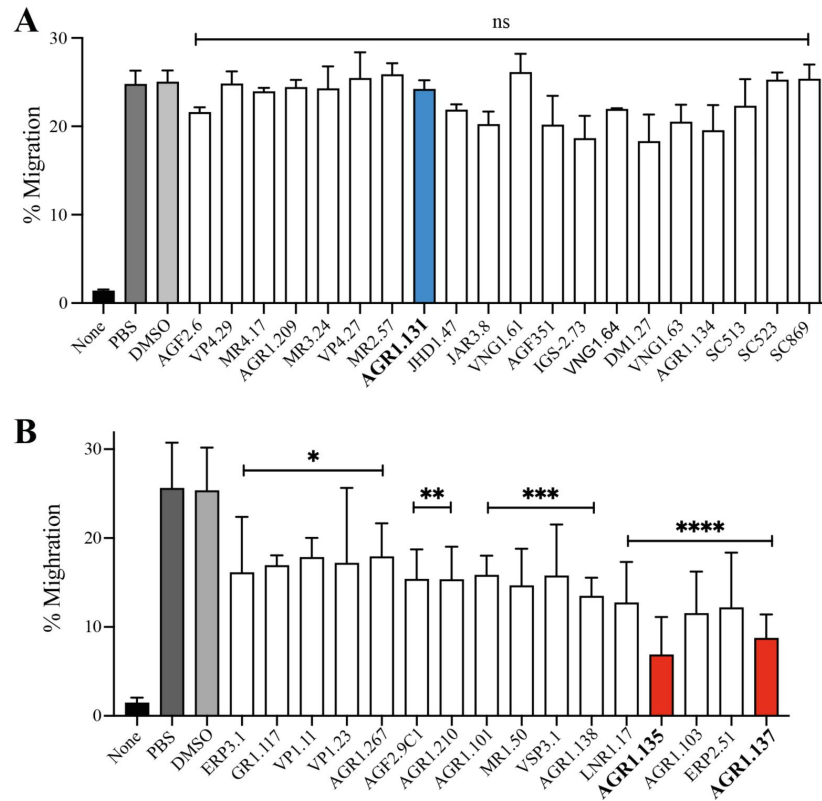




(AGR1.135)



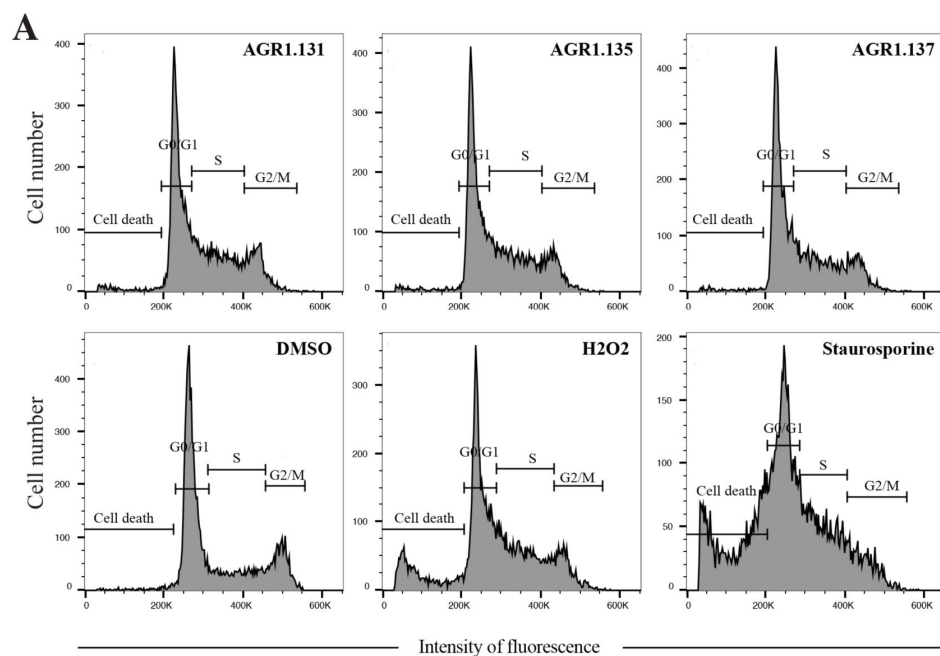




Supplementary Figure 1.

Small compound-mediated inhibition of CXCL-12-induced cell migration.

A, B). Migration of Jurkat cells, untreated or treated with the selected small compounds (50 μ M) or DMSO (vehicle) in the previous in silico screening, in response to 12.5 nM CXCL12. Data shown as percentage of migrating cells (mean $\pm$ SD; n = 5; *p $\leq$ 0.05, **p $\leq$ 0.01, ***p $\leq$ 0.001, ****p $\leq$ 0.0001, ns: not significant).



B

	Cell death	G0/G1	S	G2/M
AGR1.131	3.3	46.7	33.6	16.3
AGR1.135	3.2	47.9	34.0	14.7
AGR1.137	2.68	50.5	32.3	14.4
DMSO	3.0	57.2	21.4	18.6
H2O2	14.8	41.8	31.9	11.3
Staurosporine	28.7	37.0	24.4	9.8

Supplementary Figure 2.

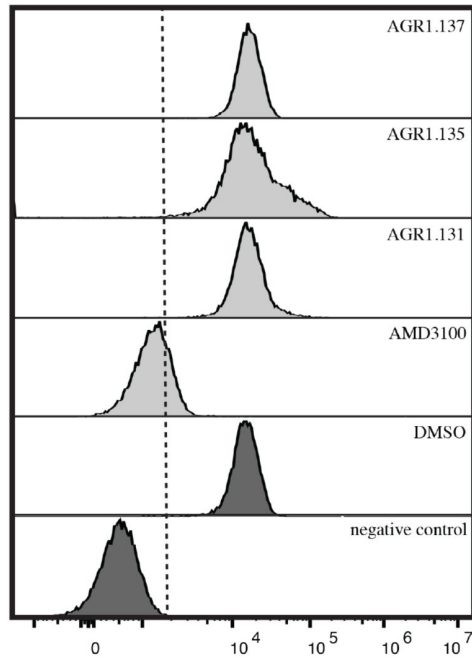
Effect of small compounds on Jurkat cell cycle.

A). Cell cycle analysis using propidium iodide incorporation and flow cytometry of Jurkat cells treated with AGR1.131, AGR1.135 and AGR1.137 (2 hours, 37°C) or with DMSO (vehicle), H₂O₂ (10%) and staurosporine (10 μM) as controls. The percentage of cells in each cycle stage is shown. **B).** Summary of the percentage of treated cells in each cell cycle phase.

Supplementary Figure 3.

Flow cytometry analysis of the binding of CXCL12-ATTO700 to JK cells.

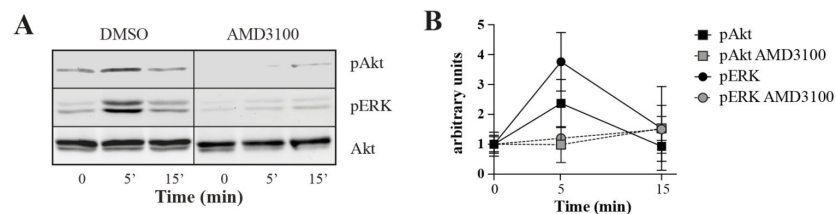
Cells were pre-incubated for 30 min at 37°C with the indicated antagonists, followed by incubation with CXCL12-ATTO700 (30 min 37°C). CXCL12 binding was determined in flow cytometry. A representative experiment of 4 performed is shown.



Supplementary Figure 4.

Western blot analysis of AMD3100-treated JK cells.

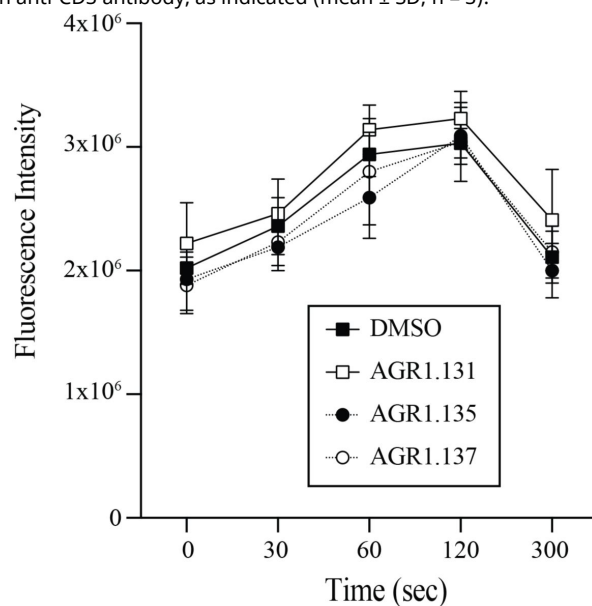
A) Cells were pre-incubated for 30 min at 37°C with AMD3100 or DMSO (vehicle) as control, stimulated with CXCL12 and analyzed by western blot using specific antibodies. **B)** Densitometry analysis of the western blot images in A) using Image J software.



Supplementary Figure 5.

AGR1.135 and AGR1.137 do not alter anti-CD3-induced actin polymerization.

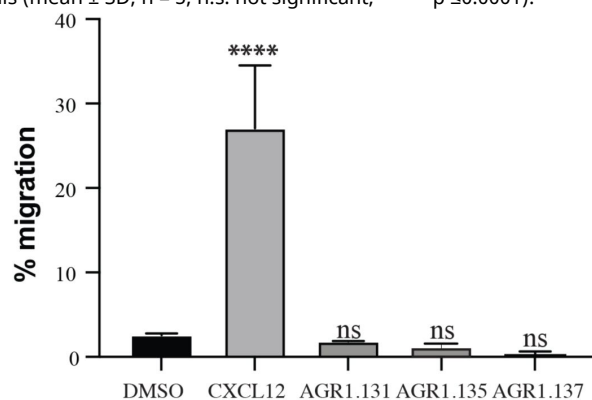
F-actin (Phalloidin-TRITC) staining of Jurkat cells treated with the indicated compounds or DMSO (vehicle), adhered to fibronectin and stimulated with anti-CD3 antibody, as indicated (mean $\pm$ SD; n = 3).



Supplementary Figure 6.

AGR1.131, AGR1.135 and AGR1.137 are not CXCR4 agonists.

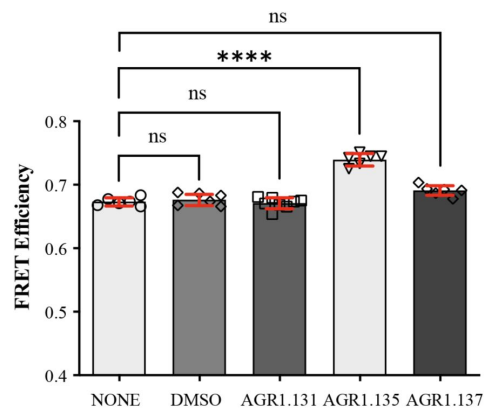
Migration of Jurkat cells in response to 12.5 nM CXCL12, 50 μ M AGR1. 131, 50 μ M AGR1. 135 or 50 μ M AGR1. 137. Data shown as percentage of migrating cells (mean $\pm$ SD; n = 5; n.s. not significant, **** p $\leq$ 0.0001).



Supplementary Figure 7.

FRET analysis of CXCR4 in the presence of AGR1.131, AGR1.135 and AGR1.137.

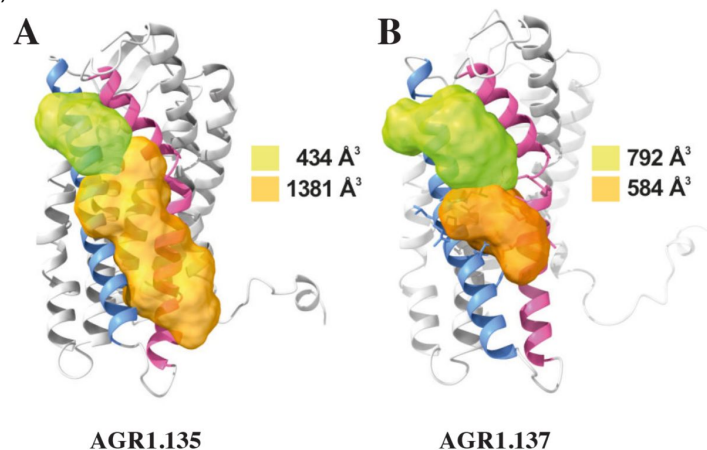
FRET efficiency in HEK-293 cells transiently transfected with CXCR4-YFP/ CXCR4-CFP (ratio 15:9), in the presence of 50 μ M AGR1.131, 50 μ M AGR1.135, 50 μ M AGR1. 137 or vehicle (DMSO). Data shows FRET efficiency (a.u.) (mean $\pm$ SD; n = 6; n.s. not significant, **** p $\leq$ 0.0001).

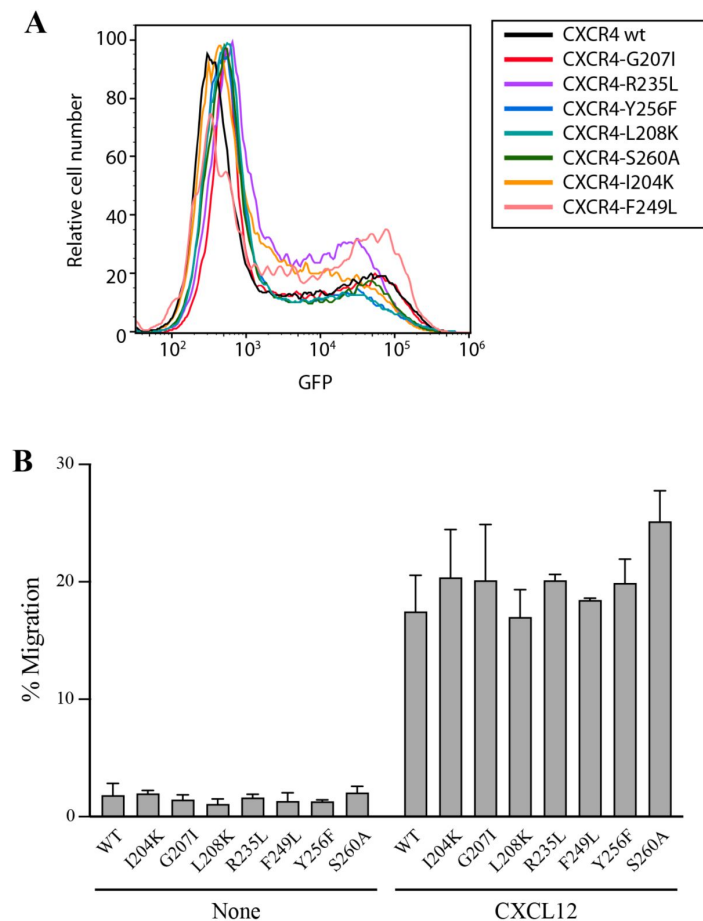


Supplementary Figure 8.

Cartoon and surface representation of CXCR4 and the clefts identified between TMV and TMVI.

The protein structure is shown in gray, with TMV and TMVI colored in blue and pink, respectively. Cavities associated to AGR1.135 (A) and AGR1.137 (B) binding were identified by SurfNet software are shown in orange and green. Volumes for each cavity are measured ($\text{\AA}^3$).





Supplementary Figure 9.

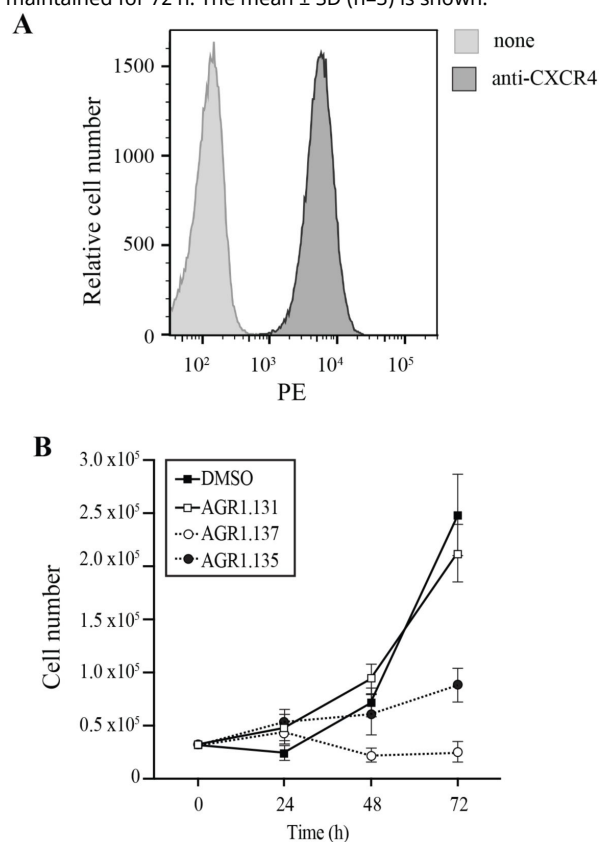
Expression and function of CXCR4 and CXCR4 mutants coupled to AcGFP in transfected JK^{-/-} cells.

A) Flow cytometry analysis of cells were transiently transfected with the indicated CXCR4 mutants or wt CXCR4, coupled to AcGFP and their expression was determined 24 hours after transfection by GFP detection in flow cytometry. A representative experiment of 4 performed is shown. **B)** CXCL12-induced migration of CXCR4-deficient Jurkat cells (JK^{-/-}) transiently transfected with CXCR4wt or its mutants, CXCR4I204K, CXCR4G207I, CXCR4L208K, CXCR4R235L, CXCR4F249L, CXCR4Y256F and CXCR4S260A. Data are shown as the mean $\pm$ SD percentage of input cells that migrate (n = 3).

Supplementary Figure 10.

Characterization of HeLa cells.

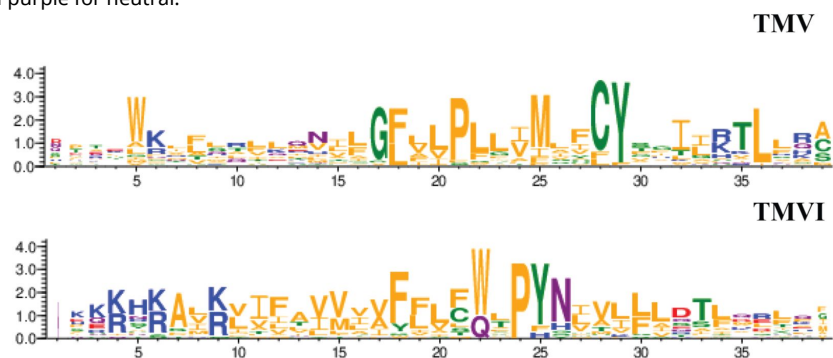
A) CXCR4 expression on the surface of HeLa cells, analyzed by flow cytometry using a specific anti-CXCR4 antibody. Data show a representative experiment of 6 performed. **B)** Effect of the indicated compounds on HeLa cells growth. Cells (0.5×10^6) were plated on 48 well plates in the presence of 100 mM of the compounds, which were added freshly every 24 hours. Cells were counted at every 24 hours and maintained for 72 h. The mean $\pm$ SD (n=3) is shown.

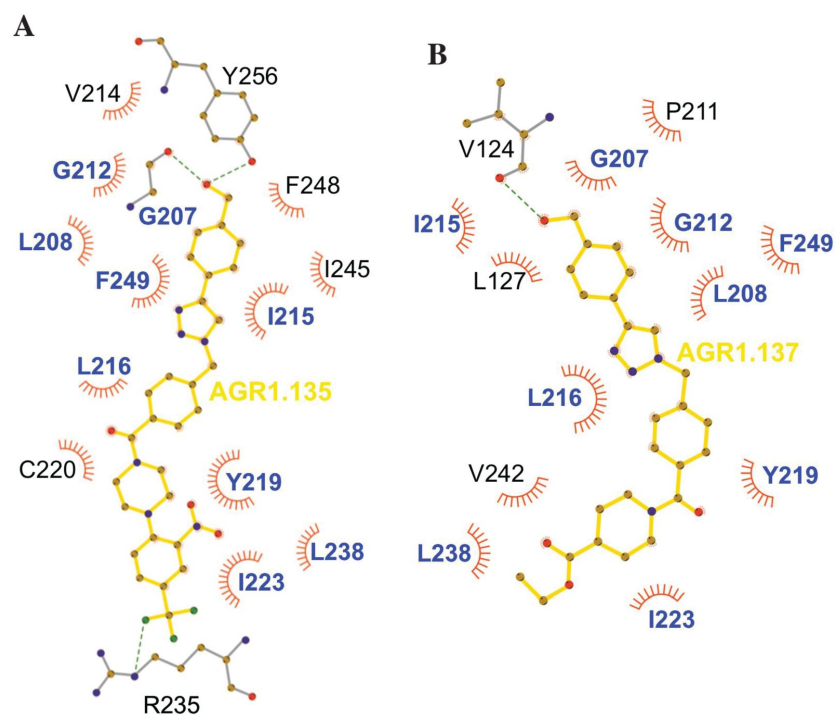


Supplementary Figure 11.

Residue conservation among all GPCR class A chemokine receptors for TMV and TMVI.

Graphs were produced in the WebLogo server after aligning all receptors sequence in the sequence alignment tool included in GPCRdb. Font size for each residue represented in one letter code stands for the degree of conservation among all chemokine receptors. Residues are colored by their chemistry being green for polar, orange for hydrophobic, red for acidic, blue for basic and purple for neutral.

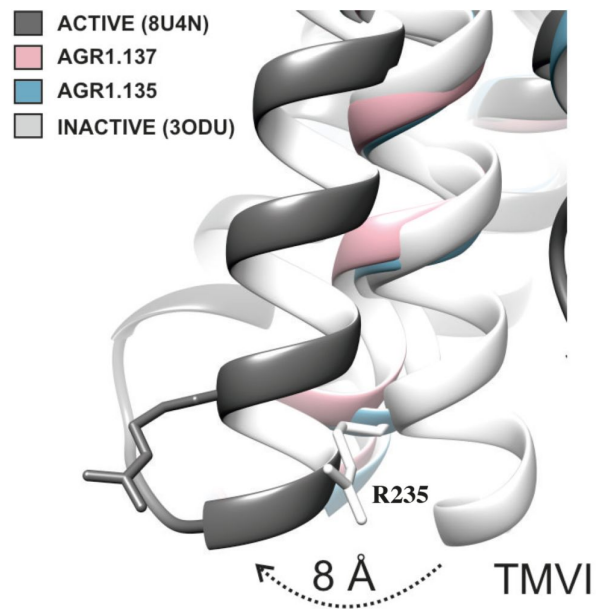




Supplementary Figure 12.

Ligplot representation of antagonists interactions with CXCR4.

Ligplot representations of AGR1. 135 (**A**) and AGR1. 137 (**B**), with hydrogen bonds represented in green dotted lines and non-bonded contacts in red spoke arcs. Residues labeled in blue font are present in the interface of both AGR1. 135 and AGR1. 137. CXCR4 residues forming a hydrogen bond with the ligands are shown in gray.



Supplementary Figure 13.

TMVI transition from inactive to active position.

Ribbon representation of TMVI of CXCR4 structures superimposed for inactive (PDB code 3ODU), intermediate active (AGR1.135, AGR1.137) and active (PDB 8U4N) upon interaction with no, negative allosteric modulators and CXCL12-induced Gi-protein interaction, respectively. Arginine 235 is depicted in stick representation as a reference for the movement of the protein.

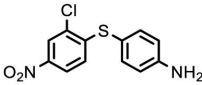
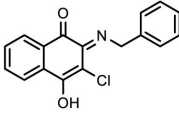
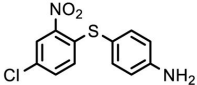
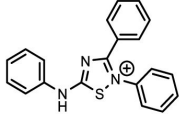
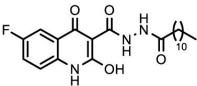
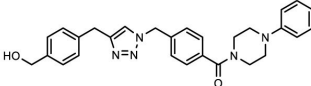
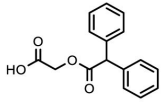
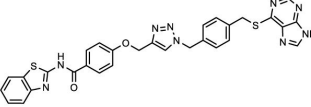
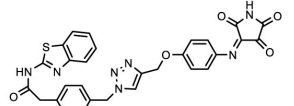
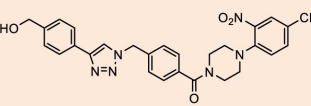
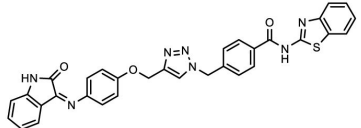
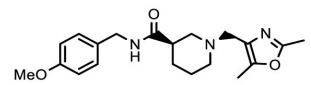
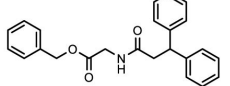
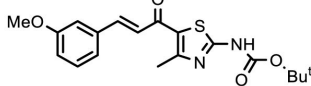
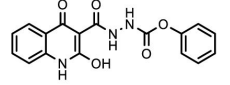
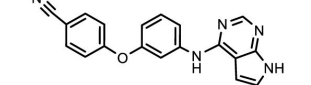
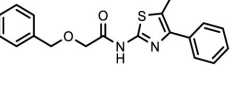
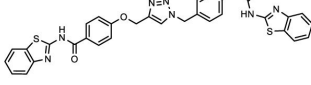
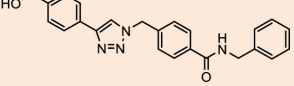
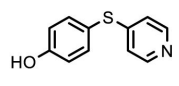
Ref.	Structure	Ref.	Structure
#1 AGF2.9C 1 % Inh: 38.01		#15 LNR1.17 % Inh: 48.71	
#2 AGF2.6 % Inh: 12.88		#16 VP1.23 % Inh: 30.69	
#3 VP4.29 % Inh: 0		#17 AGR1.13 8 % Inh: 45.68	
#4 MR4.17 % Inh: 3.43		#18 AGR1.10 3 % Inh: 53.55	
#5 AGR1.20 9 % Inh: 1.49		#19 AGR1.13 5 % Inh: 65.90	
#6 AGR1.10 1 % Inh: 36.19		#20 JHD1.47 % Inh: 11.79	
#7 MR3.24 % Inh: 2.14		#21 JAR3.8 % Inh: 21.03	
#8 VP4.27 % Inh: 0		#22 VNG1.61 % Inh: 0	
#9 MR2.57 % Inh: 0		#23 AGR1.21 0 % Inh: 40.15	
#10 AGR1.13 1 % Inh: 2.34		#24 AGF351 % Inh: 21.23	

Table S1.

Compounds with minimal interaction energy in the area of interest

Table S1. (continued)

#11 MR1.50 % Inh: 40.91		#25 IGS-2.73 % Inh: 27.36	
#12 AGR1.11 7 % Inh: 31.74		#26 VNG1.64 % Inh: 14.31	
#13 VSP3.1 % Inh: 36.43		#27 DM1.27 % Inh: 28.54	
#14 VP1.11 % Inh: 28.07		#28 ERP3.1 % Inh: 37.06	
Ref.	Structure	Ref.	Structure
#29 ERP2.51 % Inh: 52.50		#35 SC089 % Inh: 0.98	
#30 AGR1.26 7 % Inh: 30.06		#36 AGR1.13 7 % Inh: 89.82	
#31 VNG1.63 % Inh: 19.98		#37 IGS-4.32 % Inh: 0	
#32 AGR1.13 4 % Inh: 23.65		#38 IGS-2.3 % Inh: 0	
#33 SC513 % Inh: 12.94		#39 MR2.2 % Inh: 0	
#34 SC523 % Inh: 1.37		#40 JAR229 % Inh:	

Structure of the compounds employed in the study. The compounds studied in detail (AGR1.131, AGR1.135 and AGR1.137) are highlighted in green boxes. The percentage of inhibition of cell migration in response to CXCL12 is also shown.

Mutation	Forward	Reverse
CXCR4 G207I	CAC ATC ATG GTT ATT CTT ATC CTG CCT	AGG CAG GAT AAG AAT AAC CAT GAT GTG
CXCR4 R235L	GGC CAC CAG AAG TTA AAG GCC CTC AAG	CTT GAG GGC CTT TAA CTT CTG GTG GCC
CXCR4 Y256F	TGG CTG CCT TAC TTT ATT GGG ATC AGC	GCT GAT CCC AAT AAA GTA AGG CAG CCA
CXCR4 S260A	TACTACATTGGGATCGCCATCGACTCCTTCATC	GATGAAGGAGTCGATGGCGATCCCAATGTAGTA
CXCR4 L208K	<i>CACATCATGGTTGGCAAGATCCTGCCTGGTATT</i>	<i>AATACCAGGCAGGATCTTGCCAACCATGATGTG</i>
CXCR4 I204K	TTCCAGTTTCAGCACAAAGATGGTTGGCCTTATC	GATAAGGCCAACCATCTTGTGCTGAAACTGGAA
CXCR4 G212F	GCCCTTATCCTGCCTTTCATTGTCATCCTGTCC	GGACAGGATGACAATGAAAGGCAGGATAAGGCC
CXCR4 F249L	GTCATCCTCATCCTGGCTTTCTGCCTGTTGGCT GCCT	AGGCAGCCAACAGGCCAGGAAAGCCAGGATGAGGA TGAC

Supplementary Table S2.

Oligonucleotides used to generate the CXCR4 mutants

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Editors

Reviewing Editor

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Indian Institute of Science Education and Research (IISER), Pune, India

Reviewer #2 (Public Review):

Summary:

This work describes a new pharmacological targeting approach to inhibit selective functions of the ubiquitously expressed chemokine receptor CXCR4, a potential target of immunomodulatory or anti-cancer treatments. Overall, the results build a strong case for the potential of this new compound to target specific functions of CXCR4, particularly linked to tumorigenesis. However, a more thorough evaluation of the function of the compound as well as future studies in mammalian model systems are needed to better assess the promise of the compound.

Strengths:

The work elegantly utilizes in silico drug modelling to propose new small molecule compounds with specific features. This way, the authors designed compound AGR1.137, which abolishes ligand-induced CXCR4 receptor nanoclustering and the subsequent directed cell migration without affecting ligand-binding itself or some other ligand-induced signaling pathways. The authors have used a relatively broad set of experiments to validate and demonstrate the effects of the drug. Importantly, the authors also test AGR1.137 in vivo, using a zebra fish model of tumorigenesis and metastasis. A relatively strong inhibitory effect of the compound is reported.

Weaknesses:

The authors have been able to significantly strengthen their data from the first submission. The content of this manuscript is pretty solid, although studies in mammalian model systems

are naturally needed in the future to better assess the promise of the compound.

<https://doi.org/10.7554/eLife.93968.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1

(1) In the "Introduction" section, an important aspect that requires attention pertains to the discussion surrounding the heterodimerization of CXCR4 and CCR5. Notably, the manuscript overlooks a recent study (<https://doi.org/10.1038/s41467-023-42082-z>) elucidating the mechanism underlying the formation of functional dimers within these G protein-coupled receptors (GPCRs)...The inclusion of this study within the manuscript would significantly enrich the contextual framework of the work, offering readers a comprehensive understanding of the current knowledge surrounding the structural dynamics and functional implications of CXCR4 and CCR5 heterodimerization.

We thank the reviewer for his/her recommendation to enrich the contextual framework of our study. The Nature Communications paper by Di Marino et al. was published after we sent the first version of our manuscript to eLife, and therefore was not included in the discussion. As the reviewer rightly indicates, this paper elucidates the mechanism underlying the formation of functional dimers within CCR5 and CXCR4. Using metadynamics approaches, the authors emphasize the importance of distinct transmembrane regions for dimerization of the two receptors. In particular, CXCR4 shows two low energy dimer structures and the TMVI-TMVII helices are the preferred interfaces involved in the protomer interactions in both cases. Although the study uses in silico techniques, it also includes the molecular binding mechanism of CCR5 and CXCR4 in the membrane environment, as the authors generate a model in which the receptors are immersed in a 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC) phospholipid bilayer with 10% cholesterol. This is an important point in this study, as membrane lipids also interact with membrane proteins, and the lipid composition affects CXCR4 oligomerization (Gardeta S.R. et al. Front. Immunol. 2023). In particular, Di Marino et al. find a cholesterol molecule placed in-between the two CXCR4 protomers where it engages a series of hydrophobic interactions with residues including Leu132, Val214, Leu216 and Phe249. Then, the polar head of cholesterol forms an H-bond with Tyr135 that further stabilizes protomer binding. In our hands, the F249L mutation in CXCR4 reverted the antagonism of AGR1.137, suggesting that the compound binds, among others, this residue. We should, nonetheless, indicate that we analyzed receptor oligomerization and not CXCR4 dimerization, which was the main object of the Di Marino et al. study. It is therefore also plausible that other residues than those described as essential for CXCR4 dimerization might participate in receptor oligomerization. We can speculate that AGR1.137 might affect cholesterol binding to CXCR4 and, therefore, alter dimerization/oligomerization. Additionally, the CXCR4 x-ray structure with PDB code 3ODU (Wu B. et al. Science, 2010) experimentally shows the presence of two fatty acid molecules in contact with both TMV and TMVI. These molecules closely interact with hydrophobic residues in the protein, thereby stabilizing it in a hydrophobic environment. Although more experiments will be needed to clarify the mechanism involved, our results suggest that cholesterol and/or other lipids also play an important role in CXCR4 oligomerization and function, as seen for other GPCRs (Jakubik J. & ElFakahani E.E. Int J Mol Sci. 2021). However, we should also consider that other factors not included in the analysis by Di Marino et al. can also affect CXCR4 oligomerization; for instance, the co-expression of other chemokine receptors and/or other GPCRs that heterodimerize with CXCR4 might affect CXCR4 dynamics

at the cell membrane, similar to other membrane proteins such as CD4, which also forms complexes with CXCR4 (Martinez-Muñoz L. et al. Mol. Cell 2018).

The revised discussion contains references to the study by Di Marino et al. to enrich the contextual framework of our data.

(2) In "various sections" of the manuscript, there appears to be confusion surrounding the terminology used to refer to antagonists. It is recommended to provide a clearer distinction between allosteric and orthosteric antagonists to enhance reader comprehension. An orthosteric antagonist typically binds to the same site as the endogenous ligand, directly blocking its interaction with the receptor. On the other hand, an allosteric antagonist binds to a site distinct from the orthosteric site, inducing a conformational change in the receptor that inhibits the binding of the endogenous ligand. By explicitly defining the terms "allosteric antagonist" and "orthosteric antagonist" within the manuscript, readers will be better equipped to discern the specific mechanisms discussed in the context of the study.

The behavior of the compounds described in our manuscript (AGR1.35 and AGR1.137) fits with the definition of allosteric antagonists, as they bind on a site distinct from the orthosteric site, although they only block some ligand-mediated functions and not others. This would mean that they are not formally antagonists and should be not considered as allosteric compounds, as their binding on CXCR4 does not alter CXCL12 binding, although they might affect its affinity. In this sense, our compounds respond much better to the concept of negative allosteric modulators (Gao Z.-G. & Jacobson K.A. Drug Discov. Today Technol. 2013). They act by binding on a site distinct from the orthosteric site and selectively block some downstream signaling pathways but not others induced by the same endogenous agonist.

To avoid confusion and to clarify the role of the compounds described in this study, we now refer to them as negative allosteric modulators along the manuscript.

(3) In the Results section, the computational approach employed for "screening small compounds targeting CXCR4, particularly focusing on the inhibition of CXCL12-induced CXCR4 nanoclustering", requires clarification due to several points of incomprehension. The following recommendations aim to address these concerns and enhance the overall clarity of the section:

(1) Computational Approach and Binding Mode Description:

-Explicitly describe the methodology for identifying the pocket/cleft area in angstroms (Å) on the CXCR4 protein structure. Include details on how the volume of the cleft enclosed by TMV and TMVI was determined, as this information is not readily apparent in the provided reference (<https://doi.org/10.1073/pnas.1601278113>).

The identification of the cleft was based on the observations by Wu et al. (Wu B. et al. Science 2010) who described the presence of bound lipids in the area formed by TMV and VI, and those of Wescott et al. (Wescott M.P. et al. Proc. Natl. Acad. Sci. 2016) on the importance of TMVI in the transmission of conformational changes promoted by CXCL12 on CXCR4 towards the cytoplasmic surface of the receptor to link the binding site with signaling activation. Collectively, these results, and our previous data on the critical role of the N-terminus region of TMVI for CXCR4 oligomerization (Martinez-Muñoz L. et al. Mol. Cell 2018), focused our in silico screening to this region. Once we detected that several compounds bound CXCR4 in this region, the cleavage properties were calculated by subtracting the compound structure. The resulting PDB was analyzed using the PDBsum server (Laskowski R.A. et. al. Protein Sci. 2018). Volume calculations were obtained using the server analyzing surface clefts by SURFNET

(Laskowski R. A. J. Mol. Graph. 1995). The theoretical interaction surface between the selected compounds and CXCR4 and the atomic distances between the protein residues and the compounds was calculated using the PISA server (Krissinel E. & Henrick K. J. Mol. Biol. 2007) (Fig. I, only for review purposes). The analysis of the cleft occupied by AGR1.135 showed two independent cavities of 434 Å³ and 1,381 Å³ that were not connected to the orthosteric site. In the case of AGR1.137, the data revealed two distinct clefts of 790 Å³ and 580 Å³ (Fig. I, only for review purposes). These details have been included in the revised manuscript (New Fig. 1A, Supplementary Fig 8A, B).

(4) Clarify the statement regarding the cleft being "surface exposed for interactions with the plasma membrane," particularly in the context of its embedding within the membrane.

For GPCRs, transmembrane domains represent binding sites for bioactive lipids that play important functional and physiological roles (Huwiler A. & Zangemeister-Wittke U. Pharmacol. Ther. 2018). The channel between TMV and TMVI connects the orthosteric chemokine binding pocket to the lipid bilayer and is occupied by an oleic acid molecule, according to the CXCR4 structure published in 2010 (Wu B. et al. Science 2010). In addition, the target region contains residues involved in cholesterol (and perhaps other lipids) engagement (Di Marino et al. Nat. Commun. 2023). Taken together, these data support our statement that the cleft supports interactions between CXCR4 molecules and the plasma membrane.

Moreover, the data of Di Marino et al. also support that CCR5 and CXCR4 have a symmetric and an asymmetric binding mode. Therefore, either dimeric structure has the possibility to form trimers, tetramers, and even oligomers by using the free binding interface to complex with another protomer. This hypothesis suggests that the interaction of dimers to form oligomers should involve residues distinct from those included in the dimeric conformation.

The sentence has been modified in the revised manuscript to clarify comprehension.

(5) Discuss the rationale behind targeting the allosteric binding pocket instead of the orthosteric pocket, outlining potential advantages and disadvantages.

The advantages and disadvantages of using negative allosteric modulators vs orthosteric antagonists have been now included in the revised discussion.

The majority of GPCR-targeted drugs function by binding to the orthosteric site of the receptor, and are agonists, partial agonists, antagonists or inverse agonists. These orthosteric compounds can have off-target effects and poor selectivity due to highly homologous receptor orthosteric sites and to abrogation of spatial and/or temporal endogenous signaling patterns.

The alternative is to use allosteric modulators, which can tune the functions associated with the receptors without affecting the orthosteric site. They can be positive, negative or neutral modulators, depending on their effect on the functionality of the receptor (Foster D.J. & Conn P.J. Neuron 2017). For example, the use of a negative allosteric modulator of a chemokine receptor to dampen pathological signaling events, while retaining full signaling for non-pathological activities might limit adverse effects (Kohout T.A. et al. J. Biol. Chem. 2004). In this case, the negative allosteric modulator 873140 blocks CCL3 binding on CCR5 but does not alter CCL5 binding (Watson C. et al. Mol. Pharmacol. 2005). In other cases, allosteric modulators can stabilize a particular receptor conformation and block others. The mechanism of action of the anti-HIV-1, FDA-approved, CCR5 allosteric modulator, maraviroc (Jin J. et al. Sci. Signal. 2018) is attributed to its ability to modulate CCR5 dimer populations and their subsequent subcellular trafficking and localization to the cell membrane (Jin J. et al.

Sci. Signal. 2018). Two CCR5 dimeric conformations that are imperative for membrane localization were present in the absence of maraviroc; however, an additional CCR5 dimer conformation was discovered after the addition of maraviroc, and all homodimeric conformations were further stabilized. This finding is consistent with the observation that CCR5 dimers and oligomers inhibit HIV host-cell entry, likely by preventing the HIV-1 co-receptor formation.

It is well known that GPCRs activate G proteins, but they also recruit additional proteins (e.g., β -arrestins) that induce signaling cascades which, in turn, can direct specific subsets of cellular responses independent of G protein activation (Eichel K. et al. Nature 2018) and are responsible for either therapeutic or adverse effects. Allosteric modulators can thus be used to block these adverse effects without influencing the therapeutic benefits. This was the case in the design of G protein-biased agonists for the kappa opioid receptor, which maintain the desirable antinociceptive and antipruritic effects and eliminate the sedative and dissociative effects in rodent models (Brust T.F. et al. Sci. Signal 2016).

(6) Provide the PDB ID of the CXCR4 structure used as a template for modeling with SwissModel. Explain the decision to model the structure from the amino acid sequence and suggest an alternative approach, such as utilizing AlphaFold structures and performing classical molecular dynamics with subsequent clustering for the best representative structure.

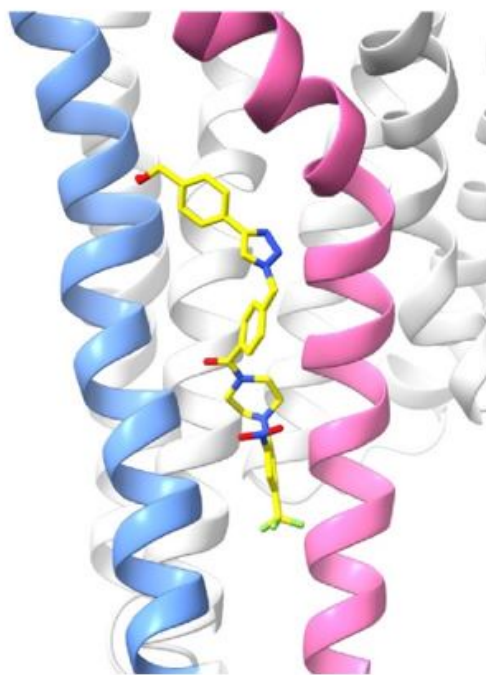
The PDB used as a template for modeling CXCR4 was 3ODU. This information was already included in the material and methods section. At the time we performed these analyses, there were several crystallographic structures of CXCR4 in complex with different molecules and peptides deposited at the PDB. None of them included a full construct containing the complete receptor sequence to provide a suitable sample for X-ray structure resolution, as the N- and C-terminal ends of CXCR4 are very flexible loops. In addition, the CXCR4 constructs contained T4 lysozyme inserted between helices TMV and TMVI to increase the stability of the protein—a common strategy used to facilitate crystallogenesis of GPCRs (Zou Y. et al. PLoS One 2012). Therefore, we generated a CXCR4 homology model using the SWISS-MODEL server (Waterhouse A. et al. Nucleic Acids Res. 2018). This program reconstructed the loop between TMV and TMVI, a domain particularly important in this study that was not present in any of the crystal structure available in PDB. The model structure was, nonetheless, still incomplete, as it began at P27 and ended at S319 because the terminal ends were not resolved in the crystal structure used as a template. Nevertheless, we considered that these terminal ends were not involved in CXCR4 oligomerization.

As AlphaFold was not available at the time we initiated this project, we didn't use it. However, we have now updated our workflow to current methods and predicted the structure of the target using AlphaFold (Jumper J. et al. Nature 2021) and the sequence available under UniProt entry P61073. We prepared the ligands using OpenBabel (O'Boyle N.M. et al., J. Cheminformatics 2011), with a gasteiger charge assignment, and generated 10 conformers for each input ligand using the OpenBabel genetic algorithm. We then prepared the target structure with Openmm, removing all waters and possible heteroatoms, and adding all missing atoms. We next predicted the target binding pockets with fPocket (Le Guilloux V. et al. BMC Bioinformatics 2009), p2rank (Krivak R. & Hoksza, J. Cheminformatics 2018), and AutoDock autosite (Ravindranath P.A. & Sanner M.F. Bioinformatics 2016). We chose only those pockets between TMV and TMVI (see answer to point 3). We merged the results of the three programs into so-called consensus pockets, as two pockets are said to be sufficiently similar if at least 75% of their surfaces are shared (del Hoyo D. et al. J. Chem. Inform. Model. 2023). From the consensus pockets, there was one pocket that was significantly larger than the others and was therefore selected. We then docked the ligand conformers in this pocket using AutoDock GPU (Santos-Martins D. et al. J. Chem. Theory Comput. 2021), LeDock (Liu N & Xu Z., IOP Conf. Ser. Earth Environ. Sci. 2019), and Vina (Eberhardt J. et al. J. Chem. Inf. Model.

2021). The number of dockings varied from 210 to 287 poses. We scored each pose with the Vina score using ODDT (Wójcikowski M. et al. J. Cheminform. 2015). Then, we clustered the different solutions into groups whose maximum RMSD was 1Å. This resulted in 40 clusters, the representative of each cluster was the one with maximum Vina score and confirmed that the selected compounds bound this pocket (Author response image 1). When required, we calculated the binding affinity using Schrodinger's MM-GBSA procedure (Greenidge P.A. et al. J. Chem. Inf. Model. 2013), in two ways: first, assuming that the ligand and target are fixed; second, with an energy minimization of all the atoms within a distance of 3Å from the ligand. This information has now been included in the revised version of the manuscript.

Author response image 1.

AGR1.135 docking in CXCR4 using the updated protocol for ligand docking. Cartoon representation colored in gray with TMV and TMVI shown in blue and pink, respectively. AGR1.135 is shown in stick representation with carbons in yellow, oxygens in red and nitrogens in blue.



(7) Specify the meaning of "minimal interaction energy" and where (if present) the interaction scores are reported in the text.

We refer to minimal interaction energy, the best docking score, that is, the best score obtained in our docking studies. These data were not included in the previous manuscript due to space restrictions but are now included in the reviewed manuscript.

(8) You performed docking studies using GLIDE to identify potential binding sites for the small compounds on the CXCR4 protein. The top-scoring binders were then subjected to further refinement using PELE simulations. However, I realize that a detailed description of the specific binding modes of these compounds was not provided in the text. Please make the description of binding poses more detailed

Firstly, to assess the reliability of this method, a PELE study was carried out for the control molecule IT1t, which is a small drug-like isothiourea derivative that has been crystallized in complex with CXCR4 (PDB code: 3ODU). IT1t is a CXCR4 antagonist that binds to the CXCL12 binding cavity and inhibits HIV-1 infection (Das D. *Antimicrob. Agents Chemother.* 2015; Dekkers S. et al. *J. Med. Chem.* 2023). From the best five trajectories, two of them had clearly better binding energies, and corresponded to almost the same predicted pose of the molecule. Although the predicted binding mode was not exactly the same as the one in the crystal structure, the approximation was very good, giving validation to the approach. Although PELE is a suitable technique to find potential binding sites, the predicted poses must be subsequently refined using docking programs.

Analyzing the best trajectories for the remaining ligands, at least one of the best-scored poses was always located at the orthosteric binding site of CXCR4. Even though these poses showed good binding energies, they were discarded as the *in vitro* biological experiments indicated that the compounds were unable to block CXCL12 binding or CXCL12-mediated inhibition of cAMP release or CXCR4 internalization. Collectively, these data indicated that the selected compounds did not behave as orthosteric inhibitors of CXCR4. The CXCL12 binding pocket is the biggest cavity in CXCR4, and so PELE may tend to place the molecules near it. However, all the compounds presented other feasible binding sites with a comparable binding energy.

AGR1.135 and AGR1.137 showed interesting poses between TMV and TMVI with very good binding energy (-51.4 and -37.2 kcal/mol, respectively). This was precisely the region we had previously selected for the *in silico* screening, as previously described (see response to point 3).

AGR1.131 showed two poses with low binding energy that were placed between helices TMI and TMVII (-43.6 kcal/mol) and between helices TMV and TMVI (-39.8 kcal/mol). This compound was unable to affect CXCL12-mediated chemotaxis and was therefore used as an internal negative control as it was selected in the *in silico* screening with the same criteria as the other compounds but failed to alter any CXCL12-mediated functions. PELE studies nonetheless provided different binding sites for each molecule, which had to be further studied using docking to obtain a more accurate binding mode. In agreement with the previous commentary, we repeated the analysis using AlphaFold and the rest of the procedure described (see our response to point 6) and calculated the binding energies for all the compounds using Schrodinger's MM-GBSA procedure (Greenidge P.A. et al. *J. Chem. Inf. Model.* 2013). Calculations were performed in two ways: first, assuming that the ligand and target are fixed; second, with an energy minimization of all the atoms within a distance of 3 Å from the ligand. The results using the first method indicated that AGR1.135 and AGR1.137 showed poses between TMV and TMVI with -56.4 and -62.4 kcal/mol, respectively and AGR1.131 had a pose between TMI and TMVII with -61.6 kcal/mol. In the second method AGR1.135 and AGR1.137 showed poses between TMV and TMVI with -57.9, and -67.6 kcal/mol, respectively, and AGR1.131 of -62.2 kcal/mol between TMI and TMVII.

This information is now included in the text.

(9) (2) Experimental Design:-Justify the choice of treating Jurkat cells with a concentration of 50 μ M of the selected compound. Consider exploring different concentrations and provide a rationale for the selected dosage. Additionally, clearly identify the type of small compound used in the initial experiment.

The revised version contains a new panel in Fig. 1B to show a more detailed kinetic analysis with different concentrations (1-100 μ M) of the compounds in the Jurkat migration experiments. In all cases, 100 μ M nearly completely abrogated cell migration, but in order to reduce the amount of DMSO added to the cells we selected 50 μ M for further experiments, as it was the concentration that inhibits 50-75% of ligand-induced cell migration. Regarding the

type of small compounds used in the initial experiments, they were compounds included in the library described in reference #24 (Sebastián-Pérez V. et al. *Med. Biol. Chem.* 2017), which contains heterocyclic compounds. We would note that we do not consider AGR1.137 a final compound. We think that there is scope to develop AGR1.137-based second-generation compounds with greater solubility in water, greater specificity or affinity for CXCR4, and to evaluate delivery methods to hopefully increase activity.

(10) Avoid reporting details in rounded parentheses within the text; consider relocating such information to the Materials and Methods section or figure captions for improved readability.

Most of the rounded parentheses within the text have been eliminated in the revised version of the manuscript to improve readability.

(11) Elaborate on the virtual screening approach using GLIDE software, specifying the targeted site and methodology employed.

For the virtual screening, we used the Glide module (SP and XP function scoring) included in the Schrödinger software package, utilizing the corresponding 3D target structure and our MBC library (Sebastián-Pérez V et al. *J. Chem. Inf. Model.* 2017). The center of the catalytic pocket was selected as the centroid of the grid. In the grid generation, a scaling factor of 1.0 in van der Waals radius scaling and a partial charge cutoff of 0.25 were used. A rescoring of the SP poses of each compound was then performed with the XP scoring function of the Glide. The XP mode in Glide was used in the virtual screening, the ligand sampling was flexible, epik state penalties were added and an energy window of 2.5 kcal/mol was used for ring sampling. In the energy minimization step, the distance-dependent dielectric constant was 4.0 with a maximum number of minimization steps of 100,000. In the clustering, poses were considered as duplicates and discarded if both RMS deviation is less than 0.5 Å and maximum atomic displacement is less than 1.3 Å.

(12) Provide clarity on the statement that AGR1.131 "theoretically" binds the same motif, explaining the docking procedure used for this determination.

In the in silico screening, AGR1.131 was one of the 40 selected compounds that showed, according to the PELE analysis (see answer to point 8), a pose with low binding energy (-39.8 kcal/mol) between TMV and TMVI helices, which is the selected area for the screening. It, nonetheless, also showed a best pose placed between helices TM1 and TM7 (-43.7 kcal/mol) using the initial workflow. In conclusion, although AGR1.131 also faced to the TMV-TMVI, the most favorable pose was in the area between TMI and TMVII. In addition, the compound was included in the biological screening, where it did not affect CXCL12-mediated chemotaxis. We thus decided to use it as an internal negative control, as it has a skeleton very similar to AGR1.135 and AGR1.137 and can interact with the TM domains of CXCR4 without promoting biological effects. This statement has been clarified in the revised text.

(13) Toxicity Testing:

-Enhance the explanation of the approach to testing the toxicity of the compound in Jurkat cells. Consider incorporating positive controls to strengthen the assessment and clarify the experimental design.

All the selected compounds in the in silico screening were initially tested for propidium iodide incorporation in treated cells in a toxicity assay, and some of them were discarded for further experiments (e.g., AGR1.103 and VSP3.1).

Further evaluation of Jurkat cell viability was determined by cell cycle analysis using propidium iodide. Supplementary Fig. 1B included the percentage of each cell cycle phase, and data indicated no significant differences between the treatments tested. Nevertheless, at the suggestion of the reviewer, and to clarify this issue, positive controls inducing Jurkat cell death (staurosporine and hydrogen peroxide) have also been included in the new Supplementary Fig. 2. The new figure also includes a table showing the percentage of cells in each cell-cycle phase.

(14) In the Results section concerning "AGR1.135 and AGR1.137 blocking CXCL12-mediated CXCR4 nanoclustering and dynamics", several points can be improved to enhance clarity and coherence: 1. Specificity of Low Molecular Weight Compounds:

-Clearly articulate how AGR1.135 and AGR1.137 specifically target homodimeric CXCR4 and provide an explanation for their lack of impact on heterodimeric CXCR4-CCR5 in that region.

First of all, we should clarify that when we talk about receptor nanoclustering, oligomers refer to complexes including 3 or more receptors and, therefore, the residues involved in these interactions can differ from those involved in receptor dimerization. Moreover, our FRET experiments did not indicate that the compounds alter receptor dimerization (see new Supplementary Fig. 7). Of note, mutant receptors unable to oligomerize can still form dimers (Martínez-Muñoz L. et al. Mol. Cell 2018; García-Cuesta E.M. et al. Proc. Natl. Acad. Sci. USA 2022). Additionally, we believe that these oligomers can also include other chemokine receptors/proteins expressed at the cell membrane, which we are currently studying using different models and techniques.

We have results supporting the existence of CCR5/CXCR4 heterodimers (Martínez-Muñoz L et al. Proc. Natl. Acad. Sci. USA 2014), in line with the data published by Di Marino et al. However, in the current study we have not evaluated the impact of the selected compounds on other CXCR4 complexes distinct from CXCR4 oligomers. Our Jurkat cells do not express CCR5 and, therefore, we cannot discuss whether AGR1.137 affects CCR5/CXCR4 heterodimers. The chemokine field is very complex and most receptors can form dimers (homo- and heterodimers) as well as oligomers (Martínez-Muñoz L., et al Pharmacol & Therap. 2011) when co-expressed. To evaluate different receptor combinations in the same experiment is a complex task, as the number of potential combinations between distinct expressed receptors makes the analysis very difficult. We started with CXCR4 as a model, to continue later with other possible CXCR4 complexes. In addition, for the analysis of CCR5/CXCR4 dynamics, it is much better to use dual-TIRF techniques, which allow the simultaneous detection of two distinct molecules coupled to different fluorochromes.

Regarding the data of Di Marino et al., it is possible that the compounds might also affect heterodimeric conformations of CXCR4. This aspect has also been broached in the revised discussion. We would again note that we evaluated CXCR4 oligomers and not monomers or dimers; this is especially relevant when we compare the residues involved in these processes as they might differ depending on the receptor conformation considered. This issue was also hypothesized by Di Marino et al. (see our response to point 4).

(15) When referring to "unstimulated" cells, provide a more detailed explanation to elucidate the experimental conditions and cellular state under consideration.

Unstimulated cells refer to the cells in basal conditions, that is, cells in the absence of CXCL12. For TIRF-M experiments, transiently-transfected Jurkat cells were plated on glass-bottomed microwell dishes coated with fibronectin; these are the unstimulated cells. To observe the

effect of the ligand, dishes were coated as above plus CXCL12 (stimulated cells). We have clarified this point in the material and methods section of the revised version.

(16) 2. Paragraph Organization

-Reorganize the second paragraph to eliminate redundancy and improve overall flow. A more concise and fluid presentation will facilitate reader comprehension and engagement.

The second paragraph has been reorganized to improve overall flow.

(17) Ensure that each paragraph contributes distinct information, avoiding repetition and redundancy.

We have carefully revised each paragraph of the manuscript to avoid redundancy.

(18) 3. Claim of Allosteric Antagonism:

-Exercise caution when asserting that "AGR1.135 and AGR1.137 behave as allosteric antagonists of CXCR4" based on the presented results. Consider rephrasing to reflect that the observed effects suggest the potential allosteric nature of these compounds, acknowledging the need for further investigations and evidence.

To avoid misinterpretations on the effect of the compounds on CXCR4, as we have commented in our response to point 2, we have substituted the term allosteric inhibitors with negative allosteric modulators, which refer to molecules that act by binding a site distinct from the orthosteric site, and selectively block some downstream signaling pathways, whereas others induced by the same endogenous or orthosteric agonist are unaffected (Gao Z.-G. & Jacobson K.A. Drug Discov. Today Technol. 2013). Our data indicate that the selected small compounds do not block ligand binding or G protein activation or receptor internalization, but inhibit receptor oligomerization and ligand-mediated directed cell migration.

(19) In the Results section discussing the "incomplete abolition of CXCR4-mediated responses in Jurkat cells by AGR1.135 and AGR1.137", several points can be refined for better clarity and completeness: 1. Inclusion of Positive Controls:

-Consider incorporating positive controls in relevant experiments to provide a comparative benchmark for assessing the impact of AGR1.135 and AGR1.137. This addition will strengthen the interpretation of results and enhance the experimental rigor.

The in vivo experiments (Fig. 7E,F) used AMD3100, an orthosteric antagonist of CXCR4, as a positive control. We also included AMD3100, as a positive control of inhibition when evaluating the effect of the compounds on CXCL12 binding (Fig. 3, new Supplementary Fig. 3). The revised version of the manuscript also includes the effect of this inhibitor on other relevant CXCL12-mediated responses such as cell migration (Fig. 1B), receptor internalization (Fig. 3A), cAMP production (Fig. 3C), ERK1/2 and AKT phosphorylation (Supplementary Fig. 4), actin polymerization (Fig. 4A), cell polarization (Fig. 4B, C) and cell adhesion (Fig. 4D), to facilitate the interpretation of the results and improve the experimental rigor.

(20) 2. Clarification of Terminology:

-Clarify the term "CXCR4 internalizes" by providing context, perhaps explaining the process of receptor internalization and its relevance to the study.

We refer to CXCR4 internalization as a CXCL12-mediated endocytosis process that results in reduction of CXCR4 levels on the cell surface. We use CXCR4 internalization in this study with two purposes: First, for CXCR4 and other chemokine receptors, internalization processes are mediated by ligand-induced clathrin vesicles (Venkatesan et al 2003) a process that triggers CXCR4 aggregation in these vesicles. We have previously determined that the oligomers of receptors detected by TIRF-M remain unaltered in cells treated with inhibitors of clathrin vesicle formation and of internalization processes (Martinez-Muñoz L. et al. Mol. Cell 2018). Moreover, we have described a mutant CXCR4 that cannot form oligomers but internalizes normally in response to CXCL12 (Martinez-Muñoz L. et al. Mol. Cell 2018). The observation in this manuscript of normal CXCL12-mediated endocytosis in the presence of the negative allosteric inhibitors of CXCR4 that abrogate receptor oligomerization reinforces the idea that the oligomers detected by TIRF are not related to receptor aggregates involved in endocytosis; Second, receptor internalization is not affected by the allosteric compounds, indicating that they downregulate some CXCL12-mediated signaling events but not others (new Fig. 3).

All these data have been included in the revised discussion of the manuscript.

(21) Elaborate on the meaning of "CXCL12 triggers normal CXCR4mut internalization" to enhance reader understanding.

We have previously described a triple-mutant CXCR4 (K239L/V242A/L246A; CXCR4mut). The mutant residues are located in the N-terminal region of TMVI, close to the cytoplasmic region, thus limiting the CXCR4 pocket described in this study (see our response to point 3). This mutant receptor dimerizes but neither oligomerizes in response to CXCL12 nor supports CXCL12-induced directed cell migration, although it can still trigger some Ca²⁺ flux and is internalized after ligand activation (Martinez-Muñoz L. et al. Mol. Cell 2018). We use the behavior of this mutant (CXCR4mut) to show that the CXCR4 oligomers and the complexes involved in internalization processes are not the same and to explain why we evaluated CXCR4 endocytosis in the presence of the negative allosteric modulators.

As we indicated in a previous answer to the reviewer, these issues have been re-elaborated in the revised version.

(22) 3. Discrepancy in CXCL12 Concentration:

-Address the apparent discrepancy between the text stating, "...were stimulated with CXCL12 (50 nM, 37{degree sign}C)," and the figure caption (Fig. 3A) reporting a concentration of 12.5 nM. Rectify this inconsistency and provide an accurate and clear explanation.

We apologize for this error, which is now corrected in the revised manuscript. With the exception of the cell migration assays in Transwells, where the optimal concentration was established at 12.5 nM, in the remaining experiments the optimal concentration of CXCL12 employed was 50 nM. These concentrations were optimized in previous works of our laboratory using the same type of experiment. We should also remark that in the experiments using lipid bilayers or TIRF-M experiments, CXCL12 is used to coat the plates and therefore it is difficult to determine the real concentration of the ligand that is retained in the surface of the plates after the washing steps performed prior to adding the cells. In addition, we use 100 nM CXCL12 to create the gradient in the chambers used to perform the directed-cell migration experiments.

(23) 4. Speculation on CXCL12 Binding:

-Refrain from making speculative statements, such as "These data suggest that none of the antagonists alters CXCL12 binding to CXCR4," unless there is concrete evidence

presented up to that point. Clearly outline the results that support this conclusion.

Figure 3B and Supplementary Figure 3 show CXCL12-ATTO700 binding by flow cytometry in cells pretreated with the negative allosteric modulators. We have also included AMD3100, the orthosteric antagonist, as a control for inhibition. While these experiments showed no major effect of the compounds on CXCL12 binding, we cannot discard small changes in the affinity of the interaction between CXCL12 and CXCR4. In consequence we have re-written these statements.

(24) 5. Corroboration of Data:

-Specify where the corroborating data from immunostaining and confocal analysis are reported, ensuring readers can access the relevant information to support the conclusions drawn in this section.

In agreement with the suggestion of the reviewer, the revised manuscript includes data from immunostaining and confocal analysis to complement Fig. 4B (new Fig. 4C). The revised version also includes some representative videos for the TIRF experiments showed in Figure 2 to clarify readability.

(25) In the Results section concerning "AGR1.135 and AGR1.137 antagonists and their direct binding to CXCR4", several aspects need clarification and refinement for a more comprehensive and understandable presentation: 1. Workflow Clarification:

-Clearly articulate the workflow used for assessing the binding of AGR1.135 and AGR1.137 to CXCR4. Address the apparent contradiction between the inability to detect a direct interaction and the utilization of Glide for docking in the TMV-TMVI cleft.

To address the direct interaction of the compounds with CXCR4, we intentionally avoided the modification of the small compounds with different labels, which could affect their properties. We therefore attempted a fluorescence spectroscopy strategy to formally prove the ability of the small compounds to bind CXCR4, but this failed because the AGR1.135 is yellow in color, which interfered with the determinations. We also tried a FRET strategy (see new Supplementary Fig. 7) and detected a significant increase in FRET efficiency of CXCR4 homodimers when AGR1.135 was evaluated, but again the yellow color interfered with FRET determinations. Moreover, AGR1.137 did not modify FRET efficiency of CXCR4 dimers. Therefore, we were unable to detect the interaction of the compounds with CXCR4.

We elected to develop an indirect strategy; in silico, we evaluated the binding-site using docking and molecular dynamics to predict the most promising CXCR4 binding residues involved in the interaction with the selected compounds. Next, we generated point mutant receptors of the predicted residues and re-evaluated the behavior of the allosteric antagonists in a CXCL12-induced cell migration experiment. Obviously, we first discarded those CXCR4 mutants that were not expressed on the cell membrane as well as those that were not functional when activated with CXCL12. Using this strategy, we eliminated the interference due to the physical properties of the compounds and demonstrated that if the antagonism of a compound is reversed in a particular CXCR4 mutant it is because the mutated residue participates or interferes with the interaction between CXCR4 and the compound, thus assuming (albeit indirectly) that the compound binds CXCR4.

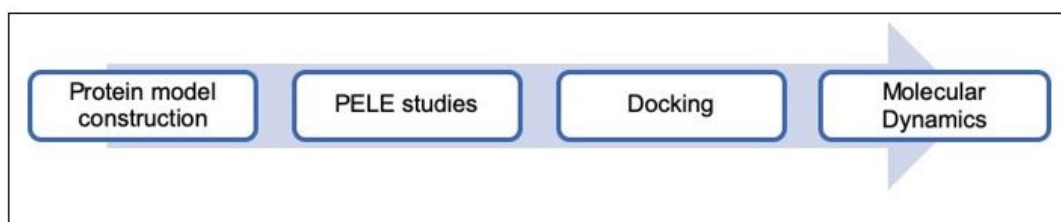
To select the specific mutations included in the analysis, our strategy was to generate point mutations in residues present in the TMV-TMVI pocket of CXCR4 that were not directly proposed as critical residues involved in chemokine engagement, signal initiation, signal propagation, or G protein-binding, based on the extensive mutational study published by Wescott MP et. al. (Wescott M.P. et. al. Proc. Natl. Acad. Sci. U S A. 2016).

(26) Provide a cohesive explanation of the transition from docking evaluation to MD analysis, ensuring a transparent representation of the methodology.

Based on the aim of this work, the workflow shown in Author response image 2, was proposed to predict the binding mode of the selected molecules. Firstly, a CXCR4 model was generated to reconstruct some unresolved parts of the protein structure; then a binding site search using PELE software was performed to identify the most promising binding sites; subsequently, docking studies were performed to refine the binding mode of the molecules; and finally, molecular dynamics simulations were run to determine the most stable poses and predict the residues that we should mutate to test that the compounds interact with CXCR4.

Author response image 2.

Workflow followed to determine the binding mode of the studied compounds.



(27) 2. Choice of Software and Techniques:

-Justify the use of "AMBER14" and the PELE approach, considering their potential obsolescence.

These experiments were performed five years ago when the project was initiated. As the reviewer indicates, AMBER14 and PELE approaches might perhaps be considered obsolescent. Thus, we have predicted the structure of the target using AlphaFold (Jumper J. et al, Nature 2021) and the sequence available under UniProt entry P61073. The complete analysis performed (see our response to point 4) confirmed that the compounds bound the selected pocket, as we had originally determined using PELE. These new analyses have been incorporated into the revised manuscript.

(28)-Discuss the role of the membrane in the receptor-ligand interaction. Elaborate on how the lipidic double layer may influence the binding of small compounds to GPCRs embedded in the membrane.

Biological membranes are vital components of living organisms, providing a diffusion barrier that separates cells from the extracellular environment, and compartmentalizing specialized organelles within the cell. In order to maintain the diffusion barrier and to keep it electrochemically sealed, a close interaction of membrane proteins with the lipid bilayer is necessary. It is well known that this is important, as many membrane proteins undergo conformational changes that affect their transmembrane regions and that may regulate their activity, as seen with GPCRs (Daemen F.J. & Bonting S.L., Biophys. Struct. Mech. 1977; Gether U. et al. EMBO J. 1997). The lateral and rotational mobility of membrane lipids supports the sealing function while allowing for the structural rearrangement of membrane proteins, as they can adhere to the surface of integral membrane proteins and flexibly adjust to a changing microenvironment. In the case of the first atomistic structure of CXCR4 (Wu B. et al. Science 2010), it was indicated that for dimers, monomers interact only at the extracellular side of helices V and VI, leaving at least a 4-Å gap between the intracellular regions, which is

presumably filled by lipids. In particular, they indicated that the channel between TMV and TMVI that connects the orthosteric chemokine binding pocket to the lipid bilayer is occupied by an oleic acid molecule. Recently, Di Marino et al., analyzing the dimeric structure of CXCR4, found a cholesterol molecule placed in between the two protomers, where it engages a series of hydrophobic interactions with residues located in the area between TMI and TMVI (Leu132, Val214, Leu216, Leu246, and Phe249). The polar head of cholesterol forms an H-bond with Tyr135 that further stabilizes its binding mode. This finding confirms that cholesterol might play an important role in mediating and stabilizing receptor dimerization, as seen in other GPCRs (Pluhackova, K., et al. PLoS Comput. Biol. 2016). In addition, we have previously observed that, independently of the structural changes on CXCR4 triggered by lipids, the local lipid environment also regulates CXCR4 organization, dynamics and function at the cell membrane and modulates chemokine-triggered directed cell migration. Prolonged treatment of T cells with bacterial sphingomyelinase promoted the complete and sustained breakdown of sphingomyelins and the accumulation of the corresponding ceramides, which altered both membrane fluidity and CXCR4 nanoclustering and dynamics. Under these conditions, CXCR4 retained some CXCL12-mediated signaling activity but failed to promote efficient directed cell migration (Gardeta S.R. et al. Front. Immunol. 2022). Collectively, these data demonstrate the key role that lipids play in the stabilization of CXCR4 conformations and in regulating its lateral mobility, influencing their associated functions. These considerations have been included in the revised version of the manuscript.

(29) 3. Stable Trajectories and Binding Mode Superimposition -Specify the criteria for defining "stable trajectories" to enhance reader understanding

There could be several ways to describe the stability of a MD simulation, based on the convergence of energies, distances or ligand-target interactions, among others. In this work, we use the expression "stable trajectories" to refer to simulations in which the ligand trajectory converges and the ligand RMSD does not fluctuate more than 0.25Å. This definition is now included in the revised text.

(30) Clarify the meaning behind superimposing the two small compounds and ensure that the statement in the figure caption aligns with the information presented in the main text.

We apologize for the error in the previous Fig. 5A and in its legend. The figure was created by superimposing the protein component of the poses for the two compounds, AGR1.135 and AGR1.137, rather than the compounds themselves. As panel 5A was confusing, we have modified all Fig. 5 in the revised manuscript to improve clarity.

(31) 4. Volume Analysis and Distances:

-Provide details on how the volume analysis was computed and how distances were accounted for. Consider adding a figure to illustrate these analyses, aiding reader comprehension.

The cleft search and analysis were performed using the default settings of SURFNET (Laskowski R.A. J. Mol. Graph. 1995) included in the PDBsum server (Laskowski R.A. et. al. Trends Biochem. Sci. 1997). The first run of the input model for CXCR4 3ODU identified a promising cleft of 870 Å³ in the lower half of the region flanked by TMV and TMVI, highlighting this area as a possible small molecule binding site (Fig. I, only for review purposes). Analysis of the cleft occupied by AGR1.135 showed two independent cavities of 434 Å³ and 1381 Å³ that were not connected to the orthosteric site. The same procedure for AGR1.137 revealed two distinct clefts of 790 Å³ and 580 Å³, respectively (Fig. I, only for review purposes). Analysis of the atomic distances between the protein residues and the

compounds was performed using the PISA server. Krissinel E. & Henrick K. J. Mol. Biol. 2007). (Please see our response to point 3 and the corresponding figure).

(32) 5. Mutant Selection and Relevance:

-Clarify the rationale behind selecting the CXCR4 mutants used in the study. Consider justifying the choice and exploring the possibility of performing an alanine (ALA) scan for a more comprehensive mutational analysis.

The selection of the residues to be mutated along the cleft was first based on their presence in the proposed cleft and the direct interaction of the compounds with them, either by hydrogen bonding or by hydrophobic interactions. Secondly, all mutated residues did not belong to any of the critical residues involved in transmitting the signal generated by the interaction of CXCL12 with the receptor. In any case, mutants producing a non-functional CXCR4 at the cell membrane were discarded after FACS analysis and chemotaxis experiments. Finally, the length and nature of the resulting mutations were designed mainly to occlude the cleft in case of the introduction of long residues such as lysines (I204K, L208K) or to alter hydrophobic interactions by changing the carbon side chain composition of the residues in the cleft. Indeed, we agree that the alanine scan mutation analysis would have been an alternative strategy to evaluate the residues involved in the interactions of the compounds.

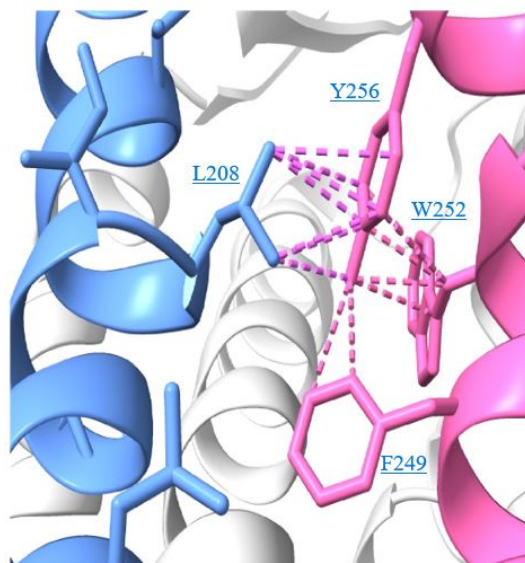
(33) Reevaluate the statement regarding the relevance of the Y256F mutation for the binding of AGR1.137. If there is a significant impact on migration in the mutant (Fig. 6B), elaborate on the significance in the context of AGR1.137 binding.

In the revised discussion we provide more detail on the relevance of Y256F mutation for the binding of AGR1.137 as well as for the partial effect of G207I and R235L mutations. The predicted interactions for each compound are depicted in new Fig. 6 C, D after LigPlot+ analysis (Laskowski R.A. & Swindells M.B. J. Chem. Inf. Model. 2011), showing that AGR1.135 interacted directly with the receptor through a hydrogen bond with Y256. When this residue was mutated to F, one of the anchor points for the compound was lost, weakening the potential interaction in the region of the upper anchor point.

It is not clear how the Y256F mutation will affect the binding of AGR1.137, but other potential contacts cannot be ruled out since that portion of the compound is identical in both AGR1.135 and AGR1.137. This is especially true for its neighboring residues in the alpha helix, F249, L208, as shown in 3ODU structure (Fig. 6D), which are shown to be directly implicated in the interaction of both compounds. Alternatively, we cannot discard that Y256 interacts with other TMs or lipids stabilizing the overall structure, which could reverse the effect of the mutant at a later stage (Author response image 3).

Author response image 3.

Cartoon representation of Y256 and its intramolecular interactions in the CXCR4 X-ray solved structure 3ODU. TMV helix is colored in blue and TMVI in pink.

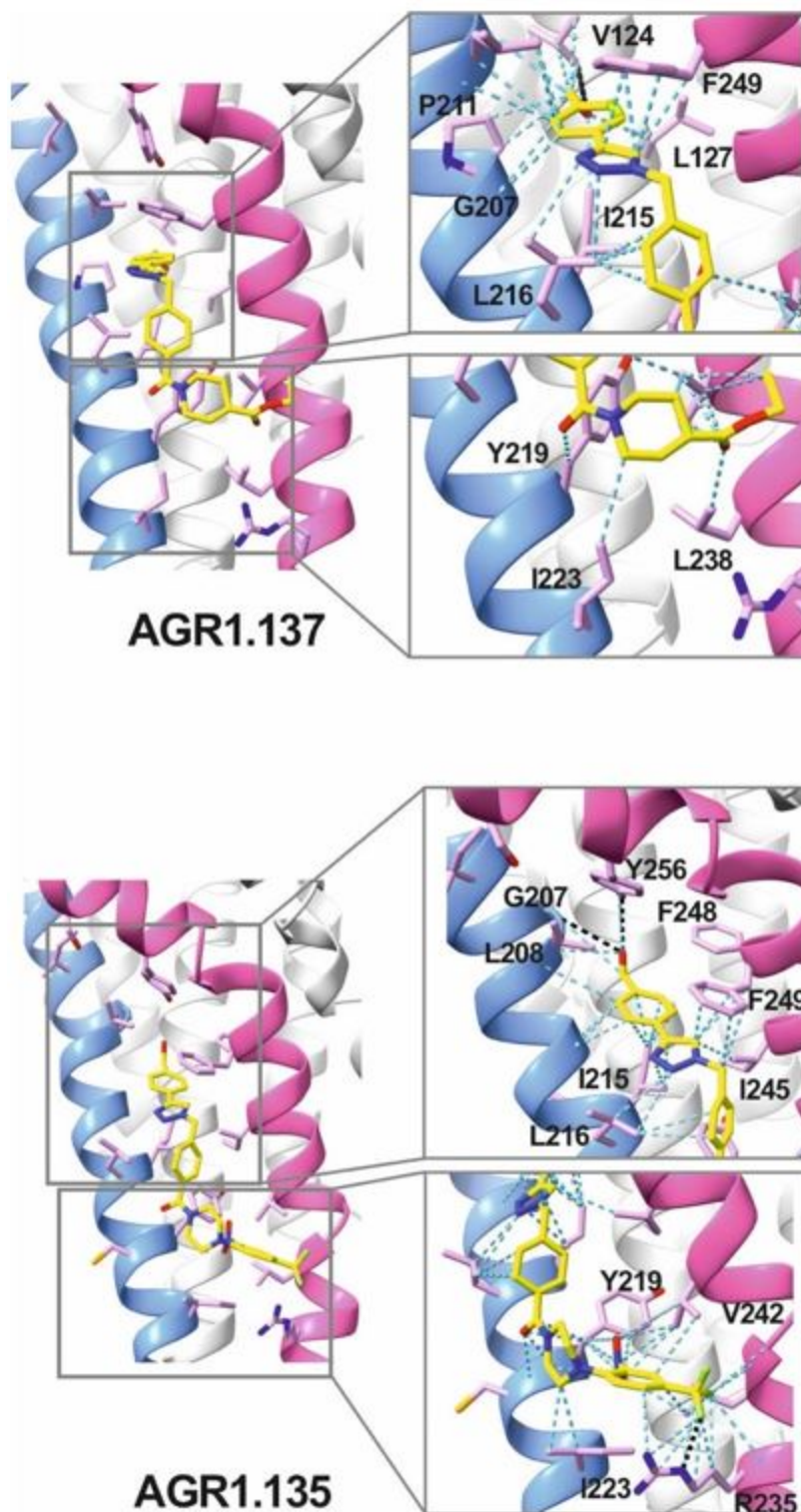


(34) Address the apparent discrepancy in residue involvement between AGR1.135 and AGR1.137, particularly if they share the same binding mode in the same cleft.

AGR1.135 and AGR1.137 exhibit comparable yet distinct binding modes, engaging with CXCR4 within a molecular cavity formed by TMV and TMVI. AGR1.135 binds to CXCR4 through three hydrogen bonds, two on the apical side of the compound that interact with residues TMV-G207 and TMVI-Y256 and one on the basal side that interacts with TMVI-R235 (Fig. 5A). This results in a more extended and rigid conformation when sharing hydrogen bonds, with both TMs occupying a surface area of 400 Å² and a length of 20 Å in the cleft between TMV and TMVI (Supplementary Fig. 8A). AGR1.137 exhibits a distinct binding profile, interacting with a more internal region of the receptor. This interaction involves the formation of a hydrogen bond with TMIIIV124, which induces a conformational shift in the TMVI helix towards an active conformation (Fig. 5B; Supplementary Fig. 13). Moreover, AGR1.137 may utilize the carboxyl group of V124 in TMIII and overlap with AGR1.135 binding in the cavity, interacting with the other 19 residues dispersed between TMV and VI to create an interaction surface of 370 Å² along 20 Å (Supplementary Fig. 8B). This is illustrated in the new Fig. 5B. AGR1.137 lacks the phenyl ring present in AGR1.135, resulting in a shorter compound with greater difficulty in reaching the lower part of TMVI where R235 sits.

Author response image 4.

AGR1.135 and AGR1.137 interaction with TMV and TMVI. The model shows the location of the compounds within the TMV-VI cleft, illustrated by a ribbon and stick representation. The CXCR4 segments of TMV and TMVI are represented in blue and pink ribbons respectively, and side chains for some of the residues defining the cavity are shown in sticks. AGR1.135 and AGR1.137 are shown in stick representation with carbon in yellow, nitrogen in blue, oxygen in red, and fluorine in green. Hydrogen bonds are indicated by dashed black lines, while hydrophobic interactions are shown in green. The figure reproduces the panels A, B of Fig. 5 in the revised manuscript.



(35) In the Results section regarding "AGR1.137 treatment in a zebrafish xenograft model", the following points can be refined for clarity and completeness: 1. Cell Line Choice for Zebrafish Xenograft Model:

-Explain the rationale behind the choice of HeLa cells for the zebrafish xenograft model when the previous experiments primarily focused on Jurkat cells. Address any specific

biological or experimental considerations that influenced this decision.

As far as we know, there are no available models of tumors in zebrafish using Jurkat cells. We looked for a tumoral cell system that expresses CXCR4 and could be transplanted into zebrafish. HeLa cells are derived from a human cervical tumor, express a functional CXCR4, and have been previously used for tumorigenesis analyses in zebrafish (Brown H.K. et al. Expert Opin. Drug Discover. 2017; You Y. et al Front. Pharmacol. 2020). These cells grow in the fish and disseminate through the ventral area and can be used to determine primary tumor growth and metastasis. Nonetheless, we first analyzed in vitro the expression of a functional CXCR4 in these cells (Supplementary Fig. 10A), whether AGR1.137 treatment specifically abrogated CXCL12-mediated direct cell migration (Fig. 7A, B), as whether it affected cell proliferation (Supplementary Fig. 10B). As HeLa cells reproduce the in vitro effects detected for the compounds in Jurkat cells, we used this model in zebrafish. These issues were already discussed in the first version of our manuscript.

(36) 2. Toxicity Assessment in Zebrafish Embryos:

-Clarify the basis for stating that AGR1.137 is not toxic to zebrafish embryos. Consider referencing the Zebrafish Embryo Acute Toxicity Test (ZFET) and provide relevant data on lethal concentration (LC50) and non-lethal toxic phenotypes such as pericardial edema, head and tail necrosis, malformation, brain hemorrhage, or yolk sac edema.

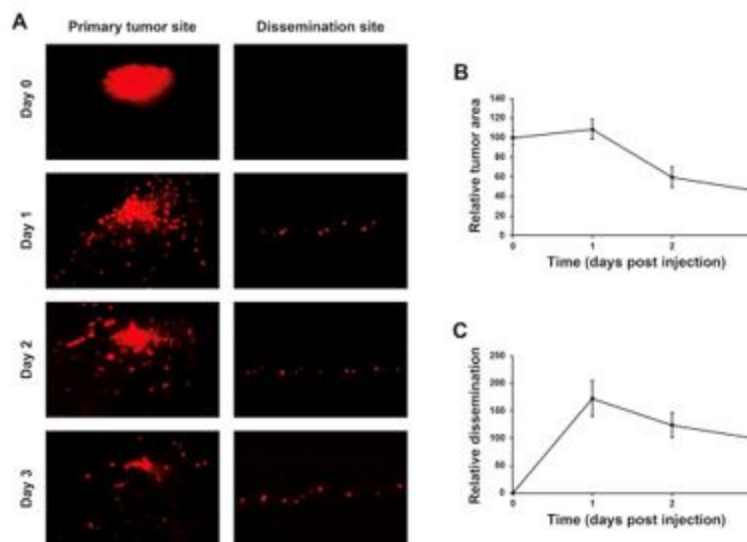
Tumor growth and metastasis kinetics within the zebrafish model have been extensively evaluated in many publications (White R. et al. Nat. Rev. Cancer. 2013; Astell K.R. and Sieger D. Cold Spring Harb. Perspect. Med. 2020; Chen X. et al. Front. Cell Dev. Biol. 2021; Weiss JM. Et al. eLife 2022; Lindhal G. et al NPJ Precis. Oncol. 2024). Our previous experience using this model shows that tumors start having a more pronounced proliferation and lower degree of apoptosis from day 4 onwards, but we cannot keep the tumor-bearing larvae for that long due to ethical reasons and also because we don't see much scientific benefit of unnecessarily extending the experiments. Anti-proliferative or pro-apoptotic effects of drugs can still be observed within the three days, even if this is then commonly seen as larger reduction (instead of a smaller growth as it is commonly seen in for example mouse tumor models) compared to controls. Initially we characterized the evolution of implanted tumors in our system and how much they metastasize over time in the absence of treatment before to test the compounds (Author response image 5).

The in vivo experiments were planned to validate efficacious concentrations of the investigated drugs rather than to derive in vivo IC50 or other values, which require testing of multiple doses. We have, however, included an additional concentration to show concentration-dependence and therefore on-target specificity of the drugs in the revised version of the manuscript (data also being elaborated in ongoing experiments). At this stage, we believe that adding the LC50 does not provide interesting new knowledge, and it is standard to only show results from the experimental endpoint (in our case 3 days post implantation). We agree that showing these new data points strengthens the manuscript and facilitates independent evaluation and conclusions to be drawn from the presented data. We have created new graphs where datapoints for each compound dose are shown.

Author response image 5.

Evolution of the tumors and metastasis along the time in the absence of any treatment. HeLa cells were labeled with 8 µg/mL Fast-DiI™ oil and then implanted in the dorsal perivitelline space of 2-days old zebrafish embryos. Tumors were imaged within 2 hours of implantation and re-imaged each 24 h for three days. Changes in tumor size was evaluated as tumor area at day 1, 2 and 3 divided by tumor area at day 0, and metastasis was evaluated as the number

of cells disseminated to the caudal hematopoietic plexus at day 1, 2 and 3 divided by the number of cells at day 3.



Regarding the statement that AGR1.137 was not toxic, this was based on visual inspection of the zebrafish larvae at the end of the experiment, which also revealed a lack of drug-related mortality in these experiments. There are a number of differences in how our experiment was run compared with the standardized ZFET. ZFET evaluates toxicity from 0 hours post-fertilization to 1 or 2 days post-fertilization, whereas here we exposed zebrafish from 2 days post-fertilization to 5 days post-fertilization. The ZFET furthermore requires that the embryos are raised at 26°C whereas kept the temperature as close as possible to a physiologically relevant temperature for the tumor cells (36°C). In the ZFET, embryos are incubated in 96-well plates whereas for our studies we required larger wells to be able to manipulate the larvae and avoid well edge-related imaging artefacts, and we therefore used 24-well plates. As such, the ZFET was for various reasons not applicable to our experimental settings. As we were not interested in rigorously determining the LD50 or other toxicity-related measurements, as our focus was instead on efficacy and we found that the targeted dose was tolerated, we did not evaluate multiple doses, including lethal doses of the drug, and are therefore not able to determine an LD50/LC50. We also did not find drug-induced non-lethal toxic phenotypes in this study, and so we cannot elaborate further on such phenotypes other than to simply state that the drug is well tolerated at the given doses. Therefore, the reference to ZFET in the manuscript was eliminated.

(37) If supplementary information is available, consider providing it for a comprehensive understanding of toxicity assessments.

The effective concentration used in the zebrafish study was derived from the in vitro experiments. That being said, and as elaborated in our response to comment 36, we have added data for one additional dose to show the dose-dependent regulation of tumor growth and metastasis.

(38) 3. Optimization and Development of AGR1.137:

-Justify the need for further optimization and development of AGR1.137 if it has a comparable effect to AMD3100. Explain the specific advantages or improvements that AGR1.137 may offer over AMD3100.

AGR1.137 is highly hydrophobic and is very difficult to handle, particularly in in vivo assays; thus, for the negative allosteric modulators to be used clinically, it would be very important to increase their solubility in water. Contrastingly, AMD3100 is a water-soluble compound. Before using the zebrafish model, we performed several experiments in mice using AGR1.137, but the inhibitory results were highly variable, probably due to its hydrophobicity. We also believe that it would be important to increase the affinity of AGR1.137 for CXCR4, as the use of lower concentrations of the negative allosteric modulator would limit potential in vivo side effects of the drug. On the other hand, we are also evaluating distinct administration alternatives, including encapsulation of the compounds in different vehicles. These alternatives may also require modifications of the compounds.

AMD3100 is an orthosteric inhibitor and therefore blocks all the signaling cascades triggered by CXCL12. For instance, we observed that AMD3100 treatment blocked CXCL12 binding, cAMP inhibition, calcium flux, cell adhesion and cell migration (Fig. 3, Fig. 4), whereas the effects of AGR1.137 were restricted to CXCL12-mediated directed cell migration. Although AMD3100 was well tolerated by healthy volunteers in a single-dose study, it also promoted some mild and reversible events, including white blood cells count elevations and variations of urine calcium just beyond the reported normal range (Hendrix C.W. et al. Antimicrob. Agents Chemother. 2000). To treat viral infections, continuous daily dosing requirements of AMD3100 were impractical due to severe side effects including cardiac arrhythmias (De Clercq E. Front Immunol. 2015). For AMD3100 to be used clinically, it would be critical to control the timing of administration. In addition, side effects after long-term administration have potential problems. Shorter-term usage and lower doses would be fundamental keys to its success in clinical use (Liu T.Y. et al. Exp. Hematol. Oncol. 2016). The use of a negative allosteric modulator that block cell migration but do not affect other signaling pathways triggered by CXCL12 would be, at least in theory, more specific and produce less side effects. These ideas have been incorporated into the revised discussion to reflect potential advantages or improvements that AGR1.137 may offer over AMD3100.

(39) 4. Discrepancy in AGR1.137 and AMD3100 Effects:

-Discuss the observed discrepancy where AGR1.137 exhibits similar effects to AMD3100 but only after 48 hours. Provide insights into the temporal dynamics of their actions and potential implications for the experimental design.

Images and data shown in Fig. 7E, F correspond to days 0 and 3 after HeLa cell implantation (tumorigenesis) and only to day 3 in the case of metastasis data. The revised version contains the effect of two distinct doses of the compounds (10 and 50 μ M, for AGR1.135 and AGR1.137 and 1 and 10 μ M for AMD3100).

(40) In the "Discussion" section, there are several points that require clarification and refinement to enhance the overall coherence and depth of the analysis: 1. Reduction of Side-Effects:

-Provide a more detailed explanation of how the identified compounds, specifically AGR1.135 and AGR1.137, contribute to the reduction of side effects. Consider discussing specific mechanisms or characteristics that differentiate these compounds from existing antagonists.

The sentence indicating that AGR1.135 and AGR1.137 contribute to reduce side effects is entirely speculative, as we have no experimental evidence to support it. We have therefore corrected this in the revised version. The origin of the sentence was that orthosteric antagonists typically bind to the same site as the endogenous ligand, thus blocking its interaction with the receptor. Therefore, orthosteric inhibitors (i.e. AMD3100) block all

signaling cascades triggered by the ligand and therefore their functional consequences. However, the compounds described in this project are essentially negative allosteric modulators, that is, they bind to a site distinct from the orthosteric site, inducing a conformational change in the receptor that does not alter the binding of the endogenous ligand, and therefore block some specific receptor-associated functions without altering others. We observed that AGR1.137 blocked receptor oligomerization and directed cell migration whereas CXCL12 still bound CXCR4, triggered calcium mobilization, did not inhibit cAMP release or promoted receptor internalization. This is why we speculated on the limitation of side effects. The statements have been nonetheless revised in the new version of the manuscript.

(41) 2. Binding Site Clarification:

-Address the apparent discrepancy between docking the small compounds in a narrow cleft formed by TMV and TMVI helices and the statement that AGR1.131 binds elsewhere. Clarify the rationale behind this assertion

After the in silico screening, a total of 40 compounds were selected. These compounds showed distinct degrees of interaction with the cleft formed by TMV and TMVI and even with other potential interaction sites on CXCR4, with the exception of the ligand binding site according to the data described by Wescott et al. (PNAS 2016 113:9928-9933), as this possibility was discarded in the initial approach of the in silico screening. According to PELE analysis, AGR1.131 was one of the 40 selected compounds that showed a pose with low binding energy, -39.8 kcal/mol, between TMV and TMVI helices, that is, it might interact with CXCR4 through the selected area for the screening. It nonetheless also showed a best pose placed between helices TMI and TMVII, -43.7 kcal/mol. In any case, the compound was included in the biological screening, where it was unable to impact CXCL12-mediated chemotaxis (Fig. 1B). We then focused on AGR1.135 and AGR1.137, as showed a higher inhibitory effect on CXCL12-mediated migration, and on AGR1.131 as an internal negative control. AGR1.131 has a skeleton very similar to the other compounds (Fig. 1C) and can interact with the TM domains of CXCR4 without promoting effects. None of the three compounds affected CXCL12 binding, or CXCL12-mediated inhibition of cAMP release, or receptor internalization. However, whereas AGR1.135 and AGR1.137, blocked CXCL12-mediated CXCR4 oligomerization and directed cell migration towards CXCL12 gradients, AGR1.131 had no effect in these experiments (Fig. 3, Fig. 4).

Next, we performed additional theoretical calculations (PELE, docking, MD) to inspect in detail the potential binding modes of active and inactive molecules. Based on these additional calculations, we identified that whereas AGR1.135 and AGR1.137 showed preferent binding on the molecular pocket between TMV and TMVI, the best pose for AGR1.131 was located between TMI and TMVII, as the initial experiments indicated. These observations and data have been clarified in the revised discussion.

(42) 3. Impact of Chemical Modifications:

-Discuss the consequences of the distinct chemical groups in AGR1.135, AGR1.137, and AGR1.131, specifically addressing how variations in amine length and chemical nature may influence binding affinity and biological activity. Provide insights into the potential effects of these modifications on cellular responses and the observed outcomes in zebrafish.

The main difference between AGR1.131 and the other two compounds is the higher flexibility of AGR1.131 due to the additional CH₂ linker, together with the lack of a piperazine ring. The additional CH₂ linking the phenyl ring increases the flexibility of AGR1.131 when compared

with AGR1.135 and AGR1.137, and the absence of the piperazine ring might be responsible for its lack of activity, as it makes this compound able to bind to CXCR4 (Fig. 1C).

AGR1.137 was chosen in a second round. The additional presence of the tertiary amine (in the piperazine ring) allows the formation of quaternary ammonium salts in the aqueous medium and its substituents to increase its solubility (Fig 1C). This characteristic might be related to the absence of toxic effects of the compound in the zebrafish model.

(43) 4. Existence of Distinct CXCR4 Conformational States:

-Provide more detailed support for the statement suggesting the "existence of distinct CXCR4 conformational states" responsible for activating different signaling pathways. Consider referencing relevant studies or experiments that support this claim.

Classical models of GPCR allostery and activation, which describe an equilibrium between a single inactive and a single signaling-competent active conformation, cannot account for the complex pharmacology of these receptors. The emerging view is that GPCRs are highly dynamic proteins, and ligands with varying pharmacological properties differentially modulate the balance between multiple conformations.

Just as a single photograph from one angle cannot capture all aspects of an object in movement, no one biophysical method can visualize all aspects of GPCR activation. In general, there is a tradeoff between high-resolution information on the entire protein versus dynamic information on limited regions. In the former category, crystal and cryo-electron microscopy (cryoEM) structures have provided comprehensive, atomic-resolution snapshots of scores of GPCRs both in inactive and active conformations, revealing conserved conformational changes associated with activation. However, different GPCRs vary considerably in the magnitude and nature of the conformational changes in the orthosteric ligand-binding site following agonist binding (Venkatakrisnan A.J.V. et al. Nature 2016). Spectroscopic and computational approaches provide complementary information, highlighting the role of conformational dynamics in GPCR activation (Latorraca N.R.V. et al. Chem. Rev 2017). In the absence of agonists, the receptor population is typically dominated by conformations closely related to those observed in inactive-state crystal structures (Manglik A. et al. Cell 2015). While agonist binding drives the receptor population towards conformations similar to those in activestate structures, a mixture of inactive and active conformations remains, reflecting “loose” or incomplete allosteric coupling between the orthosteric and transducer pockets (Dror R.O. et al. Proc. Natl. Acad. Sci. USA 2011). Surprisingly, for some GPCRs, and under some experimental conditions, a substantial fraction of unliganded receptors already reside in an active-like conformation, which may be related to their level of basal or constitutive signaling (Staus D.P. et al. J. Biol. Chem. 2019); Ye L. et al. Nature 2016). In our case, the negative allosteric modulators, (Staus DP, et al. J. Biol. Chem 2019); Ye L. et al. Nature 2016) did not alter ligand binding and had only minor effects on specific CXCL12-mediated functions such as inhibition of cAMP release or receptor internalization, among others, but failed to regulate CXCL12-mediated actin dynamics and receptor oligomerization. Collectively, these data suggest that the described compounds alter the active conformation of CXCR4 and therefore support the presence of distinct receptor conformations that explain a partial activation of the signaling cascade.

All these observations are now included in the revised discussion of the manuscript.

(44) 5. Equilibrium Shift and Allosteric Ligands:

-Clarify the statement about "allosteric ligands shifting the equilibrium to favor a particular receptor conformation". Support this suggestion with references or experimental evidence

In a previous answer (see our response to point 2), we explain why we define the compounds as negative allosteric modulators. These compounds do not bind the orthosteric binding site or a site distinct from the orthosteric site that alters the ligand-binding site. Their effect should be due to changes in the active conformation of CXCR4, which allow some signaling events whereas others are blocked. Our functional data thus support that through the same receptor the compounds separate distinct receptor-mediated signaling cascades, that is, our data suggest that CXCR4 has a conformational heterogeneity. It is known that GPCRs exhibit more than one “inactive” and “active” conformation, and the endogenous agonists stabilize a mixture of multiple conformations. Biased ligands or allosteric modulators can achieve their distinctive signaling profiles by modulating this distribution of receptor conformations. (Wingler L.M. & Lefkowitz R.J. Trends Cell Biol. 2020). For instance, some analogs of angiotensin II do not appreciably activate Gq signaling (e.g., increases in IP3 and Ca²⁺) but still induce receptor phosphorylation, internalization, and mitogen-activated protein kinase (MAPK) signaling (Wei H, et al. Proc. Natl. Acad. Sci. USA 2003). Some of these ligands activate Gi and G12 in bioluminescence resonance energy transfer (BRET) experiments (Namkung Y. et al. Sci. Signal. 2018). A similar observation was described in the case of CCR5, where some chemokine analogs promoted G protein subtype-specific signaling bias (Lorenzen E. et al. Sci. Signal 2018). Structural analysis of distinct GPCRs in the presence of different ligands vary considerably in the magnitude and nature of the conformational changes in the orthosteric ligand-binding site following agonist binding (Venkatakrishnan A.J.V. et al. Nature 2016). Yet, these changes modify conserved motifs in the interior of the receptor core and induce common conformational changes in the intracellular site involved in signal transduction. That is, these modifications might be considered distinct receptor conformations.

The revised discussion contains some of these interpretations to support our statement about the stabilization of a particular receptor conformation triggered by the negative allosteric modulators.

(45) 6. Refinement of Binding Mode:

-Clarify the workflow for obtaining the binding mode, particularly the role of GLIDE and PELE. Clearly explain how these software tools were used in tandem to refine the binding mode.

The computational sequential workflow applied in this project included, i) Protein model construction, ii) Virtual screening (Glide), iii) PELE, iv) Docking (AutoDock and Glide) and v) Molecular Dynamics (AMBER).

Glide was applied for the structure-based virtual screening to explore which compounds could fit and interact with the previously selected binding site.

After the identification of theoretically active compounds (modulators of CXCR4), additional calculations were done to identify a potential binding site. PELE was used in this sense, to study how the compounds could bind in the whole surface of the target (TMV-TMVI). By applying PELE, we avoided biasing the calculation, and we found that the trajectories with better interaction energies identified the cleft between TMV and TMVI as the binding site for AGR1.135 and AGR1.137, and not for AGR1.131. AGR1.131 showed a pose with low binding energy, -39.8 kcal/mol, between TMV and TMVI helices, that is, it might interact with CXCR4 in the selected area for the screening. But it also showed a better pose placed between helices TMI and TMVII, - 43.7 kcal/mol (see our response to point 41). These data have been now confirmed using Schrodinger's MM-GBSA procedure (see our response to points 6 and 8). In any case, the compound was included in the biological screening, where it was unable to affect CXCL12-mediated chemotaxis (Fig. 1B). Docking and MD simulations were then performed to study and refine the specific binding mode in this cavity. These data were important to choose the mutations on CXCR4 required, to test whether the compounds

reversed its behavior. In these experiments we also confirmed that AGR1.131 had a better pose on the TMI-TMVII region.

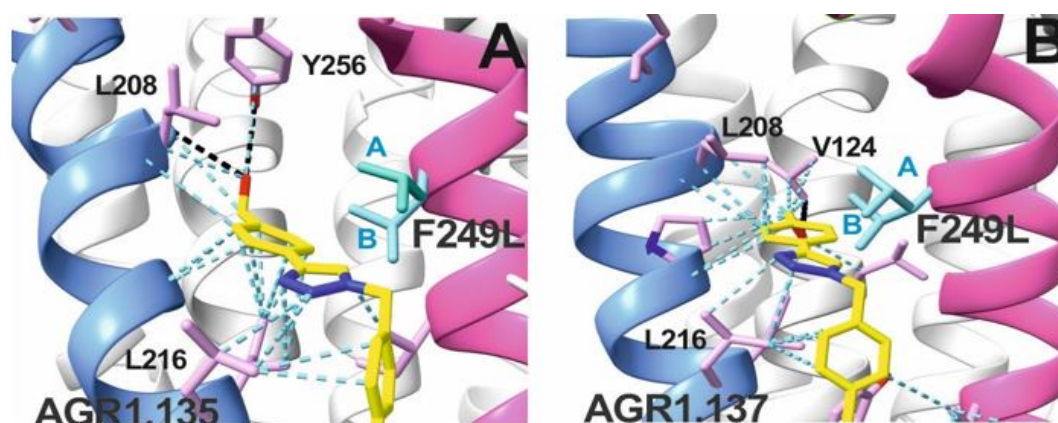
(46) 7. Impact of Compound Differences on CXCR4-F249L mutant:

-Provide visual aids, such as figures, and additional experiments to support the statement about differences in the behavior of AGR1.135 and AGR1.137 on cells expressing CXCR4-F249L mutant. Elaborate on the closer interaction suggested between the triazole group of AGR1.137 and the F249 residue

At the reviewer's suggestion, Fig. 5 has been modified to incorporate a closer view of the interactions identified and new panels in new Fig. 6 have been added to show in detail the effect of the mutations selected on the structure of the cleft between TMV and TMVI. The main difference between AGR1.135 and AGR1.137 is how the triazole group interacts with F249 and L216 (Author response image 6). In AGR1.137, the three groups are aligned in a parallel organization, which appears to be more effective: This might be due to a better adaptation of this compound to the cleft since there is only one hydrogen bond with V124. In AGR1.135, the compound interacts with the phenyl ring of F249 and has a stronger interaction at the apical edge to stabilize its position in the cleft. However, there is still an additional interaction present. When changing F249

Author response image 6.

Cartoon representation of the interaction of CXCR4 F249L mutant with AGR1.135 (A) and AGR1.137 (B). The two most probable conformations of Leucine rotamers are represented in cyan A and B conformations. Van der Waals interactions are depicted in blue cyan dashed lines, hydrogen bonds in black dashed lines. CXCR4 segments of TMV and TMVI are colored in blue and pink, respectively



to L (Fig. VIIA, B, only for review purposes) and showing the two most likely rotamers resulting from the mutation, it is observed that rotamer B is in close proximity to the compound, which may cause the binding to either displace or adopt an alternative conformation that is easier to bind into the cleft. As previously mentioned, it is likely that AGR1.135 can displace the mutant rotamer and bind into the cleft more easily due to its higher affinity.

(47) In the "Materials and Methods" section, the computational approach for the "discovery of CXCR4 modulators" requires significant revision and clarification. The following suggestions aim to address the identified issues: 1. Structural Modeling:

-Reconsider the use of SWISS-MODEL if there is an available PDB code for the entire CXCR4 structure. Clearly articulate the rationale for choosing one method over the other and explain any limitations associated with the selected approach.

The SWISS-model server allows for automated comparative modeling of 3D protein structures that was pioneered in the fields of automated modeling. At the time we started this project, it was the most accurate method to generate reliable 3D protein structure models.

As explained above, we have now predicted the structure of the target using AlphaFold (Jumper J. et al, Nature 2021) and performed several additional experiments that confirm that the small compounds bind the selected pocket as the original strategy indicated (see our response to point 6). (Fig. II, only for review purposes).

(48) 2. Parametrization of Small Compounds:

-Provide a detailed description of the parametrization process for the small compounds used in the study. Specify the force field and parameters employed, considering the obsolescence of AMBER14 and ff14SB. Consider adopting more contemporary force fields and parameterization strategies.

When we performed these experiments, some years ago, the force fields applied (ff14SB, AMBER14 used in MD or OPLS2004 in docking with Glide) were well accepted and were gold standards. It is, however, true that the force fields have evolved in the past few years, Moreover, in the case of the MD simulations, to consider the parameters of the ligands that are not contained within the force field, we performed an additional parameterization as a standard methodology. We then generated an Ab initio optimization of the ligand geometry, defining as basis sets B3LYP 6-311+g(d), using Gaussian 09, Revision A.02, and then a single point energy calculation of ESP charges, with HF 6311+g(d) on the optimized structure. As the last step of the parametrization, the antechamber module was used to adapt these charges and additional parameters for MD simulations.

(49) 3. Treatment of Lipids and Membrane:

-Elaborate on how lipids were treated in the system. Clearly describe whether a membrane was included in the simulations and provide details on its composition and structure. Address the role of the membrane in the study and its relevance to the interactions between CXCR4 and small compounds

To stabilize CXCR4 and more accurately reproduce the real environment in the MD simulation, the system was embedded in a lipid bilayer using the Membrane Builder tool (Sunhwan J. et al. Biophys. J. 2009) from the CHARMM-GUI server. The membrane was composed of 175 molecules of the fatty acid 1-palmitoyl-2-oleoyl-sn-glycero-3phosphocholine (POPC) in each leaflet. The protein-membrane complex was solvated with TIP3 water molecules. Chloride ions were added up to a concentration of 0.15 M in water, and sodium ions were added to neutralize the system. This information was previously described in detail.

(50) 4. Molecular Dynamics Protocol:

-Provide a more detailed and coherent explanation of the molecular dynamics protocol. Clarify the specific steps, parameters, and conditions used in the simulations. Ensure that the protocol aligns with established best practices in the field.

Simulations were calculated on an Asus 1151 h170 LVX-GTX-980Ti workstation, with an Intel Core i7-6500 K Processor (12 M Cache, 3.40 GHz) and 16 GB DDR4 2133 MHz RAM, equipped

with a Nvidia GeForce GTX 980Ti available for GPU (Graphics Processing Unit) computations. MD simulations were performed using AMBER14 (Case D.A. et al. AMBERT 14, Univ. of California, San Francisco, USA, 2014) with ff14SB (Maier J.A. et al. J. Chem. Theory Comput. 2015) and lipid14 (Dickson C. J. et al. J. Chem. Theory Comput. 2014) force fields in the NPT thermodynamic ensemble (constant pressure and temperature). Minimization was performed using 3500 Steepest Descent steps and 4500 Conjugate Gradient steps three times, firstly considering only hydrogens, next considering only water molecules and ions, and finally minimizing all atoms. Equilibration raises system temperature from 0 to 300 K at a constant volume fixing everything but ions and water molecules. After thermalization, several density equilibration phases were performed. In the production phase, 50 ns MD simulations without position restraints were calculated using a time step of 2 fs. Trajectories of the most interesting poses were extended to 150 ns. All bonds involving hydrogen atoms were constrained with the SHAKE algorithm (Lippert R.A. et al. J. Chem. Phys. 2007). A cutoff of 8 Å was used for the Lennard-Jones interaction and the short-range electrostatic interactions. Berendsen barostat (Berendsen H.J. et al. J. Chem. Phys. 1984) and Langevin thermostat were used to regulate the system pressure and temperature, respectively. All trajectories were processed using CPPTRAJ (Roe D.R. & Cheatham III T.E. J. Chem. Theory Comput. 2013) and visualized with VMD (Visual Molecular Dynamics) (Humphrey W. et al. J. Mol. Graphics. 1996). To reduce the complexity of the data, Principal Component Analysis (PCA) was performed on the trajectories using CPPTRAJ.

(51) Consider updating the molecular dynamics protocol to incorporate more contemporary methodologies, considering advancements in simulation techniques and software.

In our answer to points 6 and 47, we describe why we use the technology based on Swiss-model and PELE analysis and how we have now used Alphafold and other more contemporary methodologies to confirm that the small compounds bind the selected pocket.

(52) Figure 1A:

- Consider switching to a cavity representation for CXCL12 to enhance clarity and emphasize the cleft.

Fig. 1A has been modified to emphasize the cleft.

(53) Explicitly show the TMV-TMVI cleft in the figure for a more comprehensive visualization.

In Fig. 1A we have added an insert to facilitate TMV-TMVI visualization.

(54) Figure 1B:

- Clearly explain the meaning of the second DMSO barplot to avoid confusion.

To clarify this panel, we have modified the figure and the figure legend. Panel B now includes a complete titration of the three compounds analyzed in the manuscript. The first bar shows cell migration in the absence of both treatment with AMD3100 and stimulation with CXCL12. The second bar shows migration in response to CXCL12 in the absence of AMD3100. The third bar shows the effect of AMD3100 on CXCL12-induced migration, as a known control of inhibition of migration. We hope that this new representation of the data results is clearer.

(55) Figure 1C:

- Provide a clear legend explaining the significance of the green shading on the small compounds.

The legend for Fig. 1C has been modified accordingly to the reviewer's suggestion.

(56) Figure 2:

- Elaborate on the role of fibronectin in the experiment and explain the specific contribution of CD86-AcGFP.

The ideal situation for TIRF-M determinations is to employ cells on a physiological substrate complemented with or without chemokines. Fibronectin is a substrate widely used in different studies that allows cell adhesion, mimicking a physiological situation. Jurkat cells express alpha4beta1 and alpha5beta1 integrins that mediate adhesion to fibronectin (Seminario M.C. et al. J. Leuk. Biol. 1999).

Regarding the use of CD86-AcGFP in TIRF-M experiments. We currently determine the number of receptors in individual trajectories of CXCR4 using, as a reference, the MSI value of CD86-AcGFP that strictly showed a single photobleaching step (Dorsch S. et al. Nat Methods 2009).

We preferred to use CD86-AcGFP in cells instead of AcGFP on glass, to exclude any potential effect on the different photodynamics exhibited by AcGFP when bound directly to glass. In any case, this issue has been clarified in the revised version.

(57) Figure 3D:

- Include a plot for the respective band intensity to enhance data presentation

The plot showing the band intensity analysis of the experiments shown in Fig. 3D was already included in the original version (see old Supplementary Fig. 3). However, in the revised version, we include these plots in the same figure as panels 3E and 3F. As a control of inhibition of CXCL12 stimulation, we have also included a new figure (Supplementary Fig. 4) showing the effect of AMD3100 on CXCL12-induced activation of Akt and ERK as analyzed by western blot.

(58) Consider adding AMD3100 as a control for comparison.

In agreement with the reviewer's suggestion, we have added the effect of AMD3100 in most of the functional experiments performed.

(59) Figure 4:

- Address the lack of positive controls in Figure 4 and consider their inclusion for a more comprehensive analysis.

DMSO bars correspond to the control of the experiment, as they represent the effect of CXCL12 in the absence of any allosteric modulator. As previously described in this point-by-point reply, DMSO bars correspond to the control performed with the solvent with which the small compounds, at maximum concentration, are diluted. Therefore, they show the effect of the solvent on CXCL12 responses. In any case, and in order to facilitate the comprehension of the figure we have also added the controls in the absence of DMSO to demonstrate that the solvent does not affect CXCL12-mediated functions, together with the effect of the orthosteric inhibitor AMD3100. In addition, we have also included representative images of the effect of the different compounds on CXCL12-induced polarization (Fig. 4C).

(60) *In Figure 4A, carefully assess overlapping error bars and ensure accurate interpretation. If necessary, consider alternative representation.*

We have tried alternative representations of data in Fig. 4A, but in all cases the figure was unclear. We believe that the way we represent the data in the original manuscript is the most clear and appropriate. Nevertheless, we have now included significance values as a table annexed to the figure, as well as the effect of AMD3100, as a control of inhibition

(61) *Supplementary Figure 1A:*

- *Improve the clarity of bar plots for better understanding. Consider reordering them from the most significant to the least.*

This was a good idea, and therefore Supplementary Fig. 1A has been reorganized to improve clarity.

(62) *Supplementary Figure 1C:*

- *Clarify the rationale behind choosing the 12.5 nM concentration and explain if different concentrations of CXCL12 were tested.*

In old Supplementary Fig. 1C, we used untreated cells, that is, CXCL12 was not present in the assay. These experiments were performed to test the potential toxicity of DMSO (solvent) or the negative allosteric modulators on Jurkat cells. The 12.5 nM concentration of CXCL12 mentioned in the figure legend applied only to panels A and B, as indicated in the figure legend. We previously optimized this concentration for Jurkat cells using different concentrations of CXCL12 between 5 and 100 nM. Nevertheless, we have reorganized old supplementary fig. 1 and clarified the figure legend to avoid misinterpretations (see Supplementary Fig 1A, B and Supplementary Fig. 2A, B).

(63) *Explain the observed reduction in fluorescence intensity for AGR1.135.*

The cell cycle analysis has been moved from Supplementary Fig. 1C to a new Supplementary Fig. 2. It now includes the flow cytometry panels to show fluorescence intensity as a function of the number of cells analyzed (Panel 1A) as well as a table (panel B) with the percentage of cells in each phase of the cell cycle. We believe that the apparent reduction in fluorescence that the reviewer observes is mainly due to the number of events analyzed. However, we have changed the flow cytometry panels for others that are more representative and included a table with the mean of the different results. When we determined the percentage of cells in each cell cycle phase, we observed that it looks very similar in all the experimental conditions. That is, none of the compounds affected any of the cell cycle phases. We have also included the effect of H₂O₂ and staurosporine as control compounds inducing cell death and cell cycle alteration of Jurkat cells.

(64) *Supplementary Table 1:*

- *Include a column specifying the scoring for each compound to provide a clear reference for readers.*

To facilitate references to readers, we have now included the inhibitory effect of each compound on Jurkat cell migration in the revised version of this table.

(65) *Minor Points*

Page 2 - Abstract: Rephrase the first sentence of the abstract to enhance fluidity.

Although the entire manuscript was revised by a professional English editor, we appreciate the valuable comments of this reviewer and we have corrected these issues accordingly.

(66) Page 2 - Abstract: Explicitly define "CXCR4" as "C-X-C chemokine receptor type 4" the first time it appears.

We have not used C-X-C chemokine receptor type 4 the first time it appears in the abstract. CXCR4 is an acronym normally accepted to identify this chemokine receptor, and it is used as CXCR4 in many articles published in eLife. However, we introduce the complete name the first time it appears in the introduction.

(67) Page 2 - Abstract: Explicitly define "CXCL12" as "C-X-C motif chemokine 12" the first time it is mentioned.

As we have discussed in the previous response, we have not used C-X-C motif chemokine 12 the first time CXCL12 appears in the abstract, as it is a general acronym normally accepted to identify this specific chemokine, even in eLife papers. However, we introduce the complete name the first time it appears in the introduction section.

(68) Page 2 - Abstract: Explicitly define "TMV and TMVI" upon its first mention.

The acronym TM has been defined as “Transmembrane” in the revised version

(69) Page 2 - Abstract: Review the use of "in silico" in the sentence for accuracy and consider revising if necessary.

With the term “in silico” we want to refer to those experiments performed on a computer or via computer simulation software. We have carefully reviewed its use in the new version of the manuscript.

(70) Page 2 - Abstract: Add a comma after "compound" in the sentence, "We identified AGR1.137, a small compound that abolishes..."

A comma after “compound” has been added in the revised sentence.

(71) Page 2 - Significance Statement: Rephrase the first sentence of the "Significance Statement" to avoid duplication with the abstract.

The first sentence of the Significance Statement has been revised to avoid duplication with the abstract.

(72) Page 2 - Significance Statement: Break down the lengthy sentence, "Here, we performed in silico analyses..." for better readability.

The sentence starting by “Here, we performed in silico analyses...” has been broken down in the revised manuscript.

(73) Page 2 - Introduction: Replace "Murine studies" with a more specific term for clarity.

The term “murine studies” is normally used to refer to experimental studies developed in mice. We have nonetheless rephrased the sentence.

(74) Page 3 - Introduction: Rephrase the sentence for clarity: "Finally, using a zebrafish model, ..."

The sentence has been now rephrased for clarity.

(75) Results-AGR1.135 and AGR1.137 block CXCL12-mediated CXCR4 nanoclustering and dynamics:

Rephrase the sentence for clarity: "Retreatment with AGR1.135 and AGR1.137, but not with AGR1.131, substantially impaired CXCL12-mediated receptor nanoclustering."

The sentence has been rephrased for clarity.

(76) Results - AGR1.135 and AGR1.137 incompletely abolish CXCR4-mediated responses in Jurkat cells: Clarify the sentence: "In contrast to the effect promoted by AMD3100, a binding-site antagonist of CXCR4..."

The sentence has been modified for clarity.

(77) Consider using "orthosteric" instead of "binding-site" antagonist.

The term orthosteric is now used throughout to refer to a binding site antagonist.

(78) Discussion: Use the term "in silico" only when necessary.

We have carefully reviewed the use of "in silico" in the manuscript.

(79) Discussion: Clarify the sentence: "...not affect neither CXCR2-mediated cell migration...". Confirm if "CXCL12" is intended.

The sentence refers to the chemokine receptor CXCR2, which binds the chemokine CXCL2. To test the specificity of the compounds for the CXCL12/CXCR4 axis, we evaluated CXCL2-mediated cell migration. The results indicated that CXCL2/CXCR2 axis was not affected by the negative allosteric modulators, whereas CXCL12-mediated cell migration was blocked. The sentence has been clarified in the new version of the manuscript.

(80) Figure 4B: Bold the "B" in the figure label for consistency.

The "B" in Fig. 4B has been bolded.

Reviewer #2

(1) Fig 2. The SPT data is sub-optimal in its presentation as well as analysis. Example images should be shown. The analysis and visualization of the data should be reconsidered for improvements. Graphs with several hundreds, in some conditions over 1000 tracks, per condition are very hard to compare. The same (randomly selected representative set) number of data points should be shown for better visualization. Also, more thorough analyses like MSD or autocorrelation functions are lacking - they would allow enhanced overall representation of the data.

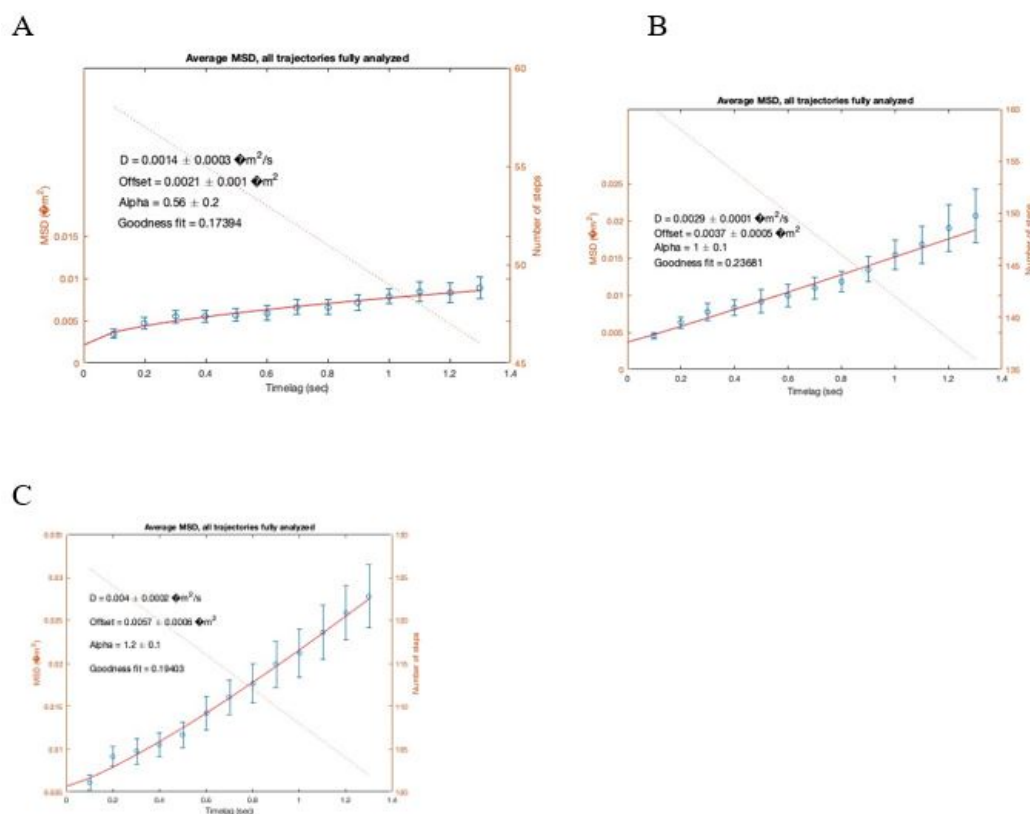
In agreement with the reviewer's commentary, we have modified the representation of Fig. 2. We have carefully read the paper published by Lord S.J. and col. (Lord S. J. et al., J. Cell Biol.

2020) and we apply their recommendations for these type of data. We have also included as supplementary material representative videos for the TIRF-M experiments performed to allow readers to visualize the original images. Regarding the MSD analyses, they were developed to determine all D1-4 values. According to the data published by Manzo & García-Parajo (Manzo C. & García-Parajo M.F. Rep.Prog. Phys. 2015) due to the finite trajectory length the MSD curve at large tlag has poor statistics and deviates from linearity. However, the estimation of the Diffusion Coefficient (D1-4) can be obtained by fitting of the short tlag region of the MSD plot giving a more accurate idea of the behavior of particles. In agreement we show D1-4 values and not MSD data.

Due to the space restrictions, it is very difficult to include all the figures generated, but, only for review purposes, we included in this point-by-point reply some representative plots of the MSD values as a function of the time from individual trajectories showing different types of motion obtained in our experiments (Author response image 7).

Author response image 7.

Representative MSD plots from individual trajectories of CXCR4-AcGFP showing different types of motion: A) confined, B) Brownian/Free, C) direct transport of CXCR4-AcGFP particles diffusing at the cell membrane detected by SPT-TIRF in resting JKCD4 cells.

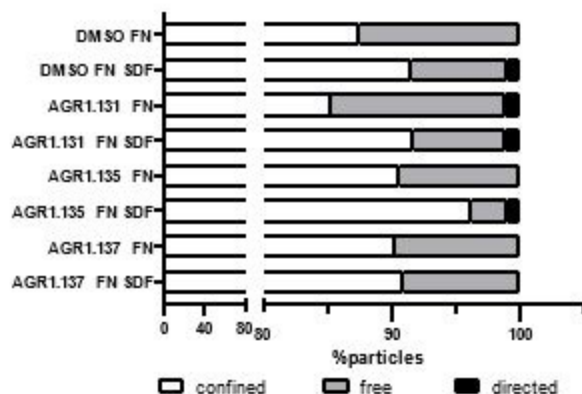


Further analysis, such as the classification based on particle motion, has not been included in this article. This classification uses the moment scaling spectrum (MSS), described by Ewers H. et al. 2005 PNAS, and requires particles with longer trajectories (>50 frames). Only for review purposes, we include a figure showing the percentage of the MSS-based particle motion classification for each condition. As expected, most of long particles are confined,

with a slight increase in the percentage upon CXCL12 stimulation in all conditions, except in cell treated with AGR1.137 (Author response image 8).

Author response image 8.

Effects of the negative allosteric modulators on the Types of Motion of CXCR4. Percentage of single trajectories with different types of motion, classified by MSS (DMSO: 58 particles in 59 cells on FN; 314 in 63 cells on FN+CXCL12; AGR1.131: 102 particles in 71 cells on FN; 258 in 69 cells on FN+CXCL12; AGR1.135: 86 particles in 70 cells on FN; 120 in 77 cells on FN+CXCL12; AGR1.137: 47 particles in 66 cells on FN; 74 in 64 cells on FN+CXCL12) n = 3.



(2) Fig 3. The figure legends have inadequate information on concentrations and incubation times used, both for the compounds and other treatments like CXCL12 and forskolin. For the Western blot data, also the quantification should be added to the main figure. The compounds, particularly AGR1.137 seem to lead to augmented stimulation of pAKT and pERK. This should be discussed

The Fig. 3 legend has been corrected in the revised manuscript. Fig. 3D now contains representative western blots and the densitometry evaluation of these experiments. As the reviewer indicates, we also detected in the western blot included, augmented stimulation of pAKT and pERK in cells treated with AGR1.137. However, as shown in the densitometry analysis, no significant differences were noted between the data obtained with each compound. As a control of inhibition of CXCL12 stimulation we have included a new Supplementary Fig. 4 showing the effect of AMD3100 on CXCL12-induced activation of Akt and ERK as analyzed by western blot.

(3) Fig. 4 immunofluorescence data on polarization as well as the flow chamber data lack the representative images of the data. The information on the source of the T cells is missing. Not clear if this experiment was done on bilayers or on static surfaces.

Representative images for the data shown in Figure 4B have been added in the revised figure (Fig. 4C). The experiments in Fig. 4B were performed on static surfaces. As indicated in the material and methods section, primary T cell blasts were added to fibronectin-coated glass slides and then were stimulated or not with CXCL12 (5 min at 37°C) prior to fix permeabilize and stain them with Phalloidin. Primary T cell blasts were generated from PBMCs isolated from buffy coats that were activated in vitro with IL-2 and PHA as indicated in the material and methods section.

(4) The data largely lacks titration of different concentrations of the compounds. How were the effective concentration and treatment times determined? What happens at

higher concentrations? It is important to show, for instance, if the CXCR12 binding gets inhibited at higher concentrations. most experiments were performed with 50 μ M, but HeLa cell data with 100 μ M. Why and how was this determined?

The revised version contains a new panel in Fig. 1B to show a more detailed kinetic analysis with different concentrations (1-100 μ M) of the compounds in the migration experiments using Jurkat cells. We choose 50 μ M for further studies as it was the concentration that inhibits 50-75% of the ligand induced cell migration.

We have also included the effect of two doses of the compounds (10 and 50 μ M) in the zebrafish model as well as AMD3100 (1 and 10 μ M) as control (new Fig. 7D, E). Tumors were imaged within 2 hours of implantation and tumor-bearing embryos were treated with either vehicle (DMSO) alone, AGR1.131 or AGR1.137 at 10 and 50 μ M or AMD3100 at 1 and 10 μ M for three days, followed by re-imaging.

Regarding the amount of CXCL12 used in these experiments, with the exception of cell migration assays in Transwells, where the optimal concentration was established at 12.5 nM, in all the other experiments the optimal concentration of CXCL12 employed was 50 nM. In the case of the directional cell migration assays, we use 100 nM to create the chemokine gradient in the device. These concentrations have been optimized in previous works of our laboratory using these types of experiments. It should also be noted that in the experiments using lipid bilayers or TIRF-M experiments, CXCL12 is used to coat the plates and therefore it is difficult to determine the real concentration that is retained in the surface after the washing steps performed prior adding the cells.

(5) The authors state that they could not detect direct binding of the compounds and the CXCR14. It should be reported what approaches were tried and discussed why this was not possible.

We attempted a fluorescence spectroscopy strategy to formally prove the ability of AGR1.135 to bind CXCR4, but this strategy failed because the compound has a yellow color that interfered with the determinations. We also tried a FRET strategy (see supplementary Fig. 7) and detected a significant increase in FRET efficiency of CXCR4 homodimers in cells treated with AGR1.135; this effect was due to the yellow color of this compound that interferes with FRET determinations. In the same assays, AGR1.137 did not modify FRET efficiency for CXCR4 homodimers and therefore we cannot assume that AGR1.137 binds on CXCR4. All these data have been considered in the revised discussion.

(6) The proliferation data in Supplementary Figure 1 lacks controls that affect proliferation and indication of different cell cycle stages. What is the conclusion of this data? More information on the effects of the drug to cell viability would be important.

Toxicity in Jurkat cells was first determined by propidium iodide incorporation. Some compounds (i.e., AGR1.103 and VSP3.1) were discarded from further analysis as they were toxic for cells. In a deeper analysis of cell toxicity, even if these compounds did not kill the cells, we checked whether they could alter the cell cycle of the cells. New Supplementary Fig. 2 includes a table (panel B) with the percentage of cells in each cell cycle phase, and no differences between any of the treatments tested were detected.

Nevertheless, to clarify this issue the revised version of the figure also includes H₂O₂ and staurosporine stimuli to induce cell death and cell cycle alterations as controls of these assays.

(7) The flow data in Supplementary Figure 2 should be statistically analysed.

Bar graphs corresponding to the old Supplementary Fig. 2 (new Supplementary Fig. 3) are shown in Fig. 3B. We have also incorporated the corresponding statistical analysis to this figure.

(8) In general, the authors should revise the figure legends to ensure that critical details are added.

We have carefully revised all the figure legends in the new version of the manuscript.

(9) Bar plots are very poor in showing the heterogeneity of the data. Individual data points should be shown whenever feasible. Superplot-type of representation is strongly advised (<https://doi.org/10.1083/jcb.202001064>).

We have carefully read the paper published by Lord S.J. and col. (Lord S. J. et al., J. Cell Biol. 2020) and we apply their recommendations for our TIRF-M data (see revised

Fig. 2).

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